# DOMINICAN COLLEGE OF TARLAC, INC.

# AGRICARE: A MOBILE APPLICATION FOR FARMING & LIVESTOCK RAISING IN THE PHILIPPINES

A Capstone Project Presented to
Dominican College of Tarlac, Inc.

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Information Technology

by:

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October 2025

# DOMINICAN COLLEGE OF TARLAC, INC.

## ADVISER'S RECOMMENDATION SHEET

This Capstone Project entitled

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And submitted in partial fulfillment of the requirements of the Bachelor of Science in Information Technology degree has been examined and is recommended for acceptance and approval

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October 2025

# DOMINICAN COLLEGE OF TARLAC, INC.

## DEAN’S ACCEPTANCE SHEET

This Capstone Project entitled

**AGRICARE: A MOBILE APPLICATION FOR FARMING & LIVESTOCK RAISING IN THE PHILIPPINES**

After having been recommended and approved is hereby accepted
by the College of Computer Studies Department
of Dominican College of Tarlac, Inc.

Rossano C. Samson, L.P.T
Dean

October 2025

# DOMINICAN COLLEGE OF TARLAC, INC.

## PANEL'S APPROVAL SHEET

This Capstone Project entitled

**AGRICARE: A MOBILE APPLICATION FOR FARMING & LIVESTOCK RAISING IN THE PHILIPPINES**

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# ABSTRACT

AgriCare is a mobile application designed to provide modern and accessible information on farming and livestock raising in the Philippines. This capstone project describes the conceptualization and development of the AgriCare application, emphasizing the importance of integrating technology into agricultural education. The documents also discuss the relevance of agriculture to the economy of the country and the role of technology in improving productivity and awareness. It also serves as a platform that promotes convenience by offering offline accessibility, user-friendly navigation, and content appropriate for local conditions. Farmers often rely on condition weather when planning an activity such as planting, harvesting, and livestock caring. While most of the aspiring farmers seeks or search for an information that helps them learn about farming. A key feature of AgriCare is its all-in-one feature, it has a weather forecast, crops and livestock information with the varieties of the mostly used in the Philippines, crops and livestock staging, calculator about crops and livestock and a scheduler that helps the user to remind in an upcoming farming activity. Lastly, AgriCare application was developed to help the aspiring farmers gain a knowledge about farming and guide them in making better agricultural decisions.

## ACKNOWLEDGEMENT

First and foremost, we give our deepest thanks to Almighty God for His divine guidance, wisdom, and protection throughout the completion of this capstone project. His blessings have strengthened us in times of difficulty and inspired us to persevere and finish this work with faith and determination.

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# 1.0 INTRODUCTION

## 1.1 Project Context

Agriculture is one of the essential foundations in producing foods such as crops, and livestock that are used to be managed by farmers. Livestock refers to raising an animal for a various purpose, such as food, clothing and other products, while crops is referring to a plant that produced a source of food and medical purposes. The productivity of agriculture is based on the understanding and techniques to ensure stable food supplies and products. There are 3 stages in farming such as pre-planting operation are operations needed to be done before sowing of the seeds, the failure of agricultural crop is most likely a result of poor pre-planting operations. These are series of operations carried out on the farm in preparation to planting seeds or planting materials. They are very important because they build a solid foundation for the success and survival of the crop. Growing is a farm activity between period after planting and before harvesting. They are important to achieve quality yield. Harvesting, this marks the end of cultivation. As soon as the crop reaches the full maturity, it should be harvested. Irrespective of any method of harvesting the farmer endorsed, it should not inflict any damage to the crop. This will increase the shelf life (life span) of the crop through storage and maintain the quality. While in livestock has the stages of pre-staging, growing stage, and breeding stage. The pre-staging phase involves selection of breeding, vaccination and diseases prevention and providing the right nutrients with appropriate housing conditions while growing stage is the monitoring of the health of animal growth without health complications. Then the last stage will be mating where the process of selection of compatible breeding pairs. [1]

Urban farming has become a growing trend in cities worldwide, offering a practical solution to food insecurity and a way to reconnect with nature. This innovative approach to agriculture involves cultivating, processing, and distributing food in urban areas, using rooftops, balconies, vacant lots, and community spaces. Urban farming not only addresses the challenges of limited space but also contributes to environmental sustainability and community development. As cities continue to expand, urban farming provides a way to produce fresh, nutritious food locally, reducing the need for long transportation routes. It’s an inspiring movement that encourages resource efficiency, fosters community engagement, and helps cities combat the effects of climate change. Whether you’re a beginner or an experienced gardener, urban farming offers endless opportunities to grow your food and make a positive impact. [2] The simplest definition of rural refers to areas that are not urban. They are characterized by open spaces, lower population densities, and a lifestyle often closely tied to nature and agriculture. Think of it as the opposite of a bustling city; instead of skyscrapers and heavy traffic, you'll find fields, forests, and a slower pace of life. Rural areas often have less developed infrastructure. This includes things like roads, public transportation, access to high-speed internet, and availability of services like hospitals and specialized retail. While this lack of infrastructure can present challenges, it also contributes to the slower pace and more self-sufficient lifestyle often associated with rural life. [3]

However, a common problem in agriculture technique is its dynamic nature which makes the ability of farmers outdated in modern farming, this problem in agricultural technique limits the ability to improve production, especially in preventing pests that can cause damage to crops and disease including infections in livestock that ensure the quality and the health of farming resources leading them to an economic loss. Many farmers continue to rely on traditional outdated techniques passed down through generations and generations. While these methods were proven to have worked in the past, these solutions often fail to address new challenges. As a result, these problems may lead to a huge loss prompting to wasting more time and resources. [4]

Agricultural burning, also known as slash-and-burn agriculture or kaingin, is a traditional practice that involves the cutting and burning of vegetation to clear land for farming. It has been a common farming practice in the Philippines for centuries, particularly among indigenous communities. Agricultural burning is a process that involves the clearing of land for farming by cutting down vegetation and burning the remains. The ash left behind provides nutrients for the soil, making it fertile for planting crops. This practice is commonly used in tropical countries with heavy rainfall and poor soil quality, where the vegetation is dense and difficult to remove manually. However, the practice has become controversial due to its negative impact on the environment. One of the major disadvantages is air pollution. Burning vegetation releases harmful pollutants such as carbon dioxide, carbon monoxide, and particulate matter into the air. Exposure to these pollutants can have adverse effects on human health, causing respiratory problems and other illnesses. [5]

Due to the tropical climate of the Philippines, it thrives in agricultural farming, providing source of income and products from farming and livestock. Rice, corn and sugarcane are popular crops in the Philippines PSB Rc18, PSB Rc82, NSIC Rc308, NSIC Rc354, NSIC Rc214, NSIC Rc298, NSIC Rc352, NSIC Rc358, NSIC Rc400, NSIC Rc402, PSB Rc10, NSIC Rc218, NSIC Rc82, NSIC Rc394, NSIC Rc302, NSIC Rc226, NSIC Rc360, NSIC Rc356, NSIC Rc224, NSIC Rc158 are examples of rice crops farmed in the Philippines. [6] Corn also has types and varieties most commonly farmed, which are white corn, which consists of different varieties (IPB Var6, IPB Var8, LB Lagkitan, IES Lagkitan, Native Corn, IPB Var7, IPB Var11, IPB Var13, Tiniguib, Putian, IPB Var12), and some of its variety is primarily used for livestock and poultry feed but is also processed into corn oil, starch, cereals and glutinous corn that are used in delicacies like "Binatog," "Kakanin," and "Nilaga.". Sweet corn is another type of corn crop farmed in the Philippines. [7] Sugarcane is popular due to its importance in industrial use and commercial production (sugar and vinegar production) in the Philippines. Popular varieties include Phil 99-1793, Phil 2000-2569, Phil 94-0913, Phil 92-1043, PHIL 56-226, VMC 70-150, VMC 76-505, Phil 63-17, PSR 97-41, VMC 95-09.

Livestock farming is a significant component of Philippine agriculture, becoming source of dairy, eggs and meat. Animals utilized in agricultures are cows, pigs and chickens. Different types of cows are used to produce meat in farming, these include the Philippine native cow, brahman cow, holstein friesian. Pig is used in dairy and meat production such as landrace pig, duroc pig, hampsphire pig, Berkshire pig, pietrain pig. Chickens are popular for poultry farming because of their egg production and meat consumption. Chickens varies and types and uses, chickens are popular for backyard farming such as banaba, bolinao, darag, joloano, and paraokan.

In the Philippines, naming of rice, corn, and sugarcane varieties represents the organizations involved in development and certification of varieties. Rice variety, NSIC stands to National Seed Industry Counsil while PSB stands to Philippine Seed Board, in corn variety, IPB stands for Institute of Plant Breeding, LB stands for local to Los Baños, and lastly, in sugarcane basically, Phil stands for Philippines, it was dedicated on Philippine sugarcane breeding program under Sugar Regulator Administrator (SRA). VMC stands for Victorias Milling Company, one of the oldest sugar mills in Negros Occidental, these provide an important role to support the agriculture sectors by certifying the seeds, and their numbers at the end of the acronym is their identifier for the variety.

People who want to start farming tends to have a problem in having some knowledge in farming due to their inexperience in planting rice, corn, and sugarcane. Without proper knowledge and technique in farming, new farmers are unable to maximize their potential and their production in farming.[8]

Aspiring farmers coped up with this problem by learning and having the knowledge of farming, this includes the implementing a machines and technology in farming such as drones, tractors, two-wheel tractor. By these technologies and machines, Farmers lessen their time making it convenient for them in farming and by having an information about farming.

With "AgriCare", this application will help farmers, especially aspiring farmers in learning modern farming methods to improve their productivity and avoid any further losses from outdated techniques, this will help them manage their resources better with its accurate insights and calculations, these farmers can increase their productivity and keep up with modern farming as this app does not just help farmers, it will also make the whole farming industry more stable and sustainable.

## 1.2 Objectives

### 1.2.1 General Objective

The general objective of the study is to develop "AgriCare: A Mobile Application for Farming & Livestock Raising in the Philippines" that will help users especially the aspiring farmers in raising livestock and crop farming.

### 1.2.2 Specific Objectives

*A. To develop a module that guides the farmers in crops and livestock farming.*

This module provides a guide to users in agriculture. It is served as a resource of learning about crops and livestock farming.

*B. To develop a module that forecasts the weather.*

This module keeps the user informed about weather condition by helping them stay prepared for potential disasters through current weather predictions. It tracks temperature, relative humidity, apparent temperature, rain, showers, wind speed, and wind direction.

*C. To develop a module that calculate the investments and profits of the farmer in livestock including the estimated seeds in a land area in crops.*

This module allows the users to calculate their total investment and profit in livestock, helping them track their financial performance and make better investment choices. Also, this module allows the user to calculate or estimate the seeds in a land area.

*D. To develop a module that creates a daily set of schedules.*

This module allows the user to set a time for watering and feeding time, including the fertilizing, helping them to be consistent on the time of tasks.

*E. To develop a module that sets up a farm staging.*

This module allows the user to track the entire farming process including the pre-planting, growing, and harvesting. It also includes the livestock management providing an organized timeline for both crop and animal growth stages.

*F. To develop a module that notifies the user of their set time for scheduling, and staging including the condition of weather.*

This module allows the user to keep updated on what they set up for scheduling and staging, helping them to remind on what are their task need to do. It also notifies the user if there are changes in the weather conditions or not.

*G. To develop a module that allows the user to choose language either English or Filipino.*

This module allows the user to select two languages either English or Filipino, this will help them select their preferred languages especially, in learning how to plant the crops and raise livestock using the provided information and guide in the application.

## 1.3 Scopes and Limitations

### 1.3.1 Scope

#### A. Home Screen

This module allows the user choose the specific field of agricultural in which they are interested in farming such as livestock and crops, also, it includes weather forecast, calculator, scheduler and translator.

#### B. Crops

This module allows users to select specific crops such as rice, corn, and sugarcane that they want to learn about through detailed text-based information. It also provides 2D images in JPG and PNG formats to help users visually familiarize themselves with the crops they are reading about. In addition, it offers a land preparation guide, detailing the essential steps to take before planting. Users will also be able to view the recommended fertilizers needed to ensure the optimal growth of their selected crops. The following are:

##### B.1. Crops Information

**Variety of Rice**

- PSB Rc10
- PSB Rc18
- PSB Rc82
- NSIC Rc158
- NSIC Rc214
- NSIC Rc218
- NSIC Rc224
- NSIC Rc226
- NSIC Rc298
- NSIC Rc302
- PSB Rc308
- NSIC Rc352

- NSIC Rc354
- NSIC Rc356
- NSIC Rc358
- NSIC Rc360
- NSIC Rc394
- NSIC Rc400
- NSIC Rc402

**Pests/Insects**

- Black Bug
- Stem Borer
- Green Leaf Hopper
- Brown Plant Hopper

**Diseases**

- Bacterial Leaf Blight
- Blast
- Brown spot
- Leaf scald
- Red stripe
- Stem rot
- Tungro
- Sheat Blight

**Fertilizers**

- Urea 46-0-0
- 14-14-14

**Variety of Corn**

- IBS
- IPB Var6
- IPB Var8

- IPB Var9
- IPB Var10
- IPB Var11
- IPB Var13
- LB Lagkitan
- Native Corn
- Putian
- Tiniguib

**Pest/Insects**

- Corn Earworms
- Corn Rootworms
- Corn Borer
- Fall Armyworms
- Cutworms
- Asian Corn Borer (ACB)
- White Grub
- Seedling Manggot

**Diseases**

- Downy mildew
- Leaf rust
- Banded leaf and stealth blight
- Bacterial stalk rot
- Corn mosaic
- Corn smut
- Leaf spot
- Ear rot

**Fertilizers**

- Ammonium
- Urea 46-0-0

- 14-14-14

**Variety of Sugarcane:**

- Phil 56-226
- Phil 63-17
- Phil 92-1043
- Phil 94-0913
- Phil 99-1793
- Phil 2000-2569
- PSR 97-41
- VMC 70-150
- VMC 76-505
- VMC 95-09

**Diseases**

- Leaf Scald
- Downy Mildew
- Sugarcane smut disease
- Yellow Spot
- Rust

**Fertilizer**

- Urea 46-0-0
- Super phosphate
- Muriate of potash

**Land Preparation**

This is a step-by-step process of preparing the land or field. It will have 2 types of preparation such as wet and dry preparation.

**Pest Control**

It has a simple guide on how to prevent pests in field of rice, corn, sugarcane.

**Disease Control**

Disease control in crops can be achieved by using resistant varieties that natural avoid and diseases. When needed, fungicides are applied to further protect plants and ensure healthy growth.

**Livestock**

This module allows users to select specific animals and breeds such as cow, pig, and chicken that they want to learn about through detailed text-based information. It also provides 2D images in JPG and PNG formats to help users visually familiarize themselves with the animals they are reading about. In addition, the module covers important care topics, including shelter, food, vaccination, common diseases and infections, and cures. Users will also be able to view the recommended vaccines needed to ensure the health of their selected animals. The following are:

**C.1 Livestock Information**

**Types of Cows**

- Brahman Cow
- Philippine Native Cow
- Holstein Friesian

**Food**

- Grass
- Silage
- Rumsol Cattle Feed

**Vaccines**

- Bovine Respiratory syncytial Virus (BRSV)

- Bovine Viral Diarrhea (BVD) types I and II
- Infectious Bovine Rhinotracheitis (IBR)
- Parainfluenza type-3 (PI-3)

**Vitamins**

- Vitamin A
- Vitamin E
- Vitamin K
- Vitamin D

**Diseases/Infections**

- Mastitis
- Foot and Mouth Disease
- Bovine Pneumonia
- Anaplasmosis
- Bovine Viral Diarrhea
- Bovine viral diarrhea (BVD)

**Types of Pig**

- Large White Pig
- Landrace Pig
- Duroc Pig
- Hampshire Pig
- Berkshire Pig
- Pietrain Pig

**Food**

- Yellow Corn
- Rice Bran
- Copra Meal
- Fish Meal
- Soybean oil meal
- Skimmed Milk

- Ipil-ipil Leaf
- Brown Sugar
- Vitamins-Mineral

**Vaccine**

- Autogenous vaccines

**Diseases/Infections**

- Swine Fever
- Swine Dysentery
- Brucellosis
- Gastroenteritis Complex
- MMA
- Roundworm Infection
- Mange

**Types of Chicken**

- Types of Native Chicken
  1. Banaba
  2. Bolinao
  3. Darag
  4. Joloano
  5. Paraoakan
- Types of Egg Layers
  1. Lohmann Layers
  2. Dekalb White Layers
  3. Babcock White Layers

**Food**

- Ground Corn
- Rice Hull
- Rice Bran
- Copra Meal
- Rice Grits

- Corn Bran

**Vaccines**

- Nobilis AE + Pox
- UNIVAX PLUS
- CEVAC EDS K

**Vitamins**

- Pampered Chicken Mama Vital Nutrients
- Multivitamins and Calcium Lactate Tablets

**Disease/Infections**

- Respiratory Disease
- Fowl Pox
- Newcastle Disease

**Cure**

It has a simple guide in curing the provided disease/infection in Cow, Pig, Chicken.

**Shelter**

This module provides an information about sheltering for a specific livestock.

**C. Weather Forecast**

This module let the user keep updated on what is the weather condition by providing a feature of temperature, relative humidity apparent temperature, Rain, Showers, wind Speed(10m).

**D. Calculator**

This module let the user to calculate their total expenses and profit, here are the formula of calculator:

**A. Livestock**

This module let the user to calculate their total expenses and profit in livestock.

- (Cow Weight(kg) x price (kg)) – Expenses = **Result**
- (Pig Weight(kg) x price (kg)) – Expenses = **Result**
- (Chicken Weight(kg) x price (kg)) – Expenses = **Result**
- Total Hens x Egg per month = Total x Price of Egg = Total – Expenses = **Result**

**B. Crops**

This module will estimate the seeds needed in a land area.

- **Land Area**

  Land Area ÷ (Plant Spacing/Meter x Row Spacing/Meter) = Result of Plant Spacing and Row Spacing.

  Land Area ÷ Result of Plant Spacing and Row Spacing = **Estimated Seeds**

**E. Farm Recording**

**A. Scheduler**

This module let the users to customize their preferred times for tasks like watering, feeding, and fertilizing plants, as well as feeding, vaccination, and providing vitamins for livestock. Users have the option to select these features based on their preferences.

**B. Farm Staging**

This module let the users input the timeline or date of when crops were planted so that they can track on when it will be harvested, including livestock.

**I. Notification**

This module sends a notification based on the user's selected schedule and the farming timeline they have set. Additionally, it will provide real-time alerts about current weather conditions by opening the location or GPS of user with an internet connection.

**F. More**

This module let the user use the language preferences such as English and Filipino for the information of crops and livestock, also it has a reference used for the information of crops and livestock.

### 1.3.2 Limitations

**A. Internet Dependency**

The application requires an internet connection to access the weather forecast.

**B. Limited Crops and Livestock**

The application only provide information for a limited number of crops and livestock. Farmers who work with non-listed crops or animals will not find relevant information in the application.

**C. No Multi-Language Support**

The application only be available in English and Filipino, limiting accessibility for farmers who speak other local dialects.

**D. Does not support video tutorial**

The application was not be able to provide a video tutorial about planting and taking care of livestock.

**E. Does not support Audio**

The application not be able to provide an audio while using the application.

**F. Disclaimer in Cures**

In curing the livestock, users should seek for professional for more information about curing the specific livestock. The application will only provide basic cures about infections and diseases.

**G. Android OS version**

The Android 8 Oreo version and above is advisable in using the application.

**H. iOSversion**

iOS 16 version and above is recommended in using the application.

**I. Fertilizer**

Fertilizer should go through soil analysis before using different types of fertilizers.

## 2.0 REVIEW AND RELATED LITERATURE

### 2.1 Related theories

#### A. Herding

Herding refers to the practices of managing groups of livestock across large areas. This method developed about 10,000 years ago, when ancient people started domesticating wild animals such as sheep and goats. Hunters learned that by controlling animals their could have consistent sources of food, including meat and milk products as well as material such as hides for tents and clothing.

Herds are group of animals that naturally live and travel together. Such as goats, sheep, and llamas live in herds for protection. Without an organized direction, they move from one fertile grassland to another. Herders traditionally have been providing a protection for the animals. However, predators like lions, wolves and coyotes are at serious risk to domestic herds.

Herders often specialize in a particular type of livestock. Shepherds, for instance, herd and tend to flock of sheep. Goatherds tend to goats, and swineherds to pigs and hogs. [4]

#### B. Crops

Crops are commonly grown plants or products as its harvested to be eaten or to be sold. Crops are generally divided into 6 categories, Food crops, feed crops, fiber crops, oil crops, ornamental crops and industrial crops, as crops now has become the primary method of feeding humans in every corner or the world till this day, ever since the hunter-gatherer societies shifted to become agricultural societies 10,000 years ago, not only drive food distribution and consumption but also fuel mostly every other industry.

Food crops are crops meant for human meant to be eaten these include fruits, grains, and tubers like potatoes. Grains which include crops like wheat, rice and, corn are the most popular crops in the world with wheat as the most widely grown crop.

Feed crops are grown, meant to feed livestock like cows, horses, pig and sheep mostly known as animal feed or fodder, similar to food crops these crops can be grown and harvested. these can also be grown in large fields or meadows where animals can forage for food, Feed crops are typically engineered to meet the live stocks specific nutritional needs of the livestock according to the FAO (Food Agriculture Organization) division of the United Nations only 33% of the world croplands are used for growing Feed crops

Fiber crops are specifically grown so they can be processed and turned into products like textiles, rope, or paper instead of being directly eaten. fiber like cotton, flax or hemp are harvested and then dried or chemically transformed to create various products, for example, bamboo can be turned into pulp which is used to make paper.

Oil crops, can grow either be for human consumption or industrial uses, edible oil crops are corn, sunflower, and olives and the most popular oil crop is Soybeans. besides food and cooking uses oil crops can be used to create paints, soap, lubricants for machines, cosmetics and, fuel

Ornamental crops are grown for the purpose of landscaping or gardening mostly consists of shade trees, flowering trees, shrubs, flowers, and grasses, these are often grown in nurseries, ornamental crops are harvested and directly sold to consumers, or for commercial purposes.

Industrial crops, these are crops not meant for consumption but are harvested and used in manufacturing. Rubber is a common example of industrial crops as Rubber are created from latex which is a substance commonly found in the Hevea tree, then processed and treated so that the rubber can be transformed into various products for a variety of industrial uses.[5]

**B. Weather Forecast**

Weather Forecasting has evolved significantly over time. In 650 B.C,. The Babylonians would use cloud patterns to predict weather patterns. Aristotle wrote Meteorological three years later, explaining how weather phenomena such as rain, hail, hurricanes, and lightning. It is one of the first attempts to discuss how the weather works in detail, but it turned out to be incorrect. Two years later, the first shipping-related weather forecast was aired public. While meteorological measurements improved over time, the massive step-change in predictions came with the use of computerized numerical modelling.

The gap is a problem. Agriculture is certainly the most weather dependent industry, employing 60% of the farmers in low-income nations. Having a accurate weather forecast can help farmers to improve their decision making. They can learn when it is best to plant their crops. They know in advance when irrigation will be most needed and when fertilizer may be affected. They can receive alerts about pest and disease outbreaks so they can either protect their crops. This indicated that having access to accurate weather forecast will allow them to make the best possible use of limited resources. [6]

### 2.2 Related Projects

**A. Magtanim: A Mobile application guide in backyard planting in the Philippines**

**Developers:** Aringo Et Al.

**Source**: Playstore

Magtanim is a mobile app that helps people in the Philippines grow plants in their backyards and it gives information on different crops like vegetables, fruits, herbs, and root crops. The app works offline, supports both English and Filipino, and includes sections on plant care


*Figure 1: Crops Module*

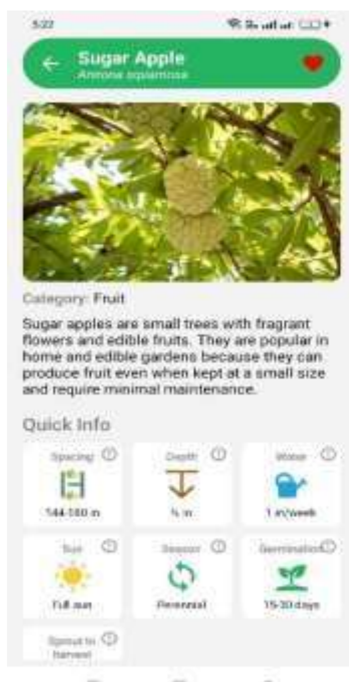


*Figure 3: Crop Information*

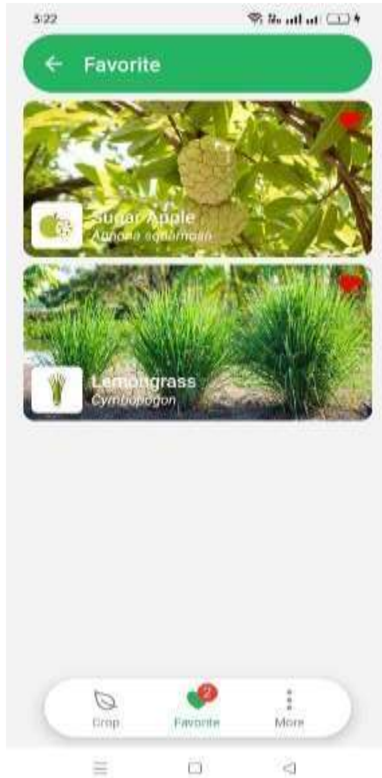


*Figure 3: Favorite Section*

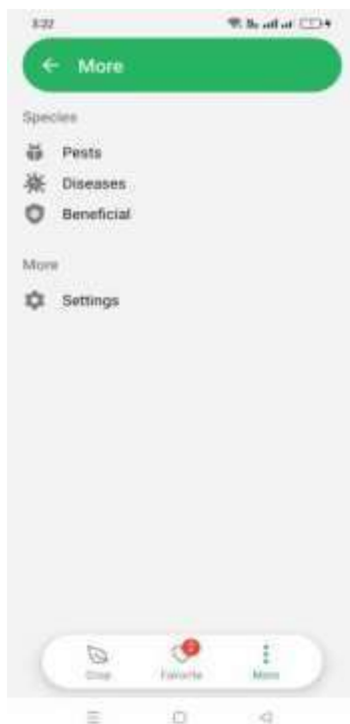


*Figure 4: More Module*

### *A.1* Similarities between AgriCare and Magtanim

AgriCare and Magtanim provide an information about crops, helping farmers with the knowledge needed to increase foods production. Also, these two provides the insects depending on the variety of crops.

### A.2 Differences between AgriCare and Magtanim

AgriCare provides an information into two categories, such as crops (rice, corn, sugarcane) and livestock (cow, pig, chicken) and has a weather forecast that notifies the farmers in what are the condition of weather. It also has a calculator and staging where user can calculate their profit and investment and tracks their crops and livestock in their growing stage and date. While Magtanim provides an information about backyard farming such as vegetables, fruit, and herb that helps farmer in what are the spaces per row in planting, depth, watering info, height and its season. It also provides pest information, diseases and beneficials depending on the crops.

### B. Outgrow: Farming Solutions App

**Publisher:** WayCool Labs

**Date Published**: May 5, 2020

**Source:** Playstore

Outgrow is a simple and easy to use farming app that helps farmers at every stage of growing their crops The app helps farmers improver their productivity and reduce losses by providing important information and real time support.

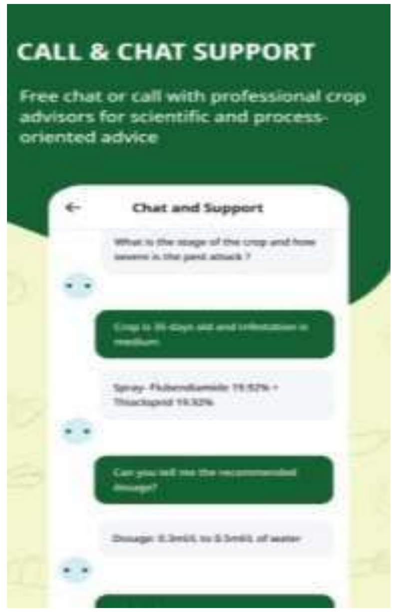


*Figure 5: Chat Support*

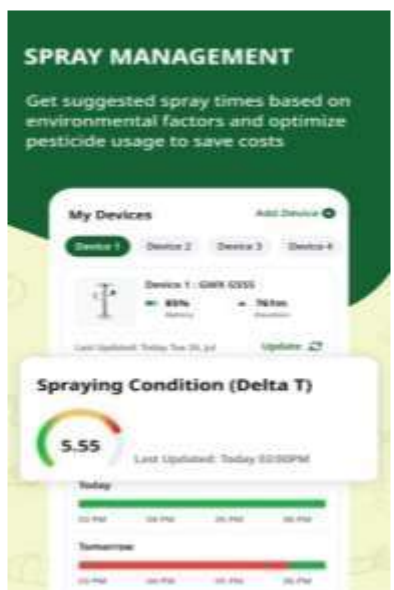


*Figure 6: Spray Management*

### A.1 Similarities between AgriCare and Outgrow

AgriCare and Outgrow provide an information about crops and helps farmers in weather updates.

### A.2 Differences between AgriCare and Outgrow

AgriCare provides an information into two categories, such as crops (rice, corn, sugarcane) and livestock (cow, pig, chicken) and has a weather forecast that notifies the farmers in what are the condition of weather. It also has a calculator and staging where user can calculate their profit and investment and tracks their crops and livestock in their growing stage and date. While "Outgrow" provides a real time market prices in crops and pest and disease detection.

## C. Learn Smart Agriculture

**Published:** CODE WORLD

**Date Published:** March 25, 2024

**Source:** Playstore

This application is designed for students, researchers and teachers who want to understand modern farming and agricultural sciences as it covers various topics related to farming and food production.


*Figure 7: Chat support*

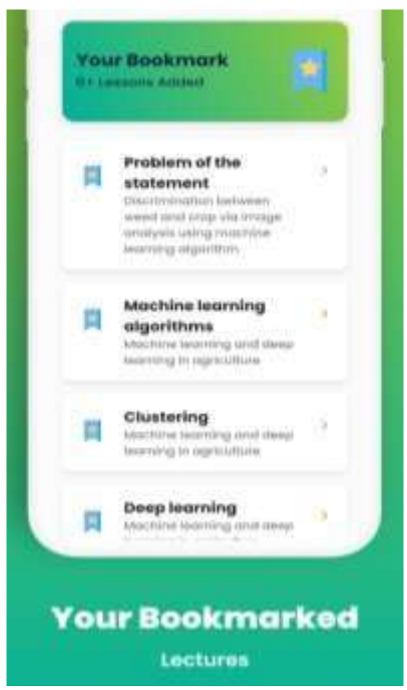


*Figure 8: Bookmark*

### *A.1* Similarities between AgriCare and Learn Smart Agriculture

AgriCare and Learn Smart Agriculture helps farmers and users in providing an information about agriculture.

### A.2 Differences between AgriCare and Learn Smart Agriculture

AgriCare provides an information into two categories, such as crops (rice, corn, sugarcane) and livestock (cow, pig, chicken) and has a weather forecast that notifies the farmers in what are the condition of weather. It also has a calculator and staging where user can calculate their profit and investment and tracks their crops and livestock in their growing stage and date. While Learn Smart Agriculture provides 29 lessons such as machine learning and deep learning in agriculture, descriptive and predictive analytics using machine learning algorithms and etc.

## 3.0 TECHNICAL BACKGROUND

### 3.1 Development

#### 3.1.1. Hardware

**A. Personal Computer / Laptop**

The proponents used personal computers and laptops to develop the proposed project; it is also used in documentation and research. This hardware can run specific tools such as IDEs and compilers with high specifications that allow the proponents to work efficiently and more compatible in installing the tools.

**B. Mobile Devices**

The proponents used mobile devices to test and to evaluate the visualization of the application. It will also help to evaluate the functionality and the performance of the application in different screen sizes, and operating systems such as Android OS and iOS.

#### 3.1.2. Software

**A. Android Studio**

The proponents used Android Studio IDE to create the program of created prototype since it is one of the best IDEs for Flutter SDK because it provides a complete set of tools for building, testing, and debugging the application. It also supports a built-in emulator and USB debugging for testing.

**B. Flutter**

The proponents used Flutter as the framework to develop a cross-platform application which is the “AgriCare” application in a single codebase. Flutter provides a built-in widget, making it easier to develop an application. This will help the proponents to work efficiently.

**C. Dart**

The proponents used Dart programming language in developing the “AgriCare” mobile application. Dart is one of the best

languages in using flutter development, this will help the developers in time and effort.

**D. Xcode**

The proponents used Xcode to implement the IOs version of the application. This helps them design the user interface using build-IOs simulator. It also allowed them to debug issues efficiently and ensure compatibility with different iOS devices.

**E. Figma**

The proponents used Figma in order to create the prototype or design of the application, this will include the user interface (UI) elements such as buttons, icons, tabs and more. Also, the designers will evaluate the prototype such as flow and the functions of the application. Canva

The proponents used Canva to create the logo of the application that will be edited by the designers and focus on farming theme that represents the target users of "AgriCare" application, making it memorable and easy to recognize across marketplace such as Google Play Store and Apple App Store.

Draw.io The proponents used Draw.io to create diagram such as fishbone diagram, use case diagram, sequence diagram, and activity diagram that will help the proponents in ideas to remain organized during the planning stage of the application development process.

**F. Git**

The proponents used Git to track the changes in code of the project, this will help the proponents in having a backup and recovery of the code.

**G. GitHub**

The proponents used GitHub where the proponents will save the codes and changes of the codes. GitHub will help the proponents in a convenient way of retrieving the codes.

**H. Weather API**

The Weather API is a lightweight XML and JSON format with 200ms average response time. Weather API is used in AgriCare application to integrate weather forecast with wind speed, humidity, and pressure.

### 3.1.2. Peopleware

#### A. Proponents

The proponents are a group or a team who works to create the application. This includes tasks such as brainstorming, analyzing, designing, documentation, and coding of the proposed project. These tasks are assigned depend on their expertise and skills to ensure smooth collaboration and efficiency, and to achieve their set goal for the capstone project.

#### B. Capstone Adviser

Mr. Jan Nicole B. Apostol, the Capstone Adviser of this proposed project, gives an essential role in guiding and supporting the proponents throughout their proposal stage through his expertise, skills, and especially the experience. Capstone Adviser is one of the crucial roles in ensuring the quality of the document, timeline, and the application, making sure the success of development of the proposed project.

#### C. Resource Person

##### C.1 Mr. Angelo Fajardo

Mr. Angelo Fajardo Helped the proponents in verifying the names and information of rice crop. Mr. Fajardo has an experience of farming in rice for 40 years which gives the proponents a confidence and credentials in verifying the names of rice.

##### C.2 Mr. Alvin Sicat

Mr Alvin Sicat also helped the proponents in verifying the names and information of crops such as rice, corn also, in livestock like cow chicken, and a pig. He has an experience of farming for 10 years.

##### C.3 Mr. Dominador Mallari

Mr. Dominador Mallari helps the proponents in verifying the names of sugarcane and corn and their information. Mr. Dominador Mallari has the experience of 60 years of farming sugarcane and corn.

**C.4 Mr. Carlos B. Tuazon**

Mr. Carlos is the head officer or Head Senior of Capas, Municipal Agricultural Office. He helped the proponents in verifying the names of crops and livestock that are mostly or common in the Philippines. He also suggested the websites of crops and livestock name and information including their new pests and their common diseases.

## 3.2 Implementation

### 3.2.1 Hardware

**A. Mobile Application**

The "AgriCare" application was implemented in a cross-platform mobile application. The minimum requirements for the hardware are the following:

**Android:**

- Android RAM – 4GB and above
- Android Storage – 32GB and above

**iOS:**

- iOS RAM - 2GB and above
- iOS Storage – 64GB and above

### 3.2.2 Software

**A. Android OS**

Android OS is the most commonly used worldwide, it is an open-source operating system developed by Google. The "AgriCare" application can be used on Android OS since many people are using Android OS and it is commonly used in daily usage. Also, in order to use the application, users may need an android version 8 Oreo or above.

**B. iOS**

iOS was integrated for this application since many people are using iPhone for their daily usage, this will ensure the accessibility for every iOS user. iOS is commonly known in high specification of processors and with these, it will be perfectly fit for the farmer users. Also, in order to use the application, users may need an iOS version 16 or above.

**C. Play Store**

The software was uploaded to the Google Play Store Application where it will be available to all Android users across the world. The Play Store offers a secure environment for downloading applications and rating system where all users can view the rate level of the application.

**D. Apple App Store**

The Apple App Store is one of the trusted platforms for iOS users to access or download different varieties of applications. The "AgriCare" application will be uploaded to the Apple App Store, where it will be available to all iOS users.

### 3.2.3 Peopleware

**A. Farmers**

This application helps and guide the farmers to improve the production of foods in rice, corn, and sugarcane by providing information on modern farming techniques. Farmers who download the application can have an access of useful features such as a calculator, scheduler, weather forecast, and so on. These features are crucial in farming, especially when dealing with seasonal changes, pest control, and financial planning.

**B. Aspiring Farmers**

This application will help and guide aspiring farmers in planting crops and livestock raising. AgriCare would provide these aspiring farmers knowledge they need so they can be more efficient and informed.

**C. Animal Husbandry**

Animal husbandry is the branch of agriculture concerned with animals that are raised for meat, fiber, milk, or other products. The application will help the Animal Husbandry in knowledge about raising livestock.

### 3.2.3 Network

**A. Wifi / Data Connection**

Users are requiredto have a Wi-Fi or Data Connection to install the application on Play Store and App Store. It also needs connection for weather forecast in order to sync or update the current condition of wea

## 4.0 METHODOLOGY, RESULTS AND DISCUSSION

### 4.1 Methodology

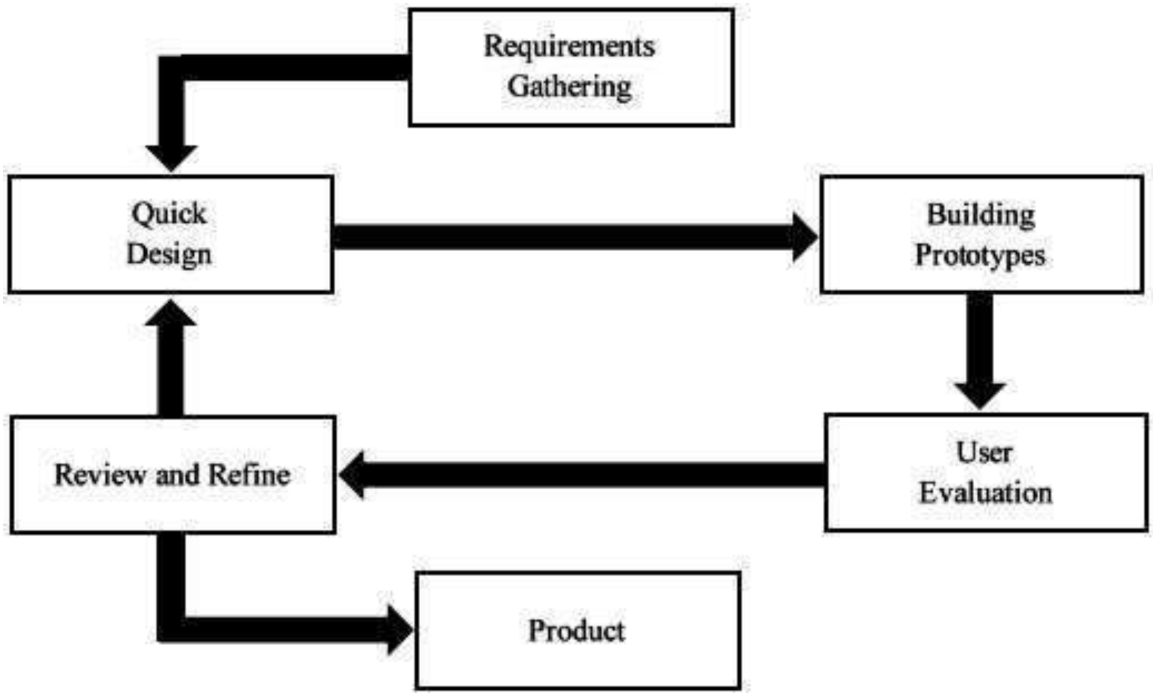


Figure 9: Prototyping Model

The proponents chose the Prototyping Model as the methodology for the development of the mobile application. Prototyping is known for its flexibility, allowing for continuous testing based on user feedback throughout the development process.

The proponents used the prototyping model that helps to identify and detect errors early, ensuring that the application is built, tested and improved through multiple iterations until its functionality was successfully achieved. This approach allows the proponents to align the application with the need of its users.

**A. Requirements Gathering**

In this phase, the proponents work together to identify and gather information about agricultural farming, including livestock and crops, using online resources. This was to prioritize the goal of the proposed project and identify its important feature.

**B. Quick Design**

Once the requirements were fully gathered, the proponents designed the user interface of the proposed project. This phase focused on creating the outline or basic structure of the proposed project. This step allowed the proponents to visualize the initial concept before moving to the next phase.

**C. Building Prototypes**

During this phase, the proponents will develop a prototype of the proposed project based on the design. This phase will serve as the initial working model that allows for testing. This prototype will also serve as a basis for gathering feedback from the next phase.

**D. User Evaluation**

Once the prototype is built, the proponents will conduct user testing and gather feedback that will help to identify the area for improvement in the application.

**E. Review and Refine**

In this phase, the proponents will make improvements to the prototype based on the user feedback. This phase will ensure that the application became fully functional and aligns with the need of its users.

**F. Product**

After the refinements were completed, the final version of the application will be deployed and uploaded on the Google Play Store and App store for users to download and use the application.

## 4.2 Requirements Specification

### 4.2.1 Operational Feasibility

*A. Fishbone Diagram*

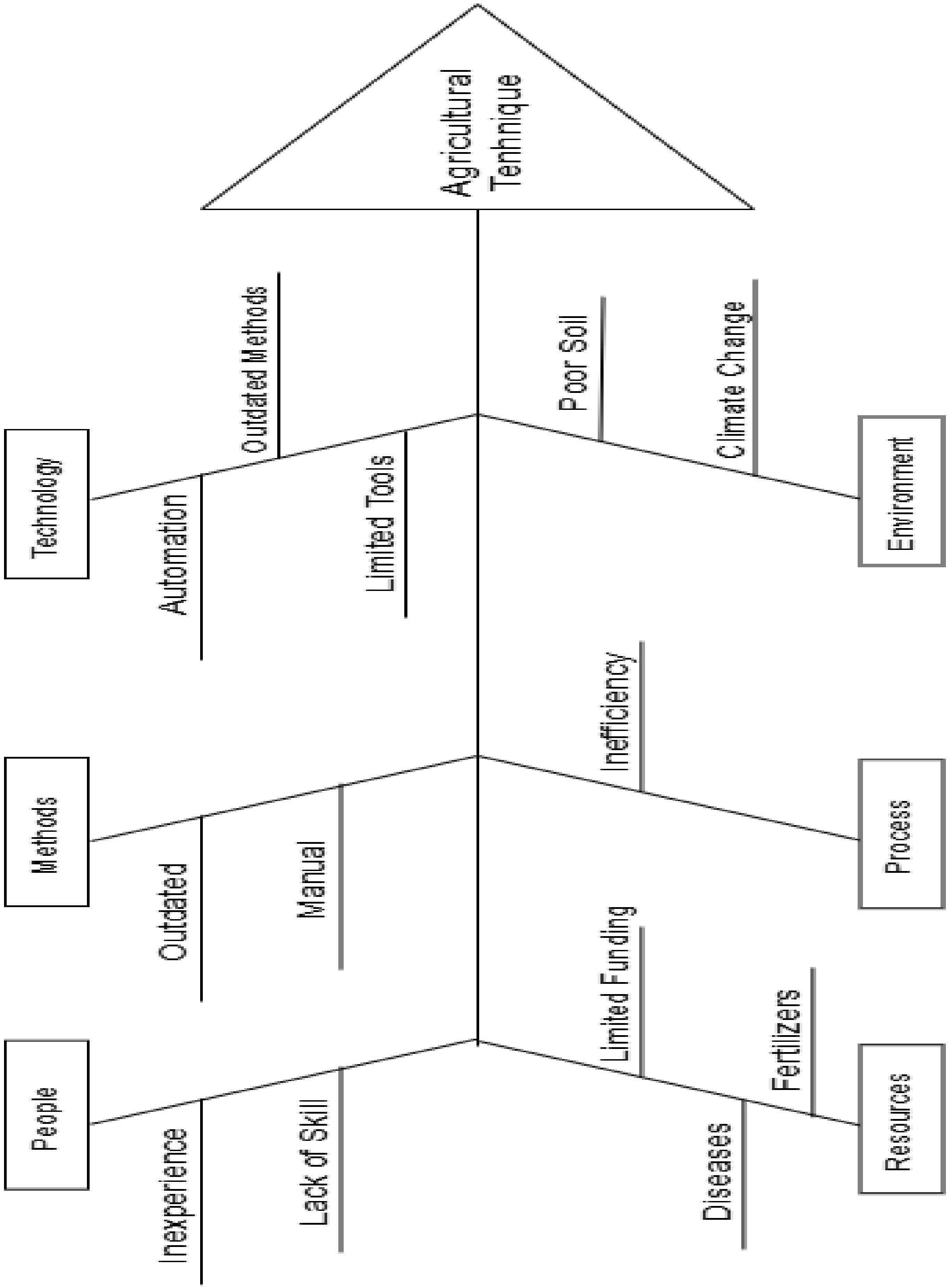


Figure 10: Fishbone Diagram

B. Functional Decomposition Diagram

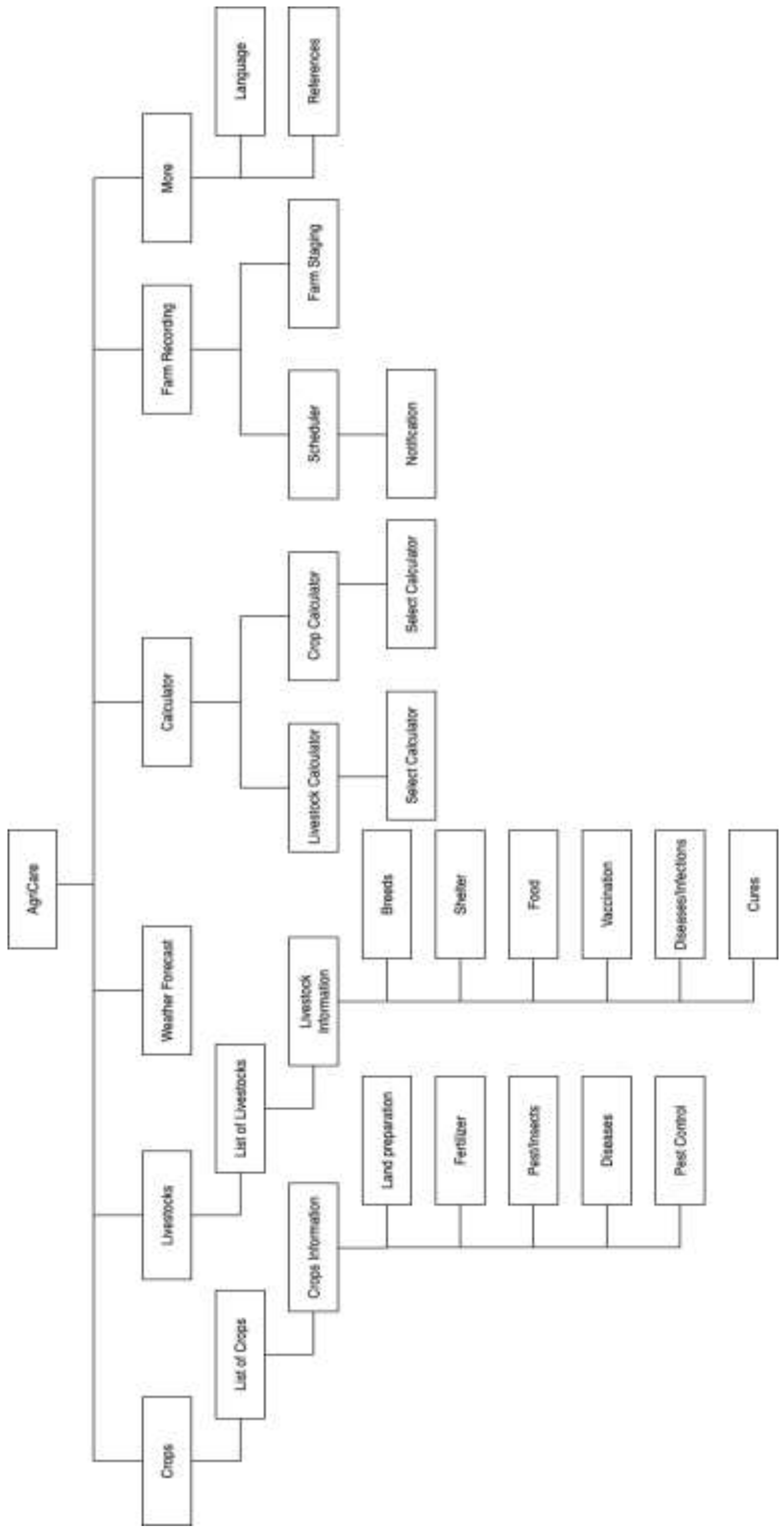


Figure 11: Functional Decomposition Diagram

**4.2.2 Technical Feasibility**

**A. Relevance of the Technology**

Many people use websites and desktop application because of its wider screen and different features that does not have mobile application; however, mobile application is more convenient and affordable and offers a broad feature in different applications depending on the specification of mobile phone. The proponents choose mobile application because of its majority number of users who own mobile phones, this will allow the user to use the application anywhere.

**4.2.3 Schedule Feasibility**

• **Gantt Chart**

Table 1: Month of December 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Groupings | | | | |
| **Requirements Gathering** | | | | |
| 2. Title Suggestion | | | | |
| 3. Research / Brainstorming | | | | |

Table 2: Month of January 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Preparation for Title Defense | | | | |
| 2. Title Defense | | | | |

| | | | | |
|---|---|---|---|---|
| 3. Preparation for Rescheduled Title Defense | | | | |
| 4. Rescheduled Title Defense | | | | |
| Assigning Adviser | | | | |

Table 3: Month of February 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Compose Chapter 1 | | | | |
| 2. Checking Chapter 1 | | | | |
| 3. Revision of Chapter 1 | | | | |
| 4. Compose Chapter 2 & 3 | | | | |
| 5. Checking of Chapter 1,2 & 3 | | | | |
| 6. Revision of Chapter 1,2 & 3 | | | | |
| 7. Gathering crops and livestock information | | | | |
| Quick Design | | | | |
| 1. Compose Chapter 4 | | | | |
| 2. Checking Chapter 1-4 | | | | |

| | | | | |
|---|---|---|---|---|
| 3. Revision of Chapter 1-4 | | | | |
| 4. Checking Chapter 1-4 | | | | |
| 5. Initial Submission for Plagiarism Check | | | | |

Table 4: Month of March 2025

| Activities | Week 1 | Week 2 | Week 3 | Week 4 |
|---|---|---|---|---|
| 1. Revision of Chapter 1 - 3 | | | | |
| 2. Submission of $2^{nd}$ Checking for Plagiarism | | | | |
| **Building Prototypes** | | | | |
| 1. Designing Prototype (Loading scree, home screen, crops and livestock module) | | | | |
| 2. Documentation of Appendices | | | | |
| 3. Checking Chapter 1 – 4 & Appendices | | | | |
| 4. Final Checking of Documents | | | | |
| 5. Designing User-Interface | | | | |
| 6. Preparation for Oral Defense | | | | |

Table 5: Month of April 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Oral Defense | | | | |
| 2. Revision of Documents | | | | |
| 3. Submission for Final Revision | | | | |
| 4. Re-Defense | | | | |

Table 6: Month of May 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Designing the User Interface (Prototype) | | | | |
| 2. Creating Loading Screen with Logo | | | | |
| 3. Designing Home Screen | | | | |
| 4. Revision of Chapter 1 | | | | |

Table 6: Month of June 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| 1. Coding (Creating Crops Module) | | | | |

| | | | | |
|---|---|---|---|---|
| 2. Coding (Creating Livestock Module) | | | | |
| 3. Coding (Creating Crops and Livestock Calculator) | | | | |
| **User Evaluation** | | | | |
| 1. Evaluation (Alpha Testing) | | | | |
| 2. Revision of Chapter 1 and 3 | | | | |

Table 6: Month of July 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| **Review and refine** | | | | |
| 1. Refining Home Screen | | | | |
| 2. Refining Crops and Livestock Module | | | | |
| 3. Revision of Chapter 3 | | | | |

Table 6: Month of August 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| **Quick Design** | | | | |

| | | | | |
|---|---|---|---|---|
| 1. Documentation of Chapter 4 | | | | |
| 2. Revision of Diagram Sequence Diagram (More Module) | | | | |
| **Building Prototypes** | | | | |
| 1. Revision UI (Color, Home Screen Design and icons) | | | | |

Table 6: Month of September 2025

| **Activities** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| **User Evaluation** | | | | |
| 1. Beta Testing | | | | |
| 2. Documentation of Chapter 4 and 5 | | | | |
| **Review and refine** | | | | |
| 1. Refining the images of crops variety | | | | |
| **Product** | | | | |
| 1. Finalizing the system and Document | | | | |
| 2. Deployment | | | | |

Table 6: Month of October 2025

| Activities | Week 1 | Week 2 | Week 3 | Week 4 |
|---|---|---|---|---|
| 1. Preparation for Re-Defense | | | | |
| 2. Final Defense | | | | |

**4.2.3 Economic Feasibility**

**A. Cost and benefits analysis**

The proponents expected cost for deploying the mobile application is approximately ₱ 2,550 pesos on App store. These expenses are just the global rate for uploading an application. The developer program for apple costs ($99/year). Additional expenses associated with the app development are not included. After the refinements were completed, the final version of the application will be deployed and uploaded on the Google Play Store and App store for users to download and use the application.

Table 7: Cost and benefits analysis

| **Platform** | **Cost** |
|---|---|
| **1. App Store** | **₱ 2,550** |
| **Total:** | **₱ 2,550** |

The Table above shows the total expenses expected for the deployment of the mobile application in both platforms, apple store.

**B. Cost recovery Scheme**

The proponents plan to recover the development cost by offering a one-time content subscription. In this plan, users can access limited content crop and livestock information for free, while access to additional content such as other varieties of rice, corn, sugarcane and other breeds of cow, chicken and pig can be unlocked through payments. This approach will gradually recover the development cost.

### 4.2.3 Requirements Modeling

- **Object Modeling**

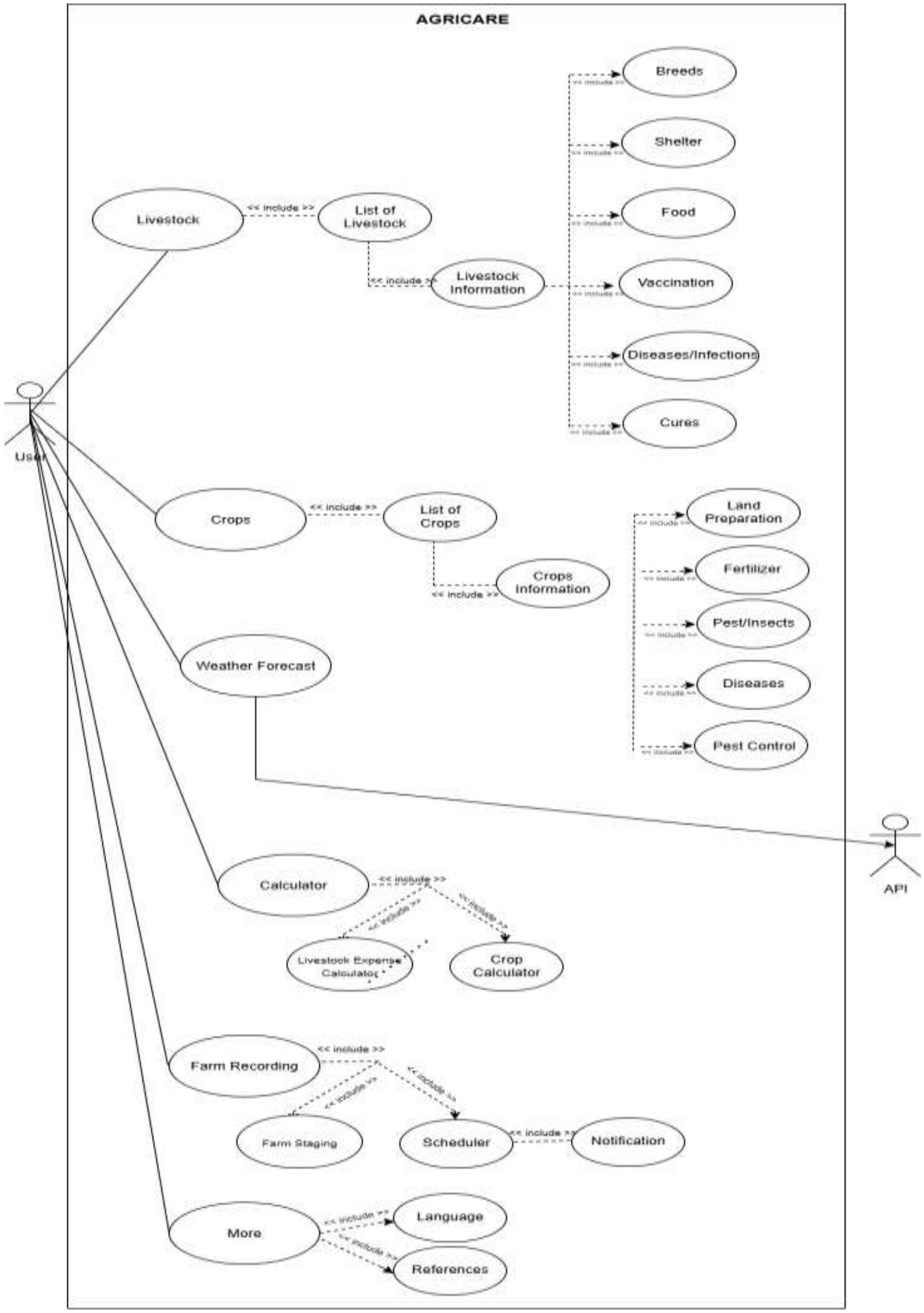


Figure 12: Use Case Diagram

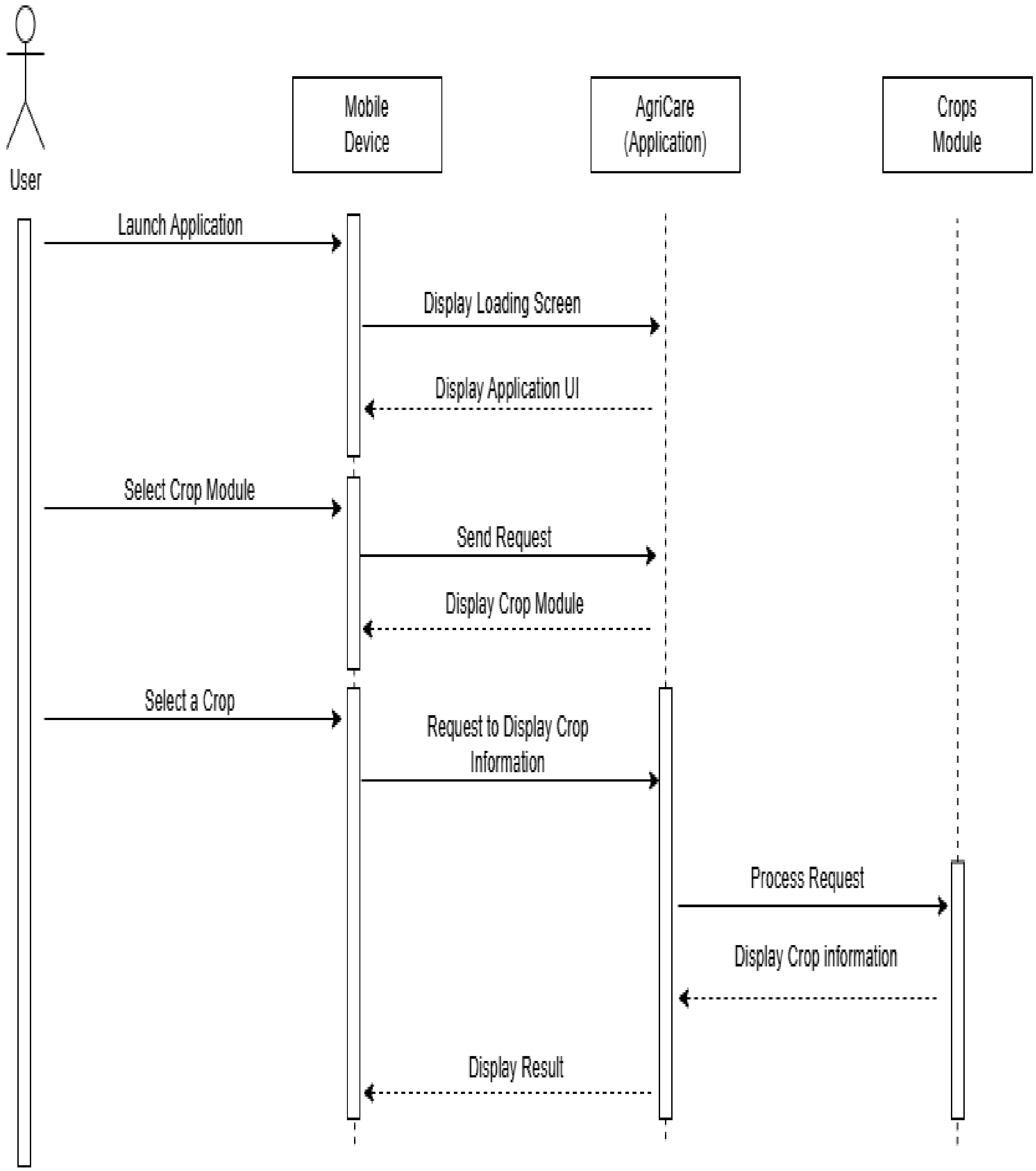


Figure 13: Sequence Diagram of Crops Module

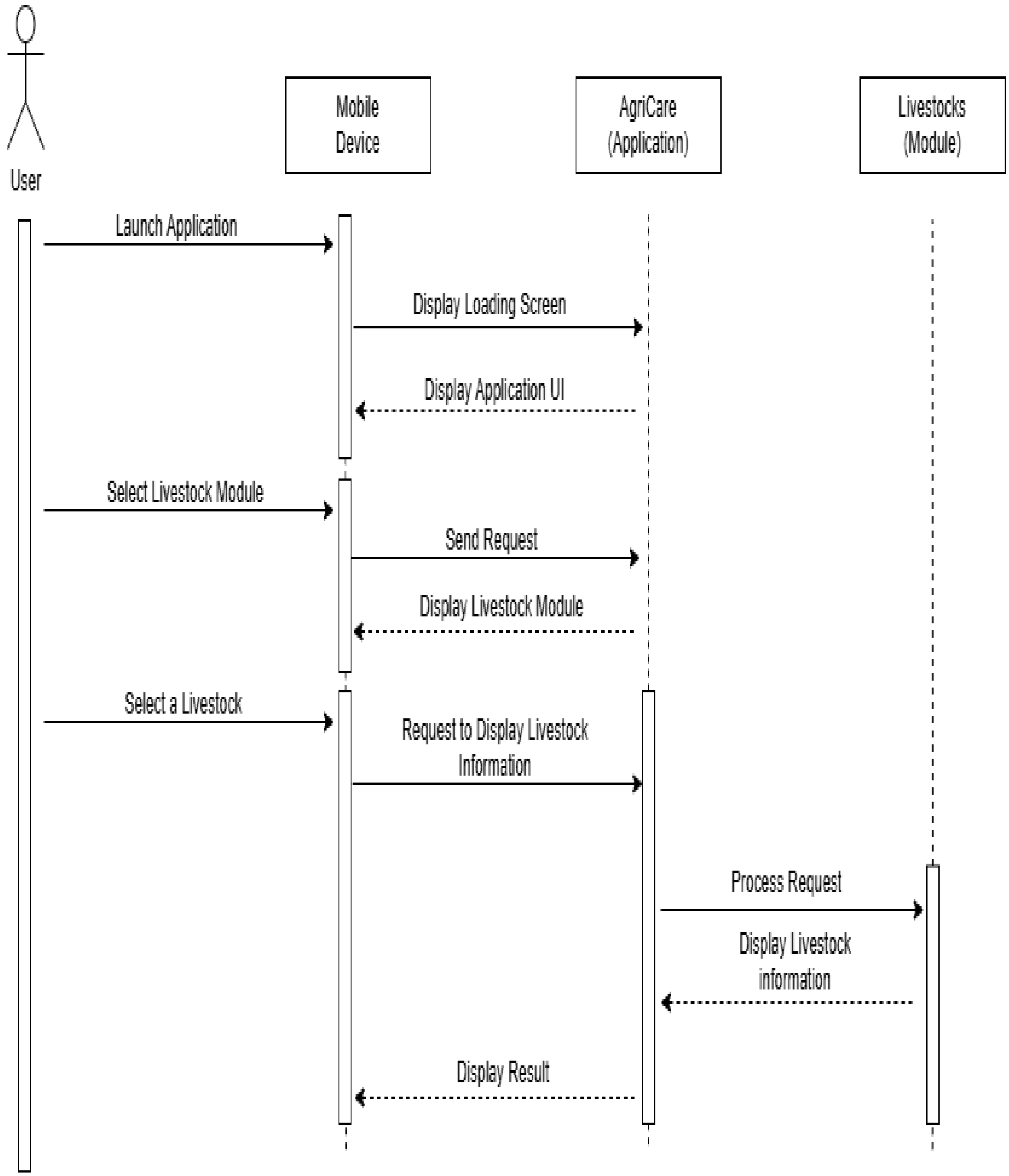


Figure 14: Sequence Diagram of Livestock

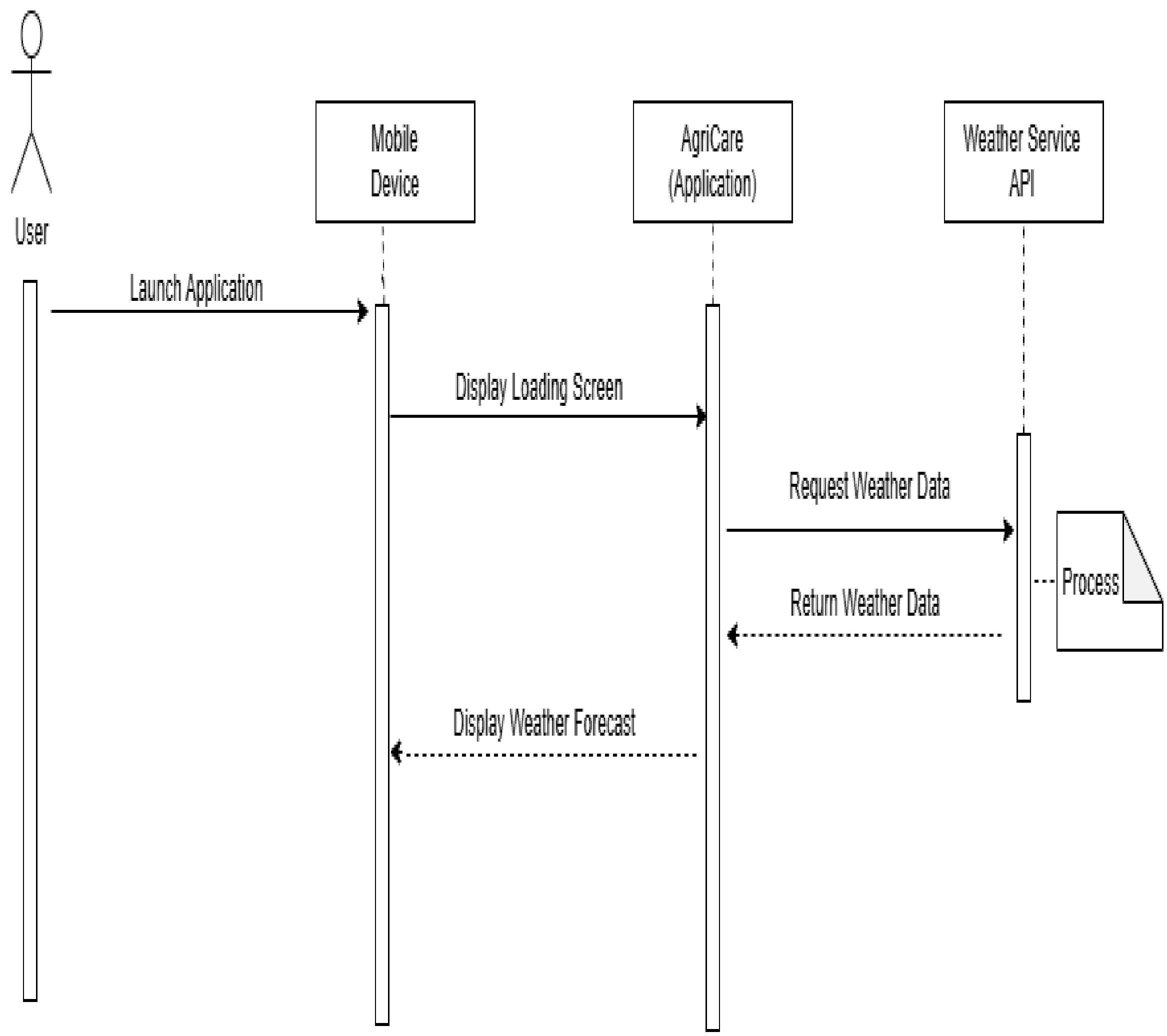


Figure 15: Sequence Diagram of Weather Forecast

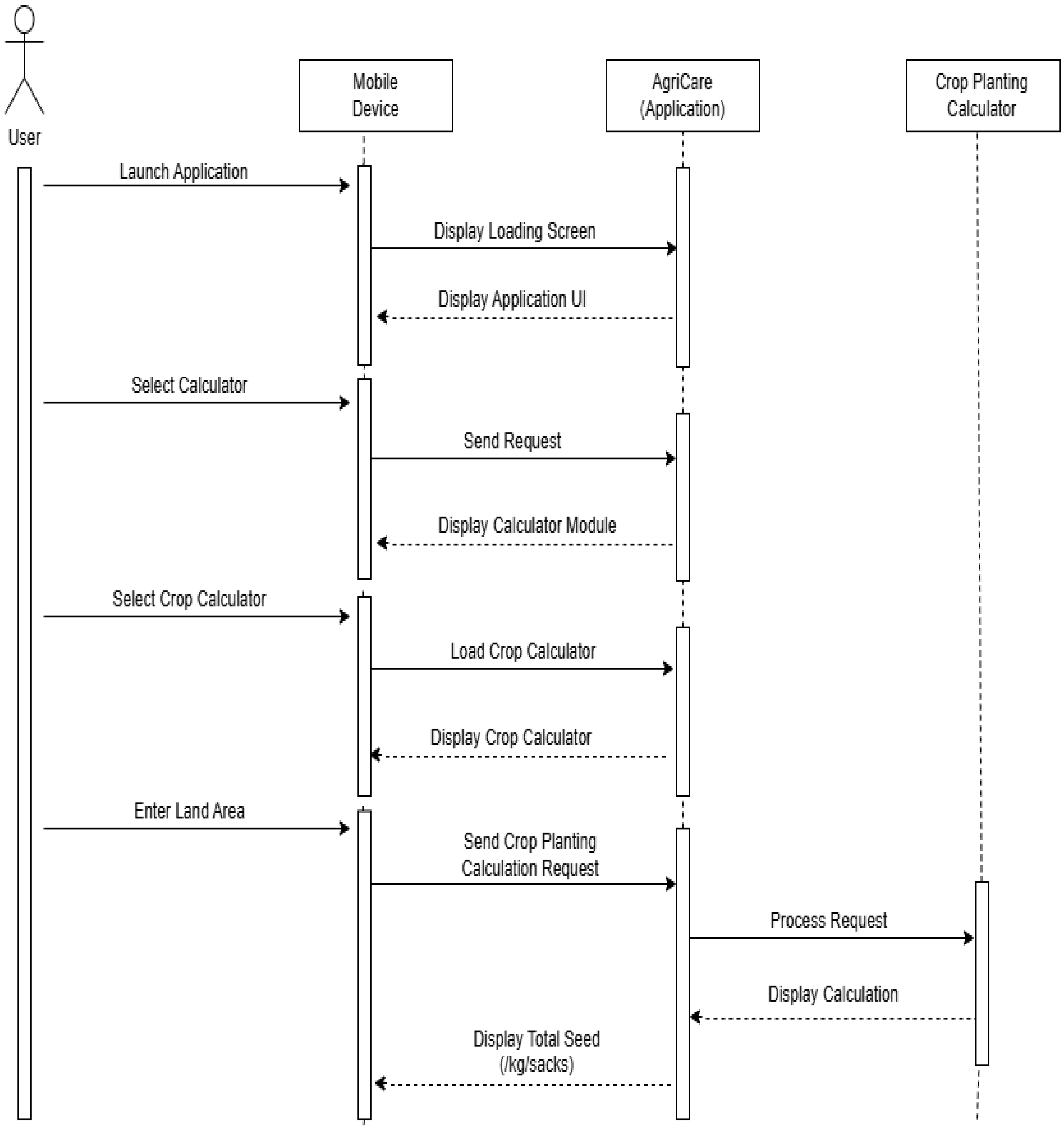


Figure 16: Sequence Diagram of Crop Calculator

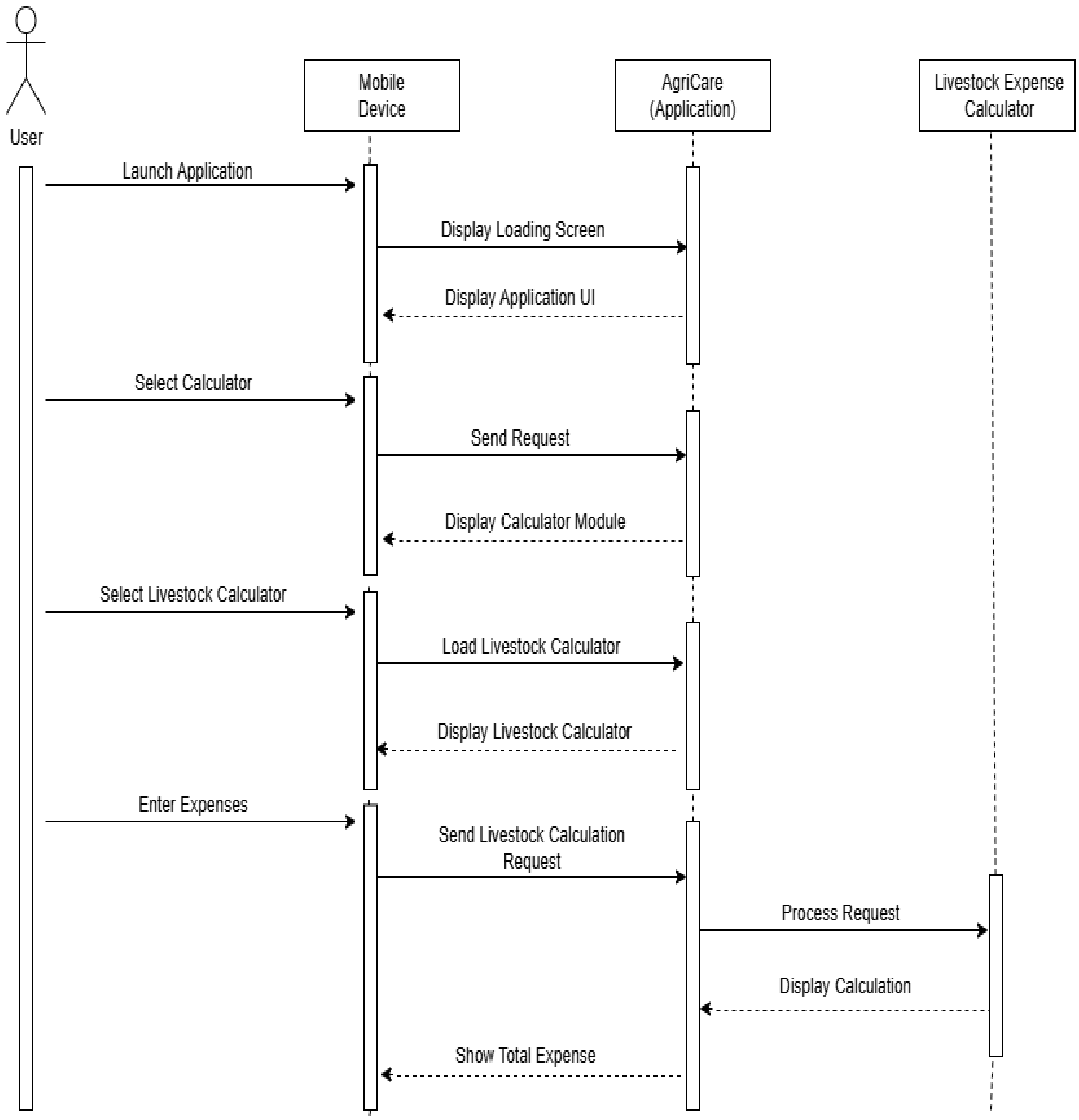


Figure 17: Sequence Diagram of Livestock Calculator

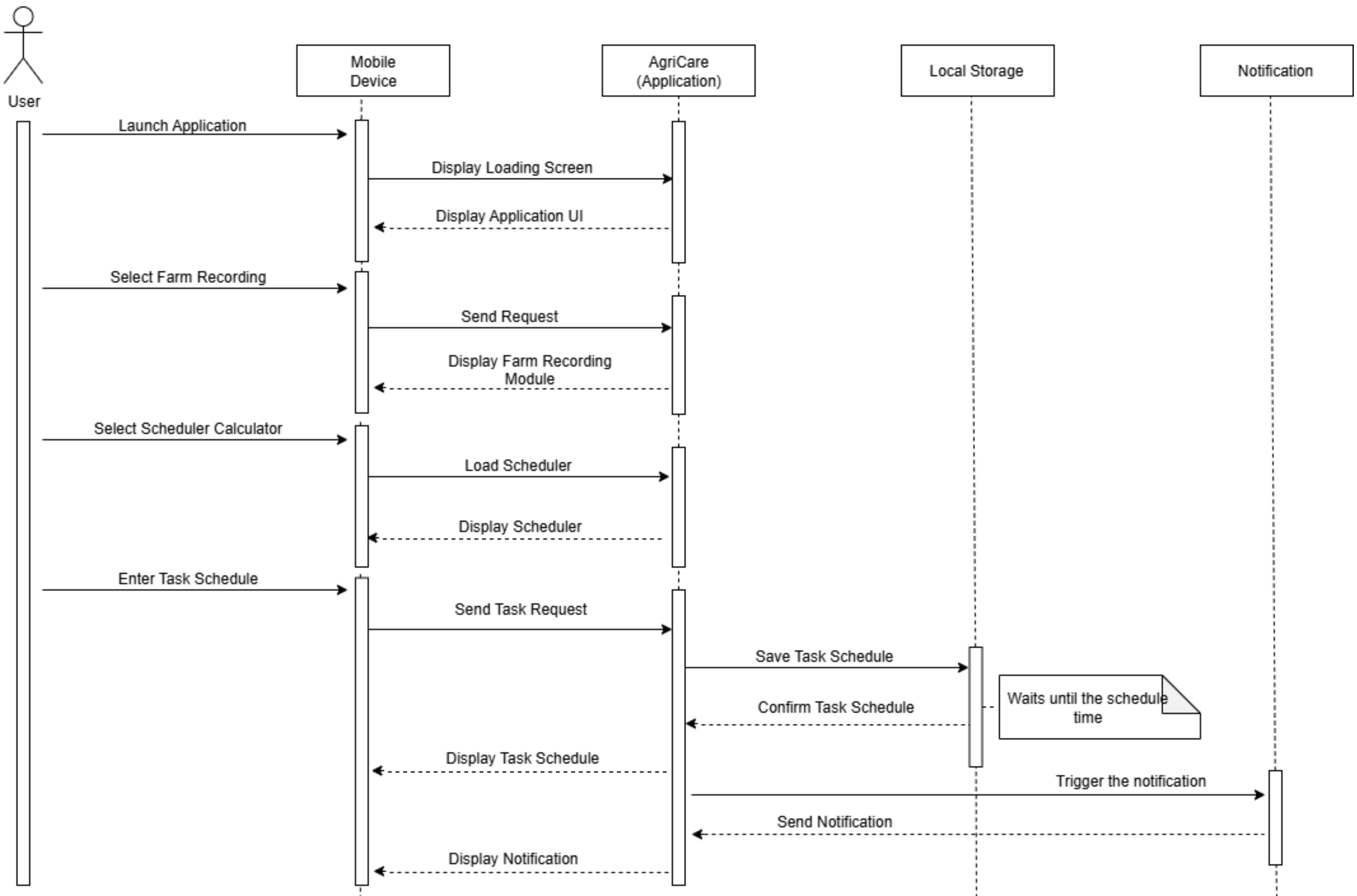


Figure 19: Sequence Diagram of Scheduler

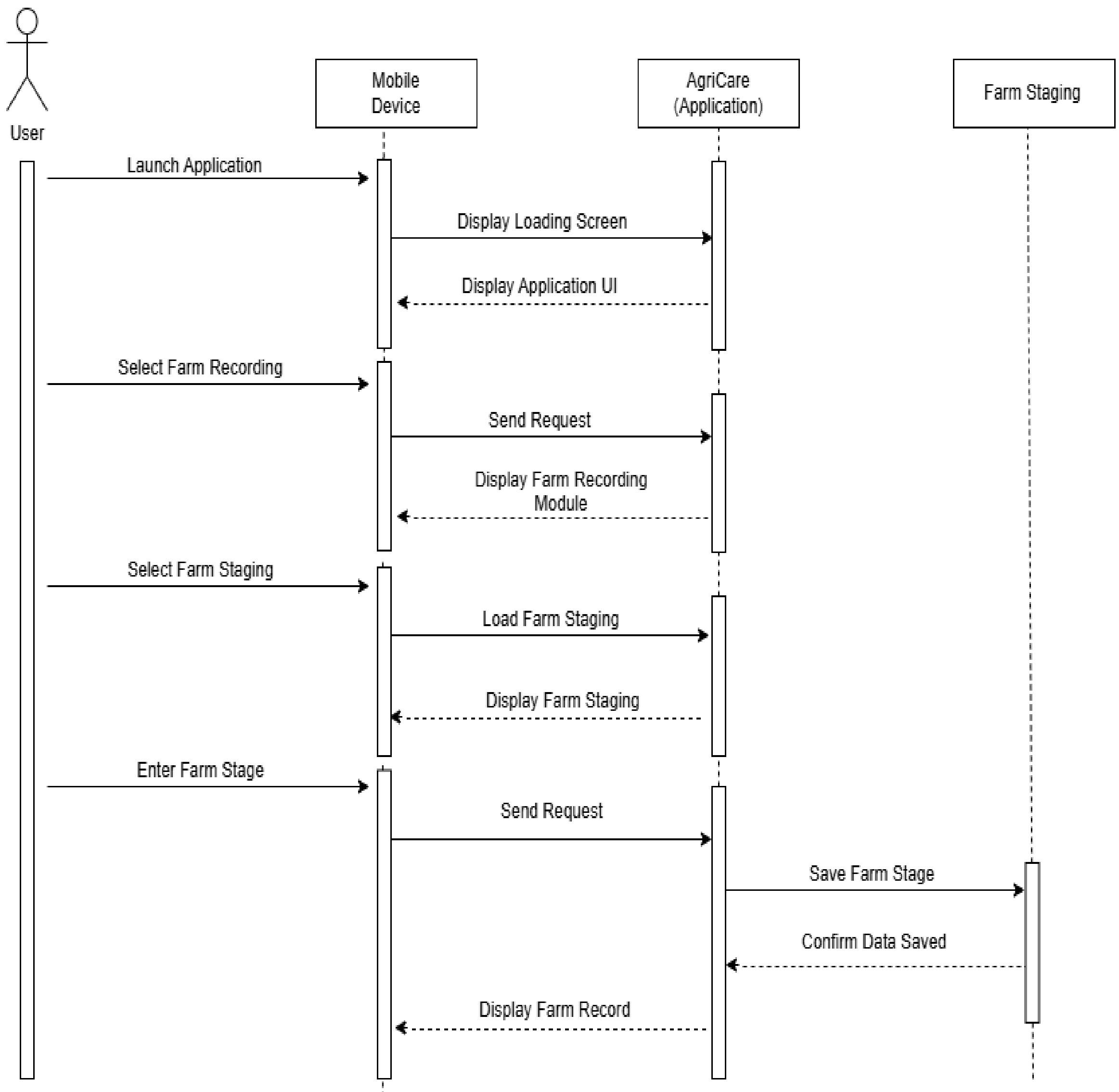


Figure 20: Sequence Diagram of Farm Staging

**Activity Diagram**

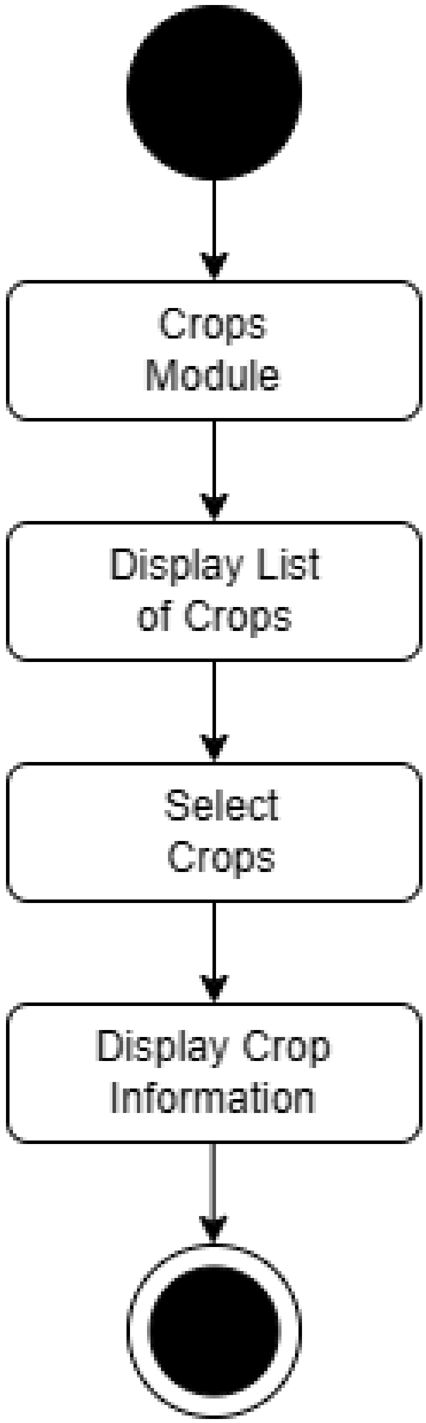


Figure 20: Activity Diagram of Crops

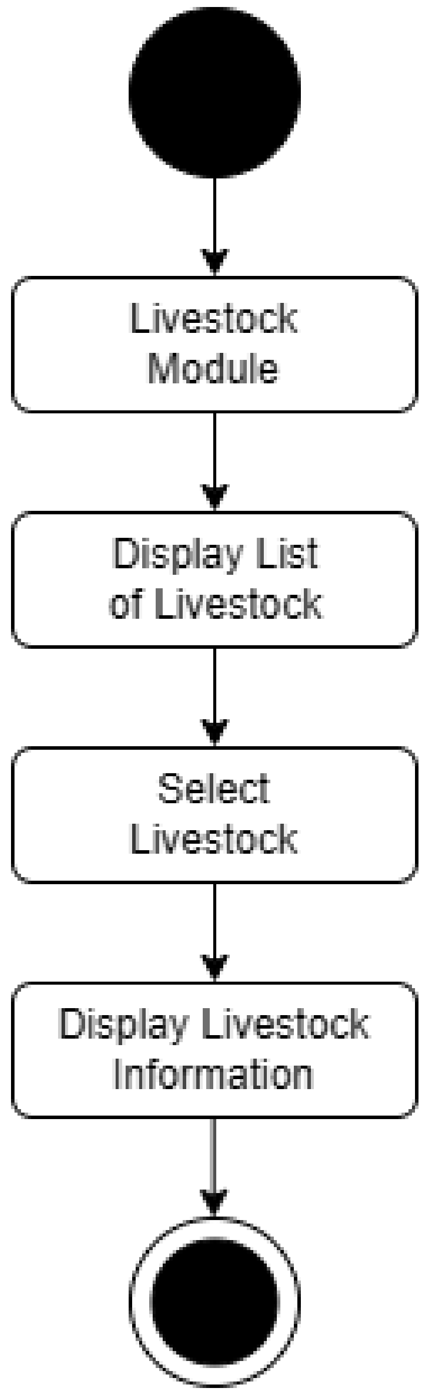


Figure 21: Activity Diagram of Livestock

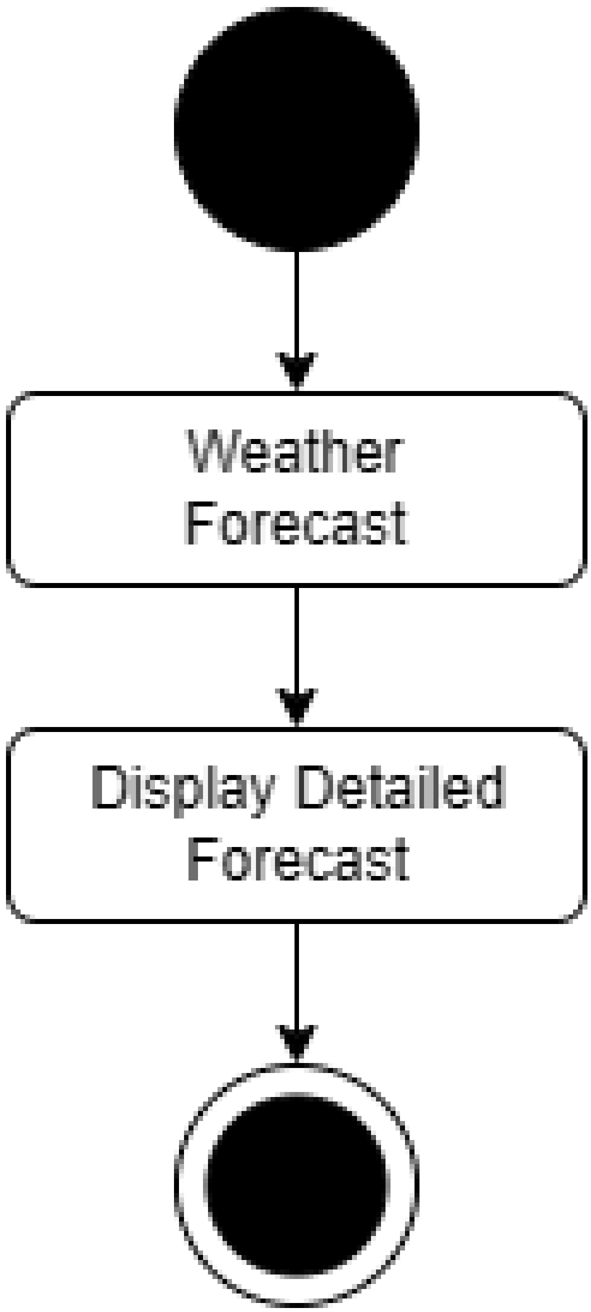


Figure 22: Activity Diagram of Weather Forecast

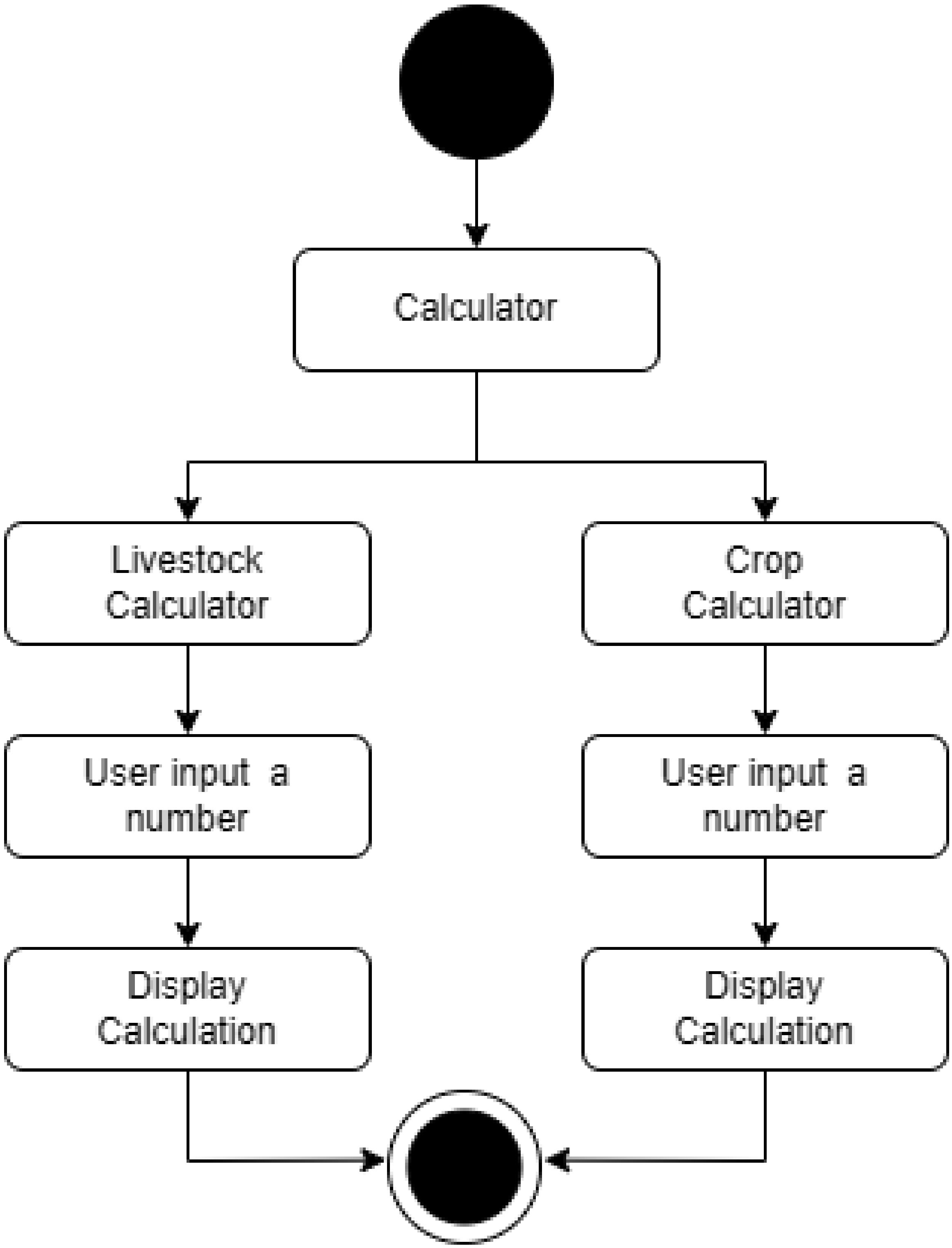

Figure 23: Activity Diagram of Calculator

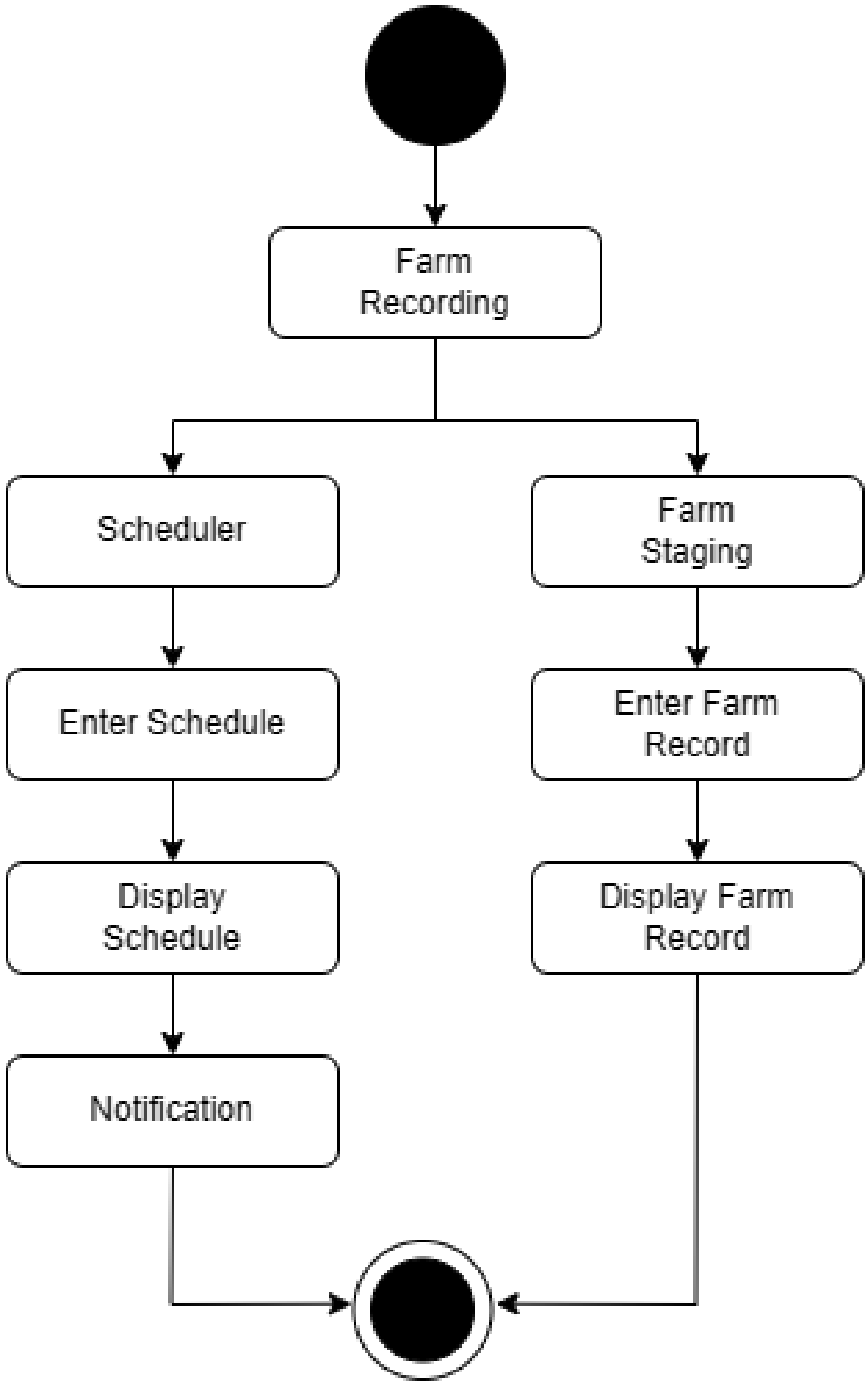


Figure 24: Activity Diagram of Farm Recording

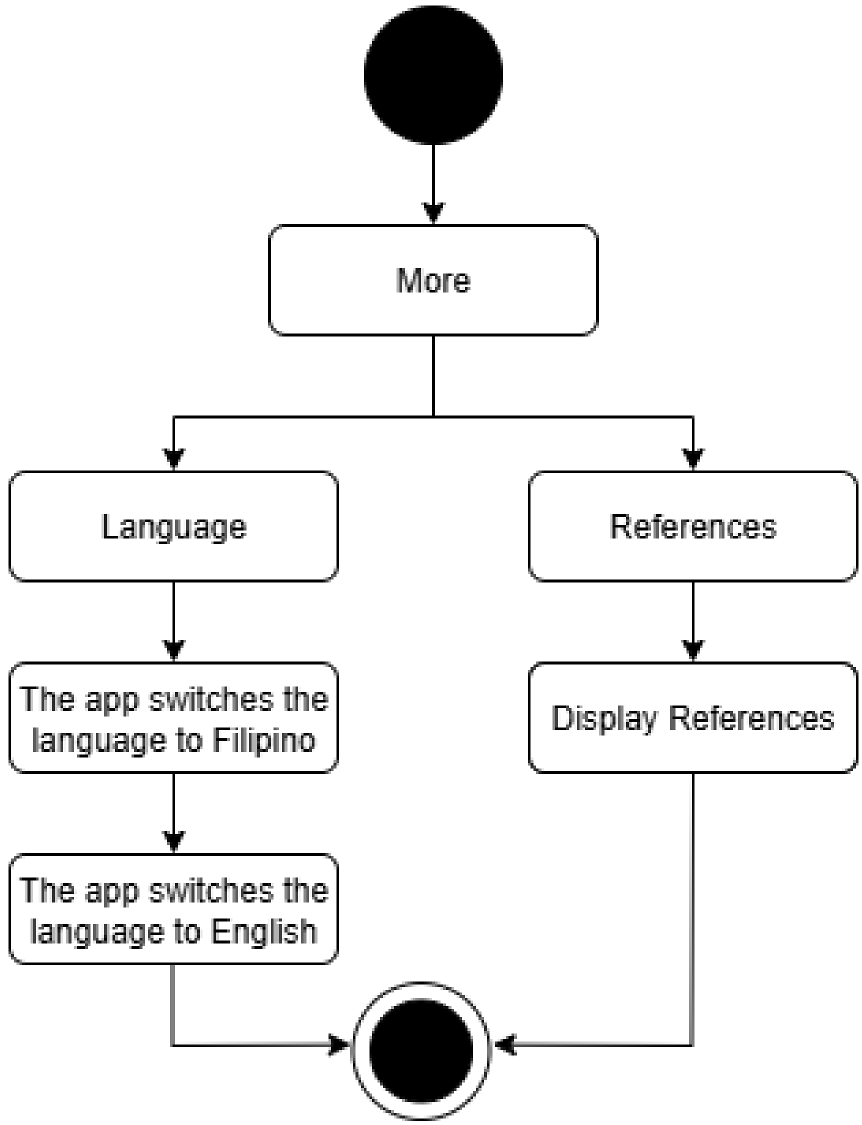


Figure 25: Activity Diagram of More

### 4.2.4 Risk Assessment/ Analysis

There are several risks that proponents have faced, first is the class schedule, the proponents are divided into 3 sections making it difficult to have a time in composing and revising the paper including the prototyping. This schedule problem was resulting in a short time of work and delay progress. Secondly, lacks of communication within the group due to the responsibilities in other subjects, this was resulted in an overload in revising the document. Third, finding a resource person that will help the proponents in having information about crops and livestock, including the name that needs to be verified. The proponents went through with these risks that helps them to keep dedicated and hard work on their capstone project by having a good communication, time management, and effort. Also, the proponents are expecting to have a problem in developing the capstone project. One of the major problems in developing the proposed project is the proponents lacks of high specification of personal computer and laptop.

## 4.3 Design

### 4.3.1 Output and User-interface Design


Figure 26: Color Palette of AgriCare

AgriCare uses the colors #4CAF50, #2E7D32, and #F5F8F5, which are shades of green that represent trust, growth, reliability, and calmness. It also incorporates neutral tones such as E8F5E9, which symbolize simplicity and cleanliness and a minimalist background. These colors enhance readability and make the information easy to understand.

The proponents chose Space Mono for its clean, modern look, which enhances readability and structure, ensuring a consistent and organized appearance in the design.

Whereas recognition of the inherent dignity

Figure 27: Typography

Whereas disregard and contempt for human rights have resulted

Figure 28: Typography 2

**Logo:**

- The logo of AgriCare consists of a stylized minimalistic design that represents the goal of the application helping Farmers with their line of work. the logo consists of a Farmer wearing a hat and holding a plant and a hoe representing Agriculture. The design is made to be simple and easily recognizable. The logo includes a plant that represents sustainability and progress, And the hoe represents a traditional farming tool bridging the gap between Agriculture and technology driven solutions into their day-to-day life.

Figure 29: Logo

**Icons**

- The proponents used icons such as rice, corn, bamboo, cow, pig, chicken, cloud, arrow, water, sun, calculator, schedule, farm, virus, trash, check, injection, bottle to easily recognize on what the user will click, this will help and guide them in using the proposed project.

Figure 30: Icons

### 4.3.2 System Architecture

#### Network Model

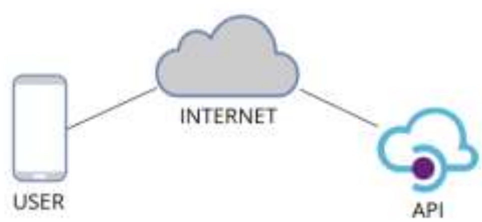


Figure 30: Network Model

## 4.4 Development

### 4.4.1 Software Specification

Table 8: Software Specification

| | |
|---|---|
| Android Studio | **Version:** Android Studio Narwhal 3 Feature<br>**Initial Release:** September 2, 2025<br><br>**First appeared:** May 16, 2013 |

| | |
|---|---|
| | **Website:** developer.android.com/studio |
| Flutter | Version: 3.24.3<br><br>**Initial Release:** May 2017<br><br>**First appeared:** May 2017 (Google I/O)<br><br>**Website:** flutter.dev |
| Dart | **Version:** 3.8.1<br><br>**Initial Release:** May 28, 2025<br><br>**First appeared:** October 10, 2011<br><br>**Website:** dart.dev |
| Xcode | **Version:** Xcode 26.0<br><br>**Initial Release:** September 15, 2025<br><br>**First appeared:** October 23, 2003<br><br>**Website:** developer.apple.com/xcode |
| Nominatim Demo | **Version:** 5.8.0<br><br>**Initial Release:** 2009<br><br>**First appeared:** 2009<br><br>**Website:** nominatim.org / demo |

### 4.4.2 Hardware Specification

The Application was develop using the following hardware specifications:

Table 9: Hardware Specification

| | |
|---|---|
| Operating System: | macOS Sequoia Version 15.6.1 |
| System Manufacturer: | **Apple** |
| System Model | iMac (24-inch, M1, 2021 |
| Firmware | EFI-based firmware |
| Processor | Apple M1 |
| OS Memory | 8 GB |
| Storage | 245.11 GB |

#### 4.4.3 Program Specification

Table 10: Program Specification of AgriCare

| **Program Specification of AgriCare**: A Mobile Application for Farming and Livestock Raising in the Philippines. |
|---|
| **Programming language: Dart** |
| **Events:** |

| |
|---|
| Display the splash screen upon application launch |
| Display the main menu featuring (6) Modules |
| Terminate the application once the back button is pressed |

| |
|---|
| **Module**: Crops |
| **Purpose**: To give an information about crops listed in the application. |
| **Events:** |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. |
| Return to Home Screen when user click the back button. |
| Display the list of rice when user pressed rice button. |
| Return to previous user interface when back button is pressed. |
| Display the information of selected rice. |
| Return to previous user interface when back button is pressed. |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. |
| Return to Home Screen when user click the back button. |
| Display the list of corn when user pressed rice button. |
| Return to previous user interface when back button is pressed. |
| Display the information of selected corn. |
| Return to previous user interface when back button is pressed. |
| Display a crops user interface where user choose 3 different types of crops such as rice, |

| |
|---|
| corn, sugarcane. |
| Return to Home Screen when user click the back button. |
| Display the list of sugarcane when user pressed sugarcane button. |
| Return to previous user interface when back button is pressed. |
| Display the information of selected sugarcane. |
| Return to previous user interface when back button is pressed. |

| |
|---|
| **Module**: Livestock |
| **Purpose**: To give user information about different livestock listed in the application |
| **Events:** |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. |
| Return to Home Screen when user click the back button. |
| Display the list of cows when user pressed cow button. |
| Return to previous user interface when back button is pressed. |
| Display the information of selected cow. |
| Return to previous user interface when back button is pressed. |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. |
| Return to Home Screen when user click the back button. |
| Display the list of pig when user pressed rice button. |
| Return to previous user interface when back button is pressed. |

| |
|---|
| Display the information of selected pig. |
| Return to previous user interface when back button is pressed. |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. |
| Return to Home Screen when user click the back button. |
| Display the list of chicken when user pressed rice button. |
| Return to previous user interface when back button is pressed. |
| Display the information of selected chicken. |
| Return to previous user interface when back button is pressed. |

| |
|---|
| **Module**: Weather Forecast |
| **Purpose**: Display a condition of weather. |
| **Events**: |
| Display weather UI that shows a weather update. |
| Show detailed forecast and the seven days forecast |
| Return to Home Screen when user click the back button |

| |
|---|
| **Module**: Calculator |
| **Purpose**: To let the user calculate their expenses and profits in crops and livestock. |
| **Event**: |
| Display a module selection of farming such as crops and livestock. |

| |
|---|
| Return to Home Screen when user click the back button |
| Display a crops calculator when user pressed crops button. |
| Display a three selection of crops. |
| Return to Calculator Selection Screen when user click the back button. |
| Display a module selection of farming such as crops and livestock |
| Return to Home Screen when user click the back button |
| Display a livestock calculator when user pressed livestock button. |
| Display a three selection of livestock. |
| Return to Calculator Selection Screen when user click the back button. |

| |
|---|
| **Module**: Farm Recording |
| **Purpose**: To allow the user to set schedules and record the stages of farm activities, including crops and livestock. |
| **Events**: |
| Display Two Modules: Scheduler and Staging |
| Return to Home Screen when user click the back button |
| **Module**: Scheduler |
| **Purpose**: To let the user set schedule |
| **Events**: |
| Display Scheduler UI |
| Show the Set Schedule form when the user clicks the Add button. |

| |
|---|
| Display the updated schedule when the user clicks the Save button. |
| Display the schedule when the user clicks the Cancel button. |
| Return to Farm Recording UI, when user click the back button. |
| **Module**: Staging |
| **Purpose**: To allow users to record and manage the stages of crops and livestock. |
| **Events**: |
| Display a module that allows the user to input stages for crops and livestock. |
| Show the recorded entry when the user clicks the Save button. |
| Allow editing by displaying the saved record when the user clicks on it. |
| Show a confirmation dialog when the user clicks the Remove button. |
| Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. |
| Return to the Staging UI if the user clicks No in the confirmation dialog. |
| Return to Farm Recording UI, when users click the back button. |

| |
|---|
| **Module**: More |
| **Purpose**: To let the user choose their preferred language and to see the sources of information of crops including the source of images. |
| **Event**: |
| Display a toggle where user can choose Filipino or English Language. |
| Display a source of information in crops and livestock including the source of images. |

**4.4.4 Programming Environment**

- **Front End**

- **Dart**

  The proponents used Android Studio along with Flutter and Dart to build the application. Their intention is to evaluate the application across multiple Android devices and screen sizes. This environment allows the system to operate effectively.

- **Back End**

  - **Nominatim Demo**

    The proponents used Nominatim Demo from OpenStreetMap was utilized for geocoding and reverse geocoding functionalities.

  - **XCode**

    The proponents used Android Studio to implement the IOs version of the application. This software helps the proponents to fully run and deploy the IOs version of the AGRICARE application.

- **Programming Considerations and Issues**

  In developing the application, several programming considerations and issues were encountered. One of the primary challenges was the availability of crop images, as some crops have very limited or hard-to-find pictures. Another issue was the accuracy of weather notifications since the data relies on external sources and location services, which are sometimes unreliable in rural areas or barangay-level coverage. Determining the user's exact location for weather updates further adds complexity, as not all areas have detailed weather information available. Lastly is the implementing the scheduler and notification system presented challenges, particularly due to iOS restrictions that delay background notifications. This limitation affects not only iOS users but also the synchronization of notifications across Android devices

### 4.4.5 Test Plan

The Proponents conducted a test plan for the android and IOs version for it is needed to meet the expected requirements of the application. The application was tested on phone devices or smartphones with different operating systems and version to see if the application is properly working. This will help the proponents to get an information about the performance and functionality of the application.

## 4.5 Verification, Validation, Testing

### 4.5.1 Unit Testing

| **Program Specification of AgriCare**: A Mobile Application for Farming and Livestock Raising in the Philippines. | | |
|---|---|---|
| | **PASSED** | **FAILED** |
| **Events:** | | |
| Display the splash screen upon application launch | **PASSED** | |
| Display the main menu featuring (6) Modules | **PASSED** | |
| Terminate the application once the back button is pressed | **PASSED** | |

| **Module**: Crops | | |
|---|---|---|
| **Purpose**: To give information about crops listed in the application. | **PASSED** | **FAILED** |
| **Events:** | **PASSED** | |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. | **PASSED** | |
| Return to Home Screen when user click the | **PASSED** | |

| | | |
|---|---|---|
| back button. | | |
| Display the list of rice when user presses rice button. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display the information of selected rice. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. | **PASSED** | |
| Return to Home Screen when user click the back button. | **PASSED** | |
| Display the list of corn when user pressed rice button. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display the information of selected corn. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. | **PASSED** | |

| | | |
|---|---|---|
| Return to Home Screen when user click the back button. | **PASSED** | |
| Display the list of sugarcane when user pressed sugarcane button. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display the information of selected sugarcane. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |

| **Module**: Livestock | | |
|---|---|---|
| **Purpose**: To give user information about different livestock listed in the application | **PASSED** | **FAILED** |
| **Events:** | **PASSED** | |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. | **PASSED** | |
| Return to Home Screen when user click the back button. | **PASSED** | |
| Display the list of cows when user pressed cow button. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |

| | | |
|---|---|---|
| Display the information of selected cow. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. | **PASSED** | |
| Return to Home Screen when user click the back button. | **PASSED** | |
| Display the list of pig when user pressed rice button. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display the information of selected pig. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. | **PASSED** | |
| Return to Home Screen when user click the back button. | **PASSED** | |
| Display the list of chicken when user pressed rice button. | **PASSED** | |
| Return to previous user interface when back | **PASSED** | |

| | | |
|---|---|---|
| button is pressed. | | |
| Display the information of selected chicken. | **PASSED** | |
| Return to previous user interface when back button is pressed. | **PASSED** | |

| | | |
|---|---|---|
| **Module**: Calculator | | |
| **Purpose**: To let the user calculate their expenses and profits in crops and livestock. | **PASSED** | **FAILED** |
| **Event**: | **PASSED** | |
| Display a module selection of farming such as crops and livestock. | **PASSED** | |
| Return to Home Screen when user click the back button | **PASSED** | |
| Display a crops calculator when user pressed crops button. | **PASSED** | |
| Display a three selection of crops. | **PASSED** | |
| Return to Calculator Selection Screen when user click the back button. | **PASSED** | |
| Display a module selection of farming such as crops and livestock | **PASSED** | |
| Return to Home Screen when user click the back button | **PASSED** | |
| Display a livestock calculator when user pressed livestock button. | **PASSED** | |

| | | |
|---|---|---|
| Display a three selection of livestock. | **PASSED** | **FAILED** |
| Return to Calculator Selection Screen when user click the back button. | **PASSED** | |

| **Module**: Farm Recording | | |
|---|---|---|
| **Purpose**: To allow the user to set schedules and record the stages of farm activities, including crops and livestock. | **PASSED** | |
| **Events**: | **PASSED** | |
| Display Two Modules: Scheduler and Staging | **PASSED** | |
| Return to Home Screen when user click the back button | **PASSED** | |
| **Module**: Scheduler | **PASSED** | |
| **Purpose**: To let the user set schedule | **PASSED** | |
| **Events**: | **PASSED** | |
| Display Scheduler UI | **PASSED** | |
| Show the Set Schedule form when the user clicks the Add button. | **PASSED** | |
| Display the updated schedule when the user clicks the Save button. | **PASSED** | |
| Display the schedule when the user clicks the Cancel button. | **PASSED** | |
| Return to Farm Recording UI, when user | **PASSED** | |

| | | |
|---|---|---|
| click the back button. | | |
| **Module**: Staging | **PASSED** | |
| **Purpose**: To allow users to record and manage the stages of crops and livestock. | **PASSED** | |
| **Events**: | **PASSED** | |
| Display a module that allows the user to input stages for crops and livestock. | **PASSED** | |
| Show the recorded entry when the user clicks the Save button. | **PASSED** | |
| Allow editing by displaying the saved record when the user clicks on it. | **PASSED** | |
| Show a confirmation dialog when the user clicks the Remove button. | **PASSED** | |
| Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. | **PASSED** | |
| Return to the Staging UI if the user clicks No in the confirmation dialog. | **PASSED** | |
| Return to Farm Recording UI, when users click the back button. | **PASSED** | |

| **Module**: More | | |
|---|---|---|
| **Purpose**: To let the user choose their preferred language and to see the sources of information of crops including the source of images. | **PASSED** | **FAILED** |

| | | |
|---|---|---|
| **Event**: | **PASSED** | |
| Display a toggle where user can choose Filipino or English Language. | **PASSED** | |
| Display a source of information in crops and livestock including the source of images. | **PASSED** | |

**4.6.1 Integration Testing**

- **Compatibility Testing**

Table 11: Compatibility Test (Android)

| **AgriCare: A Mobile Application for Farming & Livestock Raising in the Philippines** | | | | | | |
|---|---|---|---|---|---|---|
| | **Vivo Y15 Android 7 (Nougat)** | **Realme 8 Android 8 (Oreo)** | **Samsung S24 Android 15 (Vanilla Ice Cream)** | **Samsung Galaxy S23 Series Android 13 (Tiramisu)** | **Xiaomi Redmi 6A Android 8.1 (Oreo)** | **Nokia 9 PureView Android 8.1 (Oreo)** |
| Did the application install? | Failed | Passed | Passed | Passed | Passed | Passed |
| Did the application launch? | - | Passed | Passed | Passed | Passed | Passed |
| Did the screen resolution | - | Passed | Passed | Passed | Passed | Passed |

| | | | | | | |
|---|---|---|---|---|---|---|
| of the UI adjust properly? | | | | | | |
| Did the butt ons function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the crops module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the crop information module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the crop information display without an internet connection? | - | Passed | Passed | Passed | Passed | Passed |
| Did the livestock module | - | Passed | Passed | Passed | Passed | Passed |

| | | | | | | |
|---|---|---|---|---|---|---|
| function correctly? | | | | | | |
| Did the livestock information module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the livestock information display without an internet connection? | - | Passed | Passed | Passed | Passed | Passed |
| Did the weather module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the calculator module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the farm recording module | - | Passed | Passed | Passed | Passed | Passed |

| | | | | | | |
|---|---|---|---|---|---|---|
| function correctly? | | | | | | |
| Did the scheduler module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the staging module function correctly? | - | Passed | Passed | Passed | Passed | Passed |
| Did the more module function properly? | - | Passed | Passed | Passed | Passed | Passed |

Table 12: Compatibility Test (iOS)

| **AgriCare: A Mobile Application for Farming & Livestock Raising in the Philippines** | | | |
|---|---|---|---|
| | **iOS 15** | **iOS 16** | **iOS 18.5** |
| Did the application install? | - | Passed | Passed |
| Did the application launch? | - | Passed | Passed |

| | | | |
|---|---|---|---|
| Did the screen resolution of the UI adjust properly? | - | Passed | Passed |
| Did the buttons function correctly? | - | Passed | Passed |
| Did the crops module function correctly? | - | Passed | Passed |
| Did the crop information module function correctly? | - | Passed | Passed |
| Did the crop information display without an internet connection? | - | Passed | Passed |
| Did the livestock module function correctly? | - | Passed | Passed |
| Did the livestock information module function correctly? | - | Passed | Passed |
| Did the livestock information display without an internet connection? | - | Passed | Passed |
| | - | Passed | Passed |
| Did the calculator module function correctly? | - | Passed | Passed |

| | | | |
|---|---|---|---|
| Did the farm recording module function correctly? | - | Passed | Passed |
| Did the weather module function correctly? | - | Passed | Passed |
| Did the staging module function correctly? | - | Passed | Passed |
| Did the more module function properly? | - | Passed | Passed |

**Performance Testing**

- **Load Testing**

In load testing, the performance of an application is evaluated to determine how well it manages different levels of user demand. This process involves simulating many users accessing the system at the same time to measure its responsiveness.

Table 13: Load Testing

| Action | Simulated Users | Expected Result |
|---|---|---|
| **Module: Crops** | 18 users | Load in under 1 second |
| Display a crops user interface where user choose 3 different types of crops such as rice, corn, sugarcane. | 18 users | Load in under 1 second |
| Display the list of rice when user pressed rice button. | 18 users | Load in under 1 second |
| Display the information of selected rice. | 18 users | Load in under 1 second |

| | | |
|---|---|---|
| Return to previous user interface when back button is pressed. | 18 users | Load in under 1 second |
| Display a crops user interface where user chooses 3 different types of crops such as rice, corn, sugarcane. | 18 users | Load in under 1 second |
| Display the list of corn when user pressed corn button. | 18 users | Load in under 1 second |
| Display the information of selected corn. | 18 users | Load in under 1 second |
| Return to previous user interface when back button is pressed. | 18 users | Load in under 1 second |
| Display the list of sugarcane when user pressed sugarcane button. | 18 users | Load in under 1 second |
| Display the information of selected sugarcane. | 18 users | Load in under 1 second |
| Return to previous user interface when back button is pressed. | 18 users | Load in under 1 second |
| **Module: Livestock** | | |
| Display a livestock user interface where user choose 3 different types of livestock listed in the application such as cow, pig, and chicken. | 18 users | Load in under 1 second |
| Display the list of cows when user pressed cow button. | 18 users | Load in under 1 second |
| Display the information of selected cow. | 18 users | Load in under 1 second |
| Return to previous user interface when back | 18 users | Load in under 1 |

| | | |
|---|---|---|
| button is pressed. | | second |
| Display the list of pig when user pressed pig button. | 18 users | Load in under 1 second |
| Display the information of selected pig. | 18 users | Load in under 1 second |
| Return to previous user interface when back button is pressed. | 18 users | Load in under 1 second |
| Display the list of chicken when user pressed Chicken button. | 18 users | Load in under 1 second |
| Display the information of selected chicken. | 18 users | Load in under 1 second |
| Return to previous user interface when back button is pressed. | 18 users | Load in under 1 second |
| **Module: Calculator** | | |
| Display a module selection of farming, crops and livestock | 18 users | Load in under 1 second |
| Display a livestock calculator when user pressed livestock button. | 18 users | Load in under 1 second |
| Display a three selection of livestock. | 18 users | Load in under 1 second |
| Return to Calculator Selection Screen when user click the back button. | 18 users | Load in under 1 second |
| Display a Crops calculator when user pressed livestock button. | 18 users | Load in under 1 second |
| Display a three selection of Crops. | 18 users | Load in under 1 second |

| | | |
|---|---|---|
| Return to Calculator Selection Screen when user click the back button. | 18 users | Load in under 1 second |
| **Module: Farm Recording** | | |
| Display Two Modules: Scheduler and Staging | 18 users | Load in under 1 second |
| **Module: Scheduler** | | |
| Show the Set Schedule form when the user clicks the Add button. | 18 users | Load in under 1 second |
| Display the updated schedule when the user clicks the Save button. | 18 users | Load in under 1 second |
| Display the schedule when the user clicks the Cancel button. | 18 users | Load in under 1 second |
| **Module: Staging** | | |
| Display a module that allows the user to input stages for crops and livestock. | 18 users | Load in under 1 second |
| Show the recorded entry when the user clicks the Save button. | 18 users | Load in under 1 second |
| Allow editing by displaying the saved record when the user clicks on it. | 18 users | Load in under 1 second |
| Show a confirmation dialog when the user clicks the Remove button. | 18 users | Load in under 1 second |
| Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. | 18 users | Load in under 1 second |
| Return to the Staging UI if the user clicks No in the confirmation dialog. | 18 users | Load in under 1 second |
| Return to Farm Recording UI, when users | 18 users | Load in under 1 |

| | | |
|---|---|---|
| click the back button. | | second |
| **Module: More** | | |
| Display a toggle where user can choose Filipino or English Language. | 18 users | Load in under 1 second |
| Display a source of information in crops and livestock including the source of images. | 18 users | Load in under 1 second |
| **Module: Weather 18** | | |
| Display the weather condition. | 18 users | Loads in an average of 1.5 seconds |
| Display the result of search bar. | 18 users | Returns result in under 1 second |
| Display the result of refresh button. | 18 users | Returns result in under 2 seconds |

Table 14: Concurrent

| Action | Concurrent users | Expected result |
|---|---|---|
| Module: Weather | | |
| Show detailed forecast | 18 users | Loads in an average of 1.5 seconds |
| Search Bar | 18 users | Returns result in under 1 second |
| Refresh Button | 18 users | Returns result in under 2 seconds |

- **Pacing Time**

Pacing is an important factor in understanding how users interact with the AgriCare application. It refers to the time users spend between actions while using the application. This helps create realistic test scenarios that reflect actual user behavior. The proponents examine how users manage their pace when navigating the different features and modules of the app. By recording the time between actions, they gain insights into the natural flow of user activity and the time spent processing content. In this project, the proponents themselves act as respondents, using their mobile devices to manually measure and record pacing data.

Table 15: Pacing Time Result (Android)

| AgriCare: A Mobile Application for Farming and Livestock Raising in the Philippines | | |
|---|---|---|
| Module | Action | Pacing Time |
| Crop | | |
| | Display crops list when crops button was pressed. | 1.25 sec |
| | Display rice list when user pressed rice button. | 2.96 sec |
| | Display the information of specific rice when user pressed it. | 4.27 sec |
| | Display the land preparation when user pressed dry preparation or wet preparation. | 1.25 sec |
| | Display the pests/insect information when user pressed the specific pests. | 2.03 sec |
| | Display the diseases information when user pressed the | 2.30 sec |
| | Display corn list when user pressed corn button | 2.55 sec |
| | Display the information of specific corn when user pressed it. | 3.75 sec |

| | | |
|---|---|---|
| | Display sugarcane list when user pressed sugarcane button. | 1.25 sec |
| | Display the information of specific sugarcane when user pressed it. | 2.35 sec |
| Livestock | | |
| | Display livestock list when livestock button was pressed. | 1.05 sec |
| | Display cow list when user pressed cow button. | 1.26 sec |
| | Display the information of specific cow when user pressed it. | 4.02 sec |
| | Display pig list when user pressed pig button | 1.25 sec |
| | Display the information of specific pig when user pressed it. | 1.78 sec |
| | Display chicken list when user pressed chicken button | 1.05 sec |
| | Display the information of specific chicken when user pressed it. | 5.06 sec |
| Weather | | |
| | Shows the detailed forecast. | 9.08 sec |
| | Shows the result of search bar | 4.02 sec |
| | Shows the result of refresh button | 7.06 sec |
| Calculator | | |

| | | |
|---|---|---|
| | Display a selection of farming such as crops and livestock. | 2.82 sec |
| | Display the calculator of crops when user pressed crops button. | 15.92 sec |
| | Display the result when user pressed the calculate button. | 5.82 sec |
| | Display a selection of livestock such as cow, pig, chicken when user pressed livestock button in. | 2.75 sec |
| | Display the calculator of cow when user pressed it. | 10.27 sec |
| | Display the result when user pressed the calculate button | 3.27 sec |
| | Display the calculator of pig when user pressed it. | 7.28 sec |
| | Display the result when user pressed the calculate button | 4.28 sec |
| | Display the calculator of chicken when user pressed it. | 7.29 sec |
| | Display the result when user pressed the calculate button | 4.23 sec |
| | Display the history when user pressed the button | 8.23 sec |
| Farm Recording | | |
| | Display two modules when user pressed Farm Recording button | 2.48 sec |
| Scheduler | | |

| | | |
|---|---|---|
| | Show the set schedule form when the user clicks the add button. | 8.02 sec |
| | Display the updated schedule when the user clicks the save button | 4.85 sec |
| | Display the schedule when the user clicks the Cancel button. | 4.27 sec |
| | Display a confirmation if user pressed the delete button. | 2.29 sec |
| Staging | | |
| | Display a module that allows the user to input stages for crops and livestock. | 3.52 sec |
| | Show the recorded entry when the user clicks the save button. | 3.23 sec |
| | Allow editing by displaying the saved record when the user clicks on it. | 7.28 sec |
| | Show a confirmation dialog when the user clicks the Remove button. | 4.50 sec |
| | Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. | 5.23 sec |
| More | | |
| | Display a toggle where user can choose Filipino or English Language. | 6.23 sec |
| | Display a source of information in crops and livestock including the source of images. | 4.28 sec |

Table 16: Pacing Time Result (iOs)

| AgriCare: A Mobile Application for Farming and Livestock Raising in the Philippines | | |
|---|---|---|
| Module | Action | Pacing Time |
| Crop | | |
| | Display crops list when crops button was pressed. | 1.02 sec |
| | Display rice list when user pressed rice button. | 3.26 sec |
| | Display the information of specific rice when user pressed it. | 4.27 sec |
| | Display the land preparation when user pressed dry preparation or wet preparation. | 1.96 sec |
| | Display the pests/insect information when user pressed the specific pests. | 1.84 sec |
| | Display the diseases information when user pressed the | 2.80 sec |
| | Display corn list when user pressed corn button | 2.55 sec |
| | Display the information of specific corn when user pressed it. | 2.95 sec |
| | Display sugarcane list when user pressed sugarcane button. | 1.05 sec |
| | Display the information of specific sugarcane when user pressed it. | 2.05 sec |
| Livestock | | |
| | Display livestock list when livestock button was pressed. | 1.05 sec |

| | | |
|---|---|---|
| | Display cow list when user pressed cow button. | 1.56 sec |
| | Display the information of specific cow when user pressed it. | 4.82 sec |
| | Display pig list when user pressed pig button | 2.03 sec |
| | Display the information of specific pig when user pressed it. | 1.64 sec |
| | Display chicken list when user pressed chicken button | 1.35 sec |
| | Display the information of specific chicken when user pressed it. | 5.86 sec |
| Weather | | |
| | Shows the detailed forecast. | 11.04 sec |
| | Shows the result of search bar | 4.75 sec |
| | Shows the result of refresh button | 7.86 sec |
| Calculator | | |
| | Display a selection of farming such as crops and livestock. | 3.02 sec |
| | Display the calculator of crops when user pressed crops button. | 18.02 sec |
| | Display the result when user pressed the calculate button. | 5.82 sec |
| | Display a selection of livestock such as cow, pig, chicken when user pressed livestock button in. | 2.28 sec |

| | | |
|---|---|---|
| | Display the calculator of cow when user pressed it. | 10.92 sec |
| | Display the result when user pressed the calculate button | 3.92 sec |
| | Display the calculator of pig when user pressed it. | 7.82 sec |
| | Display the result when user pressed the calculate button | 3.92 sec |
| | Display the calculator of chicken when user pressed it. | 7.29 sec |
| | Display the result when user pressed the calculate button | 4.23 sec |
| | Display the history when user pressed the button | 8.23 sec |
| Farm Recording | | |
| | Display two modules when user pressed Farm Recording button | 2.93 sec |
| Scheduler | | |
| | Show the set schedule form when the user clicks the add button. | 8.02 sec |
| | Display the updated schedule when the user clicks the save button | 4.05 sec |
| | Display the schedule when the user clicks the Cancel button. | 4.27 sec |
| | Display a confirmation if user pressed the delete button. | 2.09 sec |

| | | |
|---|---|---|
| Staging | | |
| | Display a module that allows the user to input stages for crops and livestock. | 3.52 sec |
| | Show the recorded entry when the user clicks the save button. | 3.93 sec |
| | Allow editing by displaying the saved record when the user clicks on it. | 7.28 sec |
| | Show a confirmation dialog when the user clicks the Remove button. | 4.50 sec |

- **Response Time**

Response time is an important indicator of an application's performance and user experience, showing how quickly the system responds to a user's request or action. In the case of the AgriCare application, analyzing response times provides valuable insights into its efficiency across various modules and features. For this study, the proponents served as respondents, using their mobile devices to manually measure and record response time data.

Table 17: Response Time (Android)

| AgriCare: A Mobile Application for Farming and Livestock Raising in the Philippines | | |
|---|---|---|
| Module | Action | Response Time |
| Crop | | |
| | Display crops list when crops button was pressed. | 0.21 ms |
| | Display rice list when user pressed rice button. | 0.26 ms |

| | | |
|---|---|---|
| | Display the information of specific rice when user pressed it. | 0.38 ms |
| | Display the land preparation when user pressed dry preparation or wet preparation. | 0.35 ms |
| | Display the pests/insect information when user pressed the specific pests. | 0.20 ms |
| | Display the diseases information when user pressed the | 0.21 ms |
| | Display corn list when user pressed corn button | 0.19 ms |
| | Display the information of specific corn when user pressed it. | 0.21 ms |
| | Display sugarcane list when user pressed sugarcane button. | 0.19 ms |
| | Display the information of specific sugarcane when user pressed it. | 0.28 ms |
| Livestock | | |
| | Display livestock list when livestock button was pressed. | 0.14 ms |
| | Display cow list when user pressed cow button. | 0.23 ms |
| | Display the information of specific pig when user pressed it. | 0.18 ms |
| | Display pig list when user pressed pig button | 0.26 ms |
| | Display the information of specific pig when user pressed it. | 0.28 ms |

| | | |
|---|---|---|
| | Display chicken list when user pressed chicken button | 0.23 ms |
| | Display the information of specific chicken when user pressed it. | 0.28 ms |
| Weather | | |
| | Shows the detailed forecast. | 1.38 ms |
| | Shows the result of search bar | 0.96 ms |
| | Shows the result of refresh button | 1.21 ms |
| Calculator | | |
| | Display a selection of farming such as crops and livestock. | 0.36 ms |
| | Display the calculator of crops when user pressed crops button. | 0.36 ms |
| | Display the result when user pressed the calculate button. | 0.41 ms |
| | Display a selection of livestock such as cow, pig, chicken when user pressed livestock button in. | 0.21 ms |
| | Display the calculator of cow when user pressed it. | 0.21 ms |
| | Display the result when user pressed the calculate button | 0.21 ms |
| | Display the calculator of pig when user pressed it. | 0.21 ms |
| | Display the result when user pressed the calculate button | 0.21 ms |

| | | |
|---|---|---|
| | Display the calculator of chicken when user pressed it. | 0.18 ms |
| | Display the result when user pressed the calculate button | 0.18 ms |
| Farm Recording | | 0.19 ms |
| | Display two modules when user pressed Farm Recording button | 0.19 ms |
| Scheduler | | |
| | Show the set schedule form when the user clicks the add button. | 0.23 ms |
| | Display the updated schedule when the user clicks the save button | 0.23 ms |
| | Display the schedule when the user clicks the Cancel button. | 0.19 ms |
| | Display the schedule of the specific date. | 0.28 ms |
| | Display a confirmation if user pressed the delete button. | 0.26 ms |
| Staging | | |
| | Display a module that allows the user to input stages for crops and livestock. | 0.31 ms |
| | Show the recorded entry when the user clicks the save button. | 0.28 ms |

| | | |
|---|---|---|
| | Allow editing by displaying the saved record when the user clicks on it. | 0.24 ms |
| | Show a confirmation dialog when the user clicks the Remove button. | 0.26 ms |
| | Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. | 0.23 ms |
| More | | |
| | Display a toggle where user can choose Filipino or English Language. | 0.18 ms |
| | Display a source of information in crops and livestock including the source of images. | 0.16 ms |

Table 18: Response Time (iOS)

| AgriCare: A Mobile Application for Farming and Livestock Raising in the Philippines | | |
|---|---|---|
| Module | Action | Response Time |
| Crop | | |
| | Display crops list when crops button was pressed. | 0.21 ms |
| | Display rice list when user pressed rice button. | 0.21 ms |
| | Display the information of specific rice when user pressed it. | 0.14 ms |
| | Display the land preparation when user pressed dry preparation or wet preparation. | 0.21 ms |
| | Display the pests/insect information when user pressed the specific pests. | 0.21 ms |

| | | |
|---|---|---|
| | Display the diseases information when user pressed the | 0.21 ms |
| | Display corn list when user pressed corn button | 0.14 ms |
| | Display the information of specific corn when user pressed it. | 0.14 ms |
| | Display sugarcane list when user pressed sugarcane button. | 0.21 ms |
| | Display the information of specific sugarcane when user pressed it. | 0.21 ms |
| Livestock | | |
| | Display livestock list when livestock button was pressed. | 0.14 ms |
| | Display cow list when user pressed cow button. | 0.14 ms |
| | Display the information of specific pig when user pressed it. | 0.21 ms |
| | Display pig list when user pressed pig button | 0.14 ms |
| | Display the information of specific pig when user pressed it. | 0.21 ms |
| | Display chicken list when user pressed chicken button | 0.14 ms |
| | Display the information of specific chicken when user pressed it. | 0.21 ms |
| Weather | | |

| | | |
|---|---|---|
| | Shows the detailed forecast. | 1.59 sec |
| | Shows the result of search bar | 0.86 ms |
| | Shows the result of refresh button | 0.60 ms |
| Calculator | | |
| | Display a selection of farming such as crops and livestock. | 0.34 ms |
| | Display the calculator of crops when user pressed crops button. | 0.21 ms |
| | Display the result when user pressed the calculate button. | 0.41 ms |
| | Display a selection of livestock such as cow, pig, chicken when user pressed livestock button in. | 0.22 ms |
| | Display the calculator of cow when user pressed it. | 0.14 ms |
| | Display the result when user pressed the calculate button | 0.14 ms |
| | Display the calculator of pig when user pressed it. | 0.24 ms |
| | Display the result when user pressed the calculate button | 0.15 ms |
| | Display the calculator of chicken when user pressed it. | 0.14 ms |
| | Display the result when user pressed the calculate button | 0.22 ms |

| | | |
|---|---|---|
| Farm Recording | | |
| | Display two modules when user pressed Farm Recording button | 0.15 ms |
| Scheduler | | |
| | Show the set schedule form when the user clicks the add button. | 0.14 ms |
| | Display the updated schedule when the user clicks the save button | 0.15 ms |
| | Display the schedule when the user clicks the Cancel button. | 0.21 ms |
| | Display a confirmation if user pressed the delete button. | 0.21 ms |
| Staging | | |
| | Display a module that allows the user to input stages for crops and livestock. | 0.14 ms |
| | Show the recorded entry when the user clicks the save button. | 0.14 ms |
| | Allow editing by displaying the saved record when the user clicks on it. | 0.14 ms |
| | Show a confirmation dialog when the user clicks the Remove button. | 0.14 ms |
| | Display the Updated Staging UI if the user clicks Yes in the confirmation dialog. | 0.21 ms |

| | | |
|---|---|---|
| More | | |
| | Display a toggle where user can choose Filipino or English Language. | 0.14 ms |
| | Display a source of information in crops and livestock including the source of images. | 0.15 ms |

**Stress Testing**

- **Stress Testing**

The stress test for AgriCare checked how well the app works when many actions are done at the same time, like entering data repeatedly, moving quickly between modules, and using different features often. The results show that AgriCare can handle heavy use and still give stable and reliable performance for farmers.

Table 19: Stress Testing

| | Load Applied | Response Time | Result | Observation |
|---|---|---|---|---|
| **Module: Calculator** | | | | |
| Calculates crops and livestock | 20 calculation entries using different input values. | 1-2 seconds per save | Successful | Calculations saved quickly without issues |
| **Module: Scheduler** | | | | |
| Set schedules | 10 repeat tasks scheduled on separate dates. | Under 1 second per save | Successful | The process was smooth and stable. |
| **Module: Staging** | | | | |

| | | | | |
|---|---|---|---|---|
| Record farm stages | Saving records locally | Under 1 second per save | Successful | Records were stored properly with no delays |

### 4.6.2 Acceptance Testing

For the AgriCare application, acceptance testing is done to check if the app meets the requirements and expectations of its users. This testing ensures that the application works correctly, is user-friendly, and provides the functions that were planned. The main goal is to confirm that the app is ready for actual use and can satisfy the needs of its intended users.

#### 4.6.2.1 Alpha Testing

The proponents conducted alpha testing to assess the overall performance of the AgriCare application by simulating real user interactions. The primary goal of this phase was to identify and resolve any issues related to functionality, usability, and interface design. This allowed us to ensure that the application delivers a consistent, efficient, and user-friendly experience.

#### 4.6.2.2 Beta Testing

The proponents conducted beta testing using the final product of the application after the alpha testing. The proponents did a survey by using google form to easily see the feedback of the respondents such as Department of Agriculture staffs and famers.

Table 13: Feedback for Alpha Testing

| Test Scenario | Input Data | Expected Result | Result (Passed/Failed) |
|---|---|---|---|
| **Livestock Calculator** | | | |
| Missing Required Fields (Pig Weight (kg)) | Pig Weight (kg):<br>Price (kg): 230<br>Add Expenses: 1400 | Display Error. “Please enter valid numbers for all fields”. | Passed |
| **Crop Calculator** | | | |
| Missing Required Fields (Enter Hectare/square meter) | Enter Hectare/square meter: Enter plant spacing/meter: 3 Enter row spacing: 5 | Display Error. “Please enter valid numbers for all fields”. | Passed |
| **Scheduler** | | | |

| | | | |
|---|---|---|---|
| Incomplete setup in scheduling. | Description: Empty | Display Error. "Please enter a description for your schedule." | Passed |
| **Staging** | | | |
| Incomplete input | Select Date:<br>Select Activity: Name: | Display Error. "Please fill In all required fields." | Passed |

Table 20: Feedback for Beta Testing

| Item | Frequency | | | | | |
|---|---|---|---|---|---|---|
| | 1 Strongly Disagree | 2 Somewhat Disagree | 3 Disagree | 4 Agree | 5 Somewhat Agree | 6 Strongly Agree |
| **Content** | | | | | | |
| The content is clear. | | | | 3 (16.7%) | 4 (22.2%) | 12 (66.7%) |
| The content is easy to understand. | | | | 4 (22.2%) | 3 (16.7%) | 12 (66.7%) |
| The content is related to Agriculture topic. | | | | 1 (5.6%) | 3 (16.7%) | 14 (77.8%) |
| The content in AgriCare is interesting. | | | | 3 (16.7%) | 1 (5.6%) | 14 (77.8%) |
| Module | | | | | | |
| Crop | | | | 5 (27.8%) | 4 (22.2%) | 11 (61.1%) |
| Livestock | | | | 4 (22.2%) | 4 (22.2%) | 10 (55.6%) |
| Weather | | 1 (5.6%) | | 3 (16.7%) | 5 (27.8%) | 10 (55.6%) |
| Calculator | | | | 3 (16.7%) | 5 (27.7%) | 10 (55.6%) |
| Farm Recording | | | | 3 (16.7%) | 3 (16.7%) | 12 (66.7%) |
| More | | | | 3 (16.7%) | 5 (16.7%) | 10 (55.6%) |
| Multimedia Element | | | | | | |
| Appropriate font type | | | 1 (5.6%) | 3 (16.7%) | 2 (11.1%) | 12 (66.7%) |
| Appropriate font size | | | 1 (5.6%) | 4 (22.2%) | 2 (11.1%) | 11 (61.1%) |
| Appropriate graphics | | | | 4 (22.2%) | 2 (11.1%) | 12 (66.7%) |
| Appropriate button | | | | 2 (11.1%) | 3 (16.7%) | 14 (77.8%) |
| Appropriate color | | | 1 (5.6%) | 4 (22.2%) | 3 (16.7%) | 10 (55.6%) |

| | | | | | | |
|---|---|---|---|---|---|---|
| Navigation | | | | | | |
| Navigation is easy | | | | 3 (16.7%) | 1 (5.6%) | 14 (77.8%) |
| Navigation is clear and concise | | | | 3 (16.7%) | 3 (16.7%) | 12 (66.7%) |
| Number of buttons / links reasonable | | | 1 (5.6%) | 2 (11.1%) | 3 (16.7%) | 10 (55.6%) |
| **Usefulness** | | | | | | |
| AgriCare is useful for Visual students (Picture) | | | | 2 (11.1%) | 3 (16.7%) | 13 (72.2%) |
| AgriCare is useful for Auditory students (Sound) | | | | 4 (22.2%) | 3 (16.7%) | 11 (61.1%) |
| AgriCare is useful for Kinesthetic students (Touch) | 1 (5.6) | 1 (5.6) | 1 (5.6%) | 3 (16.7%) | 2 (11.1%) | 11 (61.1%) |

**4.6**

**Implementation plan**

This section outlines the hardware, environmental conditions, and infrastructure which can impact the usability of the application and its usage

**4.6.3 Physical Environment**

. The AgriCare application is designed to be compatible with Android devices using 8.0 (Oreo) and above, requiring approximately 109 MB of storage. For the iOS devices, the application works with version 16 and above in which the estimated required storage will be about 104 MB to download the application. Due to the application using a small amount of storage, it is sufficiently small to download and install, regardless of the device whether a budget device or a high end device.

**4.6.4 Interface**

The inputs of the AgriCare application come from the users which provided data from farm schedules, farm recording, and calculator options. The AgriCare application also produces output and visual information displayed on the application or users. The application system also produces push notifications as reminders to notify the users about their scheduled task, Farming stages, or weather conditions.

#### 4.6.3 Functionally

The AgriCare application is built to provide information on aspiring farmers which does not require an internet connection to access information regarding crops and livestock. However, the weather forecast does require an internet connection as well as the user's current location or GPS to determine weather conditions. Furthermore, AgriCare has a scheduled user setup task as a drive for the users to remind them of their task when a push notification is received from the application.

#### 4.6.7 Data

The AgriCare application can accept both numerical and textual input with numerical data being the values for the calculator, scheduled tasks, and crop spacing measurements. User data, such as schedules, staging records and calculator entries, is stored on the device so that it is accessible at all times, even without an internet connection. Weather data requires an internet connection only to sync and store the information necessary for notifications.

### 4.6 Installation

The following instructions provide users with a step-by-step guide for installing the AgriCare mobile application:

#### 4.6.1 Installing AgriCare Application on Android Devices

1. Open the Google Play Store on your Android devices.
2. Tap on the search bar at the top of the screen.
3. Type “AgriCarePH” into the search bar.
4. From the search result, select the AgriCarePH with the official logo.
5. Tap the” Install” button to begin downloading the application.
6. Once the installation is complete, tap Open to launch the application.

#### 4.6.2 Installing AgriCare Application on iOS Devices

1. Open the App Store on your iOS devices.
2. Tap the search tab at the bottom of the screen.
3. Type “AgriCarePH” into the search field.
4. From the search results, select AgriCarePH with the official logo.
5. Tap the Get button, and then confirm the download using Face ID, Touch

ID or Apple ID password.

6. Once the installation is complete, tap Open to launch the app.

## 5.0 Conclusion and Recommendations

### 5.1 Conclusion

In Conclusion, the AgriCare application serves as learning to farming crops and livestock to help the aspiring farmers with the common varieties in the Philippines. The interface of the application has been positively received, making it easy for aspiring farmers to explore and use the features of the application.

Moreover, the integration of weather forecast that provides a notification about the humidity, wind speed, pressure, especially the rainfall tomorrow with the 7 days forecast. user feedback indicates strong agreement on the app's usability, with 58.26% strongly agreeing and 17.13% somewhat agreeing that the interface is user-friendly. Additionally, 83.9% of respondents agree that the application supports their learning needs, while only a small percentage expressed disagreement (2.92% somewhat disagree and 19.4% disagree) with the total of 18 respondents.

These results show that AgriCare is both useful and educational, making it helpful for aspiring farmers. The application helps improve farming knowledge and skills, leading to better farming and make better decisions with current weather updates. This support allows farmers to adjust to changing weather conditions, which can lead to better crop and livestock production. Overall, AgriCare is an important tool for promoting sustainable farming and helping farmers in the Philippines become more knowledgeable.

### 5.2 Recommendation

Based on the conclusion and findings of this study, the proponents recommend the following changes and improvements for the AgriCare application in the future or for future researchers who plan to pursue similar projects:

**1. Additional Varieties of Crop and Livestock**

AgriCare currently offer a limited selection of crops and livestock, which may not fully meet the needs of aspiring farmers, to enhance the application, it is recommended to expand the range of varieties.

**2. Adding more types of calculators of crops and livestock**

AgriCare offers only a simple calculation for the aspiring farmers, so the proponents recommend the future researchers to enhance the calculator of the AgriCare application.

**3. Integration of video tutorials and audio reading**

To help users use the app more easily, it is recommended to future researchers to add video tutorials and audio reading features. This will help users understand farming better and convenient and supports those who have trouble reading.

**4. Community forum or Support Feature**

Adding a community forum or chat support within the application that allows the user to ask questions, share experiences, receive advises from expert and fellow farmers.

**5. Enhance Weather forecast feature**

To provide more detailed and localized weather information, the proponents recommended to improve the weather forecast of AgriCare such as users can use personalized notification by making it more customizable based on their crops and livestock needs.

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**APPENDICES**

**Appendix A**

**Work Assignment**

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*Capas, Tarlac*

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| **Title Project:** | **Date:** October 2025 |
|---|---|
| **AGRICARE: A MOBILE APPLICATION FOR FARMING & LIVESTOCK RAISING IN THE PHILIPPINES** | |
| **Name of the Proponent:** Baluyut, Carl Patrick M. | **Signature:** |

**Project Participations:**

- UI Designer
- Participated in Brainstorming
- Participated in Documentation of Chapter 1
- Participated in Documentation of Chapter 2
- Participated in constructing Appendix B
- Participated in constructing Appendix D

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| **Name of the Proponent:** David, Natasha C. | **Signature:** |
|---|---|

**Project Participations:**

- Programmer
- Developed the AgriCare application
- Participated in Prototyping
- Participated in Brainstorming
- Participated in Chapter 1 documentation
- Participated in Chapter 2 documentation
- Participated in Chapter 3 documentation
- Participated in Chapter 4 documentation
- Composed Appendix A
- Participated in constructing Appendix C

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| | |
|---|---|
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**Project Participations:**

- UI designer
- Participated in Brainstorming
- Participated in Chapter 1 documentation
- Participated in Chapter 3 documentation
- Participated in Chapter 4 documentation

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| | |
|---|---|
| **Name of the Proponent:** Lenon, Robbie G. | **Signature:** |

**Project Participations:**

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- Project Manager
- Participated in Brainstorming
- Participated in Chapter 1 documentation
- Composed Chapter 2
- Composed Chapter 3
- Composed Chapter 5
- Participated in Chapter 4 documentation
- Participated in constructing Appendix A
- Participated in constructing Appendix B
- Participated in constructing Appendix D

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| **Title Project:** | **Date:** October 2025 |
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| **Name of the Proponent:** Lising, Calvin Lloyd T. | **Signature:** |

**Project Participations:**

- Project Manager
- UI Designer
- Participated in Brainstorming
- Participated in Chapter 1 documentation
- Participated in Chapter 2 documentation
- Participated in Chapter 4 documentation
- Participated in constructing Appendix B

# DOMINICAN COLLEGE OF TARLAC

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| **Title Project:** | **Date:** October 2025 |
|---|---|
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| **Name of the Proponent:** Talavera, Bryan Jay G. | **Signature:** |
|---|---|

**Project Participations:**

- Technical Writer
- Participated in Brainstorming
- Participated in Chapter 1 documentation
- Participated in Chapter 2 documentation
- Participated in Chapter 4 documentation
- Participated in constructing Appendix D

## Appendix B

### Definition of Term

**Agriculture -** A practice of growing crops and raising various animals for food and resources.

**Anthrax -** A deadly bacterial disease affecting both animals and humans.

**Armyworm -** A caterpillar pest of grass pastures and cereal crops

**Aspiring farmers -** refers to individuals who want to become farmers or are just beginning their journey in agriculture.

**Banded leaf and Stealth blight -** A serious fungal disease in rice

**Bakanae -** A fungal disease of rice caused by an insect.

**Bacterial Blight -** A common disease that affects many plants, including beans, carrots, lilac, and rice

**Bacterial leaf streak -** A bacterial disease that affects a variety of crops.

**Bacterial stalk rot -** A disease that causes the stalk of a plant to rot which leads the plant to fall over.

**Black Bug -** A sap-feeding insect that attacks rice plants.

**Blackleg -** is a clostridial disease that primarily affects young cattle raised on pasture. A clostridial disease is one caused by anaerobic bacteria in the soil.

**Blast -** A disease that affects rice plants.

**Bovine Brucellosis -** A bacterial disease in cattle, leading to reproductive issues and abortion

**Bovine Foot -** refers to the anatomical structure of a cow's hoof and lower limb, which supports movement and weight-bearing

**Bovine mastitis -** Inflammation of the udder in cows, usually caused by bacterial infection.

**Bovine Respiratory disease** (Shipping Fever) **-** A respiratory infection in cattle, often due to stress and bacterial infection.

**Bovine Tuberculosis -** A chronic bacterial disease in cattle, affecting the lungs and other organs.

**Bovine diral Diarrhea** (BVD) **-** A viral disease affecting cattle, causing diarrhea and reproductive losses.

**Brown spot -** a disease that affects plants skin condition.

**Coccidiosis -** A parasitic disease in poultry, leading to diarrhea and poor growth.

**Corn Borer -** A larvae feed upon and bore into corn.

**Corn Earworms -** A larva is that is destructive to corn, tomatoes, tobacco.

**Corn mosaic -** A plant disease which causes light and dark green mottles or stripes on corn leaves.

**Corn Rootworms -** A larvae feeding on the roots and causing damage

**Corn smut -** A fungal disease on corn plants that causes galls.

**Crops -** plants that are usually grown for food and various other uses.

**Cutworms -** A moth larvae that eat plant stems and can damage seedlings and crops.

**Disease -** a condition that harms the health of humans, plants, and animals.

**Downy mildew -** A disease of the foliage caused by a fungus like organism.

**Ear rot -** A fungal disease that affects corn.

**Early Shoot Borer -** A significant pest of sugarcane crops, particularly during the early stages.

**Egg Layers -** refer to the transparent protein membranes between the eggshell and egg white.

**External Parasite -** Mites, lice, and fleas that infest poultry, causing discomfort and disease transmission.

**Fall Armyworms -** A larvae that feeds on a wide variety of plants causing damage to crops.

**Field Corn -** A type of livestock feed.

**Field Cricket -** An insect that burrows in grassland and has a musical birdlike chirp.

**Fowl Cholera -** A bacterial disease leading to high mortality in poultry.

**Fowl Pox -** A viral disease-causing lesions on chickens' skin and mouth

**Farming -** the process of taking care of land and raising various animals for food.

**Grassy shoot -** A disease in sugarcane causing excessive tillering and stunted growth.

**Hemorrhagic Septicemia** (**HS**) **-** A bacterial infection in cattle and buffalo, causing rapid death.

**Infection -** an invasion of harmful microorganisms that causes diseases to plants, humans and animals.

**Infection Bronchitis -** A viral disease in poultry causing respiratory symptoms.

**Integration** – generally means combining two or more parts into a unified whole so they work together effectively.

**Internode Borer** - A pest that bores into plant stems, disrupting growth and reducing crop yields.

**IPB** – Institute of Plant Breeding

**Leaf scald** - A disease that affects leaves of rice leaving transparent tips.

**Leaf spot -** A leaf disease that causes discolored areas or lesions on leaves.

**Livestock** - various animals like cows, pig and chickens raised for food and other products.

**Mastitis** - Mastitis is breast inflammation that can cause infection.

**Modern -** refers to the present time.

**Native Chicken -** are indigenous, local, or traditional poultry.

**Native Cow -** An indigenous cattle breed adapted to local environments.

**Nature -** refers to a physical world full of various living organisms.

**Nematodes -** Long cylinder-shaped parasitic worms in plants, animals, or soil.

**Newcastle** disease **-** A contagious viral disease in poultry.

**NSIC** – National Seed Industry Council

**Parasitic disease -** Illness caused by parasites like worms, ticks, or protozoa.

**Pest** - organisms that damage crops and reduce yields.

**Popcorn -** a type of corn kernel that expands when heated.

**Poultry** - Birds like chickens or ducks raised for meat or eggs.

**Rats -** A small, long-tailed rodent.

**Red rot -** A fungal disease in sugarcane causing tissue decay.

**Red stripe -** A disease that usually occurs when plants reach reproductive stage.

**Red-stripped disease -** A bacterial disease in sugarcane with red streaks.

**Rice Bug** - A type of insect feeding on developing grains.

**Rice Skipper -** Larvae that feed on rice leaves.

**Rice stripe virus disease -** A disease causing striping and malformation of leaves.

**Rice Thrips -** Thrips feeding on rice plants.

**Rice Whorl Maggot -** A semi-aquatic insect that feeds on rice plants.

**Rust -** A fungal disease-causing spore masses on leaves.

**Slugs** - Soft-bodied animals similar to snails without shells.

**Smut -** A fungal disease producing black spores in plants.

**Spider Mite -** An active plant-eating mite.

**SRA** – Sugar Regulatory Administrator

**Stem rot -** A plant disease causing stem decay.

**Sustainable -** Something that can be maintained over time without harm.

**Sweet Corn -** Corn variety with high sugar kernels.

**Technique -** A method or way of doing something effectively.

**Termite -** Social insects feeding on wood and cellulose.

**Tropical Climate -** A warm, humid climate with frequent rainstorms.

**Variety -** A range of different types of something.

**VMC** – Victorias Milling Company

**Weather Condition -** the current state of the atmosphere at a specific time and place.

**Whitefly -** Sap-sucking insects that damage plants.

**Wilt -** A disease causing drooping of plant leaves.

**Worm infestation -** Internal parasites in poultry.

**Yellow leaf disease -** A viral disease in sugarcane causing leaf yellowing.

**Appendix C**

**Screen Design**

**Screen No. 1**

**Screen Name:** Launch Module

**Narrative Overview:** It will be displayed when the application is opened.

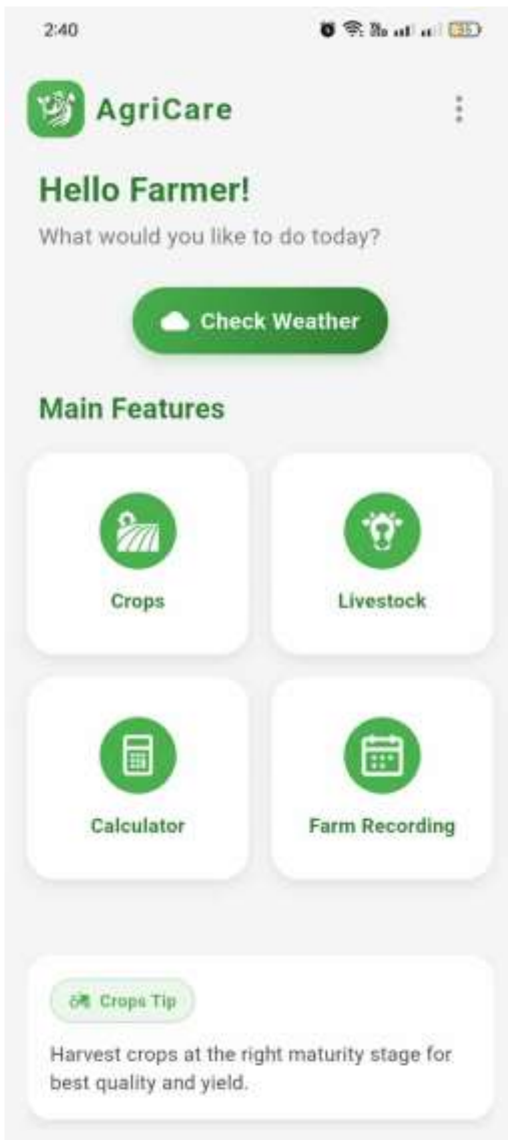
**Screen Layout:**

**Screen No. 2**

**Screen Name:** Home screen Module

**Narrative Overview:** It will display after the launch screen.

**Screen Layout:**



**Screen No. 3**

**Screen Name:** Weather forecast Module

**Narrative Overview:** It will be displayed when the user presses the weather forecast in home screen.

**Screen Layout:**

**Screen No. 4**

**Screen Name:** List of livestock Module

**Narrative Overview:** It will display when user presses the livestock below homescreen.

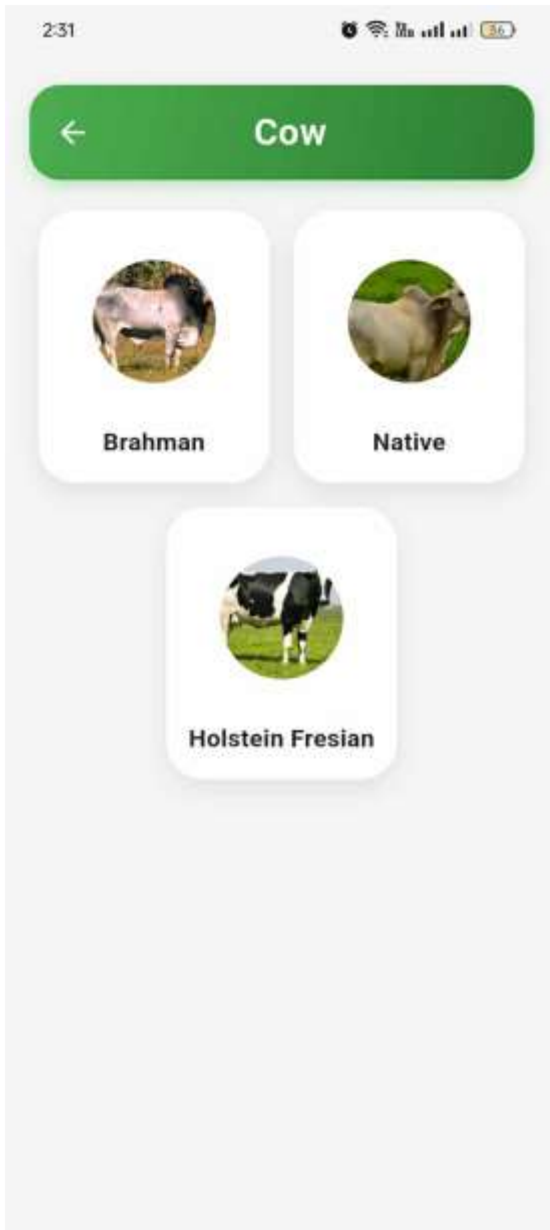
**Screen Layout:**

**Screen No. 5**

**Screen Name:** Cow Module

**Narrative Overview:** It will be displayed when the user presses the cow button in list of livestock module.

**Screen Layout:**



**Screen No. 6**

**Screen Name:** Livestock Information Module

**Narrative Overview:** It will be displayed when the user presses the specific cow breed in cow menu module.

**Screen Layout:**

**Screen No. 7**

**Screen Name:** List of Crops Module

**Narrative Overview:** It will display when user presses the Crops below home screen.

**Screen Layout:**

**Screen No. 8**

**Screen Name:** rice Module

**Narrative Overview:** It will be displayed when the user presses the rice button in list of rice module.

**Screen Layout:**

**Screen No. 9**

**Screen Name:** Field corn information Module

**Narrative Overview:** It will be displayed when the user presses the field corn button in list of Crops module.

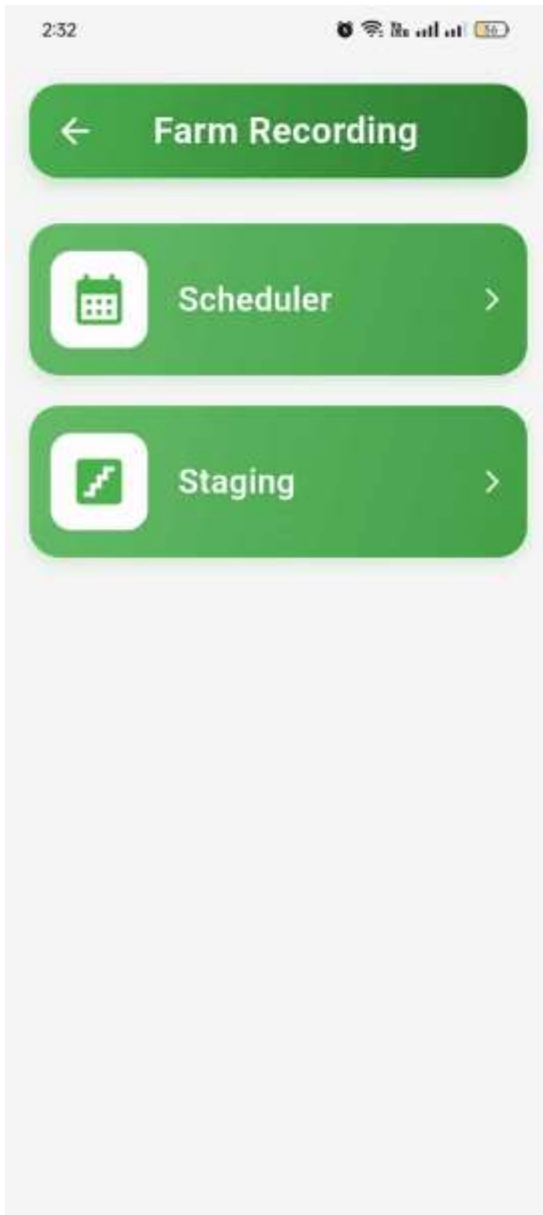
**Screen Layout:**

**Screen No. 10**

**Screen Name:** Farm Recording Module

**Narrative Overview:** It will be displayed when the user presses the farm recording in the home screen.

**Screen Layout:**

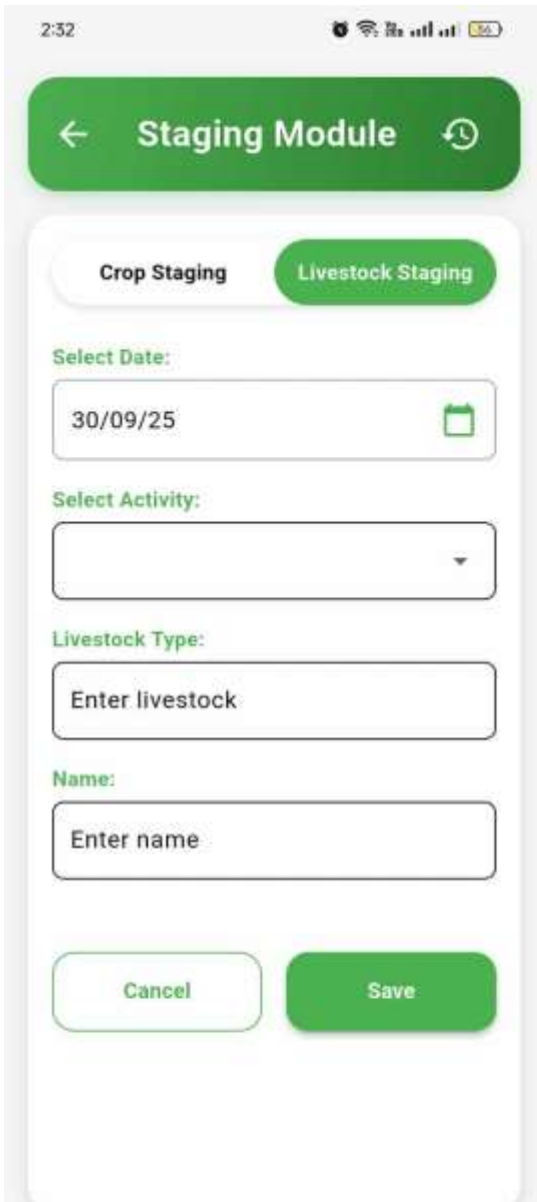


**Screen No. 11**

**Screen Name:**Staging Module

**Narrative Overview:** It will be displayed when the user presses the staging in the Farm recording module.

**Screen Layout:**

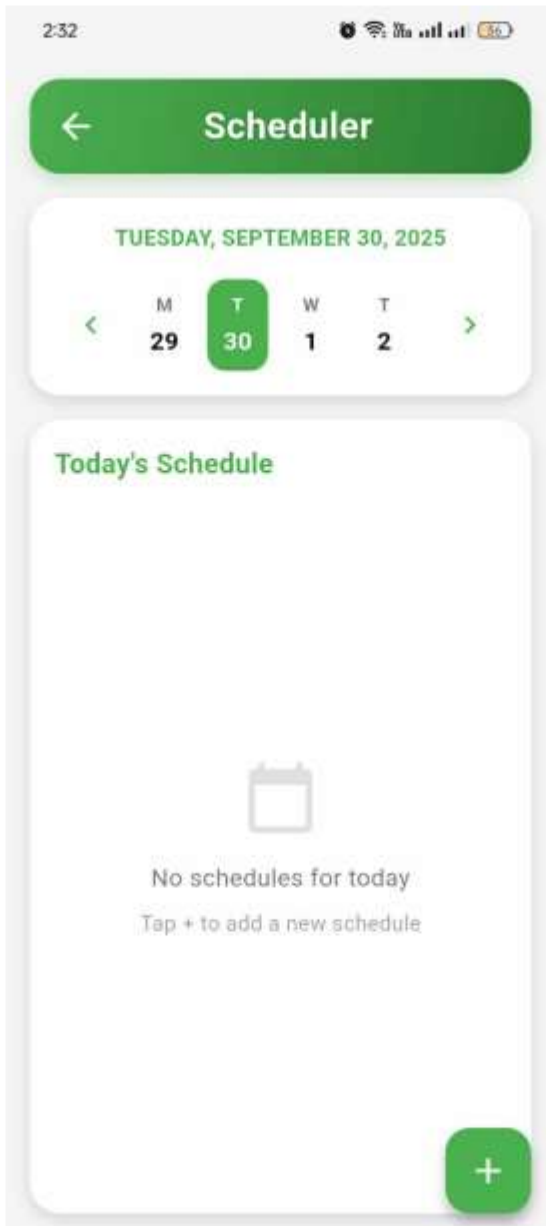


**Screen No. 12**

**Screen Name:** Scheduler Module

**Narrative Overview:** It will be displayed when the user presses the schedulerin the Farm recording module.

**Screen Layout:**



**Appendix D**

**Program Listing**

```
import 'package:http/http.dart' as http;
import 'dart:convert';
class WeatherService {
static const String apiKey = '78bc35ff7c814e039ba150846251908'; // Your API key
static const String baseUrl = 'https://api.weatherapi.com/v1';
// Fetch weather by location coordinates with forecast
static Future<Map<String, dynamic>> fetchWeatherByLocation(double latitude, double
longitude) async {
try {
finalresponse = await http.get(
Uri.parse('$baseUrl/forecast.json?key=$apiKey&q=$latitude,$longitude&days=7&aqi
=no&alerts=no'),
);
if(response.statusCode == 200) {
return json.decode(response.body);
} else {
throw Exception('Failed to load weather data: ${response.statusCode}');
}
} catch (e) {
throw Exception('Failed to load weather data: $e');
}
}
// Fetch weather by city name with forecast
static Future<Map<String, dynamic>> fetchWeatherByCity(String city) async {
try {
finalresponse = await http.get(
Uri.parse('$baseUrl/forecast.json?key=$apiKey&q=$city&days=7&aqi=no&alerts=n
o'),
);
if(response.statusCode == 200) {
return json.decode(response.body);
} else {
```

```
throw Exception('Failed to load weather data: ${response.statusCode}');

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}
} catch (e) {
throw Exception('Failed to load weather data: $e');
}
}
}
void _processForecastData(Map<String, dynamic> data) {
// Process hourly forecast
if (data['forecast'] != null &&
data['forecast']['forecastday'] != null &&
data['forecast']['forecastday'].isNotEmpty) {
// Get today's hourly forecast
final todayForecast = data['forecast']['forecastday'][0];
if (todayForecast['hour'] != null) {
hourlyForecast = List<Map<String, dynamic>>.from(todayForecast['hour']);
// Filter to show only future hours (or current time if needed)
final now = DateTime.now();
hourlyForecast = hourlyForecast.where((hour) {
final hourTime = DateTime.parse(hour['time']);
return hourTime.isAfter(now.subtract(const Duration(hours: 1)));
}).toList();
// Limit to next 12 hours or so
hourlyForecast = hourlyForecast.take(12).toList();
}
// Process daily forecast (next 7 days)
if (data['forecast']['forecastday'] != null) {
dailyForecast = List<Map<String, dynamic>>.from(data['forecast']['forecastday']);
}
}
}
```

```
void _startLocationUpdates() {
positionStreamSubscription = LocationService.getLocationStream().listen(
(Position position) {
if (position.accuracy < (locationAccuracy ?? 100)) {
setState(() {
currentPosition = position;
locationAccuracy = position.accuracy;
});
}
},
onError: (error) {
```



```
print('Location stream error: $error');
},
);
}
Future<void> _fetchWeatherByCity(String city) async {
setState(() {
isLoading = true;
errorMessage = '';
accurateLocationName = '';
hourlyForecast = [];
dailyForecast = [];
});
try {
final data = await WeatherService.fetchWeatherByCity(city);
setState(() {
weatherData = data;
_processForecastData(data);
isLoading = false;
selectedCity = city;
});
```

```
// Trigger immediate notifications for current conditions
_triggerImmediateWeatherNotifications(data);
// Also schedule future notifications
_scheduleWeatherNotifications(data);
} catch (e) {
setState(() {
errorMessage = 'Failed to get weather for $city: $e';
isLoading = false;
});
}
}
String _getWeatherAdvice(Map<String, dynamic> data) {
final condition = data['current']['condition']['text'].toString().toLowerCase();
final temp = data['current']['temp_c']?.toDouble() ?? 0.0;
final humidity = data['current']['humidity']?.toInt() ?? 0;
final wind = data['current']['wind_kph']?.toDouble() ?? 0.0;
// 1. Rain check
if (condition.contains('rain') || condition.contains('drizzle') ||
condition.contains('shower')) {
return 'Avoid pesticide application during rain — it will wash off and be ineffective.';
}
// 2. Strong winds
```



```
if (wind > 20) {
return 'Winds are strong today. Avoid spraying pesticides to prevent drift and
wastage.';
}
// 3. Extreme temperature
if (temp >= 32) {
return 'It is very hot today. Apply pesticides only in the early morning or late
afternoon to avoid evaporation and crop stress.';
}
```

```
if (temp <= 15) {
return 'It is quite cold. Pesticides may act slower than usual.';
}
// 4. Humidity-based advice
if (humidity >= 85) {
return 'High humidity increases the risk of fungal diseases. Monitor crops and apply
fungicides if needed.';
}
if (humidity <= 40) {
return 'Low humidity may reduce pesticide effectiveness. Irrigate crops first before
spraying.';
}
// 5. Default good conditions
return 'Weather conditions are favorable for pesticide application today.';
}
Widget _buildContent() {
return SingleChildScrollView(
padding: const EdgeInsets.all(16.0),
child: Column(
children: [
// Unified weather information container
Container(
padding: const EdgeInsets.all(20),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(16),
boxShadow: [
BoxShadow(
color: Colors.grey.withOpacity(0.2),
blurRadius: 8,
offset: const Offset(0, 4),
),
],
),
```

```
child: Column(
children: [
```



```
// Location and date
Text(
selectedCity.isNotEmpty
? selectedCity
: accurateLocationName.isNotEmpty
? accurateLocationName
: weatherData!['location']['name'],
style: TextStyle(
fontSize: 24,
fontWeight: FontWeight.bold,
color: darkGreen,
),
textAlign: TextAlign.center,
),
const SizedBox(height: 8),
Text(
DateFormat('EEEE | MMMM d').format(DateTime.now()),
style: TextStyle(
fontSize: 16,
color: Colors.grey[600],
),
),
const SizedBox(height: 16),
// Temperature and condition
Row(
mainAxisAlignment: MainAxisAlignment.center,
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Text(
```

```
'${weatherData!['current']['temp_c'].round()}°',
style: TextStyle(
fontSize: 64,
fontWeight: FontWeight.w300,
color: darkGreen,
),
),
],
),
const SizedBox(height: 8),
Row(
mainAxisAlignment: MainAxisAlignment.center,
children: [
Image.network(
'https:${weatherData!['current']['condition']['icon']}',
width: 32,
height: 32,
errorBuilder: (context, error, stackTrace) =>
```



```
Icon(_getWeatherIcon(weatherData!['current']['condition']['text']),
color: darkGreen, size: 32),
),
const SizedBox(width: 8),
Text(
weatherData!['current']['condition']['text'],
style: TextStyle(
fontSize: 18,
color: Colors.grey[700],
),
),
],
),
const SizedBox(height: 16),
```

```
// Weather details
Row(
mainAxisAlignment: MainAxisAlignment.spaceAround,
children: [
Column(
children: [
Text(
'${weatherData!['current']['wind_kph'].toString()} km/h',
style: TextStyle(
fontSize: 16,
fontWeight: FontWeight.bold,
color: darkGreen,
),
),
const SizedBox(height: 4),
Text(
'Wind',
style: TextStyle(
fontSize: 14,
color: Colors.grey[600],
),
),
],
),
Column(
children: [
Text(
'${weatherData!['current']['pressure_mb'].toString()} mbar',
style: TextStyle(
fontSize: 16,
fontWeight: FontWeight.bold,
color: darkGreen,
),
```

```
),
const SizedBox(height: 4),
Text(
'Pressure',
style: TextStyle(
fontSize: 14,
color: Colors.grey[600],
),
),
],
),
Column(
children: [
Text(
'${weatherData!['current']['humidity'].toString()}%',
style: TextStyle(
fontSize: 16,
fontWeight: FontWeight.bold,
color: darkGreen,
),
),
const SizedBox(height: 4),
Text(
'Humidity',
style: TextStyle(
fontSize: 14,
color: Colors.grey[600],
),
),
],
),
],
```

```
),
],
),
),
const SizedBox(height: 16),
// Hourly forecast header
Padding(
padding: const EdgeInsets.symmetric(horizontal: 8.0),
child: Row(
mainAxisAlignment: MainAxisAlignment.spaceBetween,
children: [
Text(
'Today',
style: TextStyle(
```



```
fontSize: 18,
fontWeight: FontWeight.bold,
color: darkGreen,
),
),
GestureDetector(
onTap: _show7DayForecast,
child: Text(
'7-Day Forecast',
style: TextStyle(
fontSize: 14,
color: primaryGreen,
fontWeight: FontWeight.bold,
),
),
),
],
```

```
),
),
const SizedBox(height: 8),
// Hourly forecast with real data
SizedBox(
height: 130,
child: hourlyForecast.isEmpty
? Center(
child: Text(
'Loading hourly forecast...',
style: TextStyle(color: Colors.grey[600]),
),
)
: ListView(
scrollDirection: Axis.horizontal,
children: hourlyForecast.map((hour) {
final time = DateTime.parse(hour['time']);
final formattedTime = DateFormat('HH:mm').format(time);
final isNow = DateTime.now().hour == time.hour;
return _buildHourlyForecast(
isNow ? 'Now' : formattedTime,
hour['condition']['text'],
'${hour['temp_c']?.round()}°',
_getRainChance(hour),
hour['condition']['icon'],
);
}).toList(),
),
```



```
),
const SizedBox(height: 16),
// Advice section
```

```
Container(
padding: const EdgeInsets.all(16),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(16),
boxShadow: [
BoxShadow(
color: Colors.grey.withOpacity(0.2),
blurRadius: 8,
offset: const Offset(0, 4),
),
],
),
child: Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
Text(
'Agricultural Advice',
style: TextStyle(
fontSize: 18,
fontWeight: FontWeight.bold,
color: darkGreen,
),
),
const SizedBox(height: 8),
Text(
_getWeatherAdvice(weatherData!),
style: TextStyle(
fontSize: 16,
color: Colors.grey[600],
),
),
],
),
```

```
),
const SizedBox(height: 20),
],
) );
}
@override
Widget build(BuildContext context) {
```



```
return Scaffold(
backgroundColor: backgroundGray,
body: SafeArea(
child: Column(
children: [
// Platform-specific header
isIOS ? _buildIOSHeader() : _buildAndroidHeader(),
Expanded(
child: isLoading
? _buildLoading()
: errorMessage.isNotEmpty
? _buildError()
: weatherData != null
? _buildContent()
: Center(
child: Text(
'No weather data available',
style: TextStyle(color: Colors.grey[600]),
),
),
),
],
),
),
```

```
floatingActionButton: FloatingActionButton(
onPressed: _fetchWeatherByLocation,
backgroundColor: primaryGreen,
child: const Icon(Icons.refresh, color: Colors.white),
),
);
}
Widget _buildHourlyForecast(String time, String condition, String temp, String
rainChance, String iconUrl) {
return Container(
width: 90,
margin: const EdgeInsets.symmetric(horizontal: 8),
padding: const EdgeInsets.all(12),
decoration: BoxDecoration(
color: Colors.white,
borderRadius: BorderRadius.circular(16),
boxShadow: [
BoxShadow(
color: Colors.grey.withOpacity(0.2),
blurRadius: 4,
offset: const Offset(0, 2),
),
```

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```
],
),
child: Column(
mainAxisAlignment: MainAxisAlignment.spaceEvenly,
children: [
Text(
time,
style: TextStyle(
fontSize: 14,
```

```
fontWeight: FontWeight.w500,
color: Colors.grey[700],
),
textAlign: TextAlign.center,
),
Image.network(
'https:$iconUrl',
width: 24,
height: 24,
errorBuilder: (context, error, stackTrace) =>
Icon(_getWeatherIcon(condition), color: primaryGreen, size: 24),
),
Text(
temp,
style: TextStyle(
fontSize: 14,
fontWeight: FontWeight.bold,
color: darkGreen,
),
textAlign: TextAlign.center,
),
Text(
'$rainChance%',
style: TextStyle(
fontSize: 12,
color: int.parse(rainChance) > 30 ? Colors.blue : Colors.grey[600],
fontWeight: int.parse(rainChance) > 30 ? FontWeight.bold : FontWeight.normal,
```

**Appendix E**

**Curriculum Vitae**

# BALUYUT, CARL PATRICK M.

UI DESIGNER

## Contact

09639824031

carlpatrickmanucduc@gmail.com

Carl Patrick Manucduc Baluyut

## Information

Sta. Rosa, Conception, Tarlac

22 Years Old

February, 11, 2003

## About Me

With a strong commitment to academic excellence, I remain focused and composed during demanding situations. I am driven by a passion for lifelong learning and continuously advancing my expertise in Information Technology.

## Educational

2022-2026

**Dominican College of Tarlac**

# FAJARDO, ARJAY G.

UI DESIGNER

## Contact

09517524010

Fajardiarjay@gmail.com

Arjay Fajardo

## Information

0781 Dulong St Lawy, Capas , Tarlac

21 Years Old

Dec 22, 2003

## About Me

Constantly seeking growth, I stay motivated in the face of challenges and embrace opportunities to enhance my IT expertise. My academic dedication and adaptability help me thrive in dynamic environments.

## Educational

2022-2026

**Dominican College of Tarlac**

# DAVID, NATASHA C.

PROGRAMMER

## Contact

09615441116

natashacruz184@gmail.com

Natasha David

## Information

Blk 14 Lot 13 Brgy. Banaba, Dapdap, Bamban, Tarlac

20 Years Old

July 31, 2004

## About Me

I approach academic and technical challenges with a problem-solving mindset, staying composed in high-pressure situations. My commitment to continuous learning fuels my growth in the field of Information Technology.

## Educational

2022-2026

**Dominican College of Tarlac**

# LENON, ROBBIE G.

CO-PROGRAMMER

## Contact


09614603219

lenonrobbie18@gmail.com

Robbie Lenon


## Information

Blk 49, Lot 58 Cristo Rey, Capas, Tarlac

21 Years Old

August 19, 2003

## About Me

I am dedicated in my academic obligations and stay motivated and calm at difficult times. I am also committed on learning to continuously learn and enhance my skills in I.T field

## Educational

2022-2026

**Dominican College of Tarlac**

# LISING CALVIN LLOYD T.

PROJECT MANAGER

## Contact

09151265948

lisingcalvinlloyd@gmail.com

Calvin Lloyd

## Information

Old Anupul, Bambam, Tarlac

20 Years Old

May 27, 2004

## About Me

With a deep passion for technology, I embrace challenges with a calm and determined mindset. I am dedicated to learning, growing, and staying ahead in the ever-evolving IT landscape.

## Educational

2022-2026

**Dominican College of Tarlac**

# BRYAN JAY, GUTIERRES TALAVERA

UI DESIGNER

## Contact

09612145476

bryanjaytalavera22@gmail.com

Bryan Talavera

## Information

321 Brgy Manlapig, Capas, Tarlac

22 Years Old

June, 22, 2002

## About Me

Dedicated to academic success and continuous learning in IT, I thrive under pressure and stay focused on enhancing my skills for future challenges.

## Educational

2022-2026

**Dominican College of Tarlac**

**Appendix F**

## Transcript of Interviews

**ALVIN SICAT**

**Farmer**

1. Anong pangalan po ninyo?

- Alvin Sicat

2. Kelan po kayo nag simula na mag farm?

- Nag simula ako nung 2015

3. Gaano katagal naman po kayo nag fafarm?

- Simula nung 2015 di na ako huminto mag farm, ngayon eto nag tatanim padin ng palay, eto din kase hanap buhay naming e.

-

4. Saang lugar po yung lupa nyo o yung farm nyo?

- Dito sa Manlapig lang din

5 . Pang Negosyo po ba yung sa farm nyo?

- Oo, pang source of food din naming

6. Ano ano po yung mga klase ng pananim ang mga tinatanim po ninyo?

- Palay, Mais

7. Sa palay po ba ano ano yung mga tinatanim nyo ang pagkaka alam din po kasi naming madaming klase ng palay?

- kami kasi ang mga tinatanim naming triple 222, second crop, 160, 534, sa 160 tsaka 534 kapag sinabi na yan alam na agad nila yan

8. sa mais naman po ano yung mga tinatanim?

- madalas talaga tinatanim jan yung mga sweetcorn kami sweetcorn lang tinatanim naming e pero sa mga ibang pang malakihan na farm sa mais tintanim nila mga pagkain sa mga alaga nilang hayop okaya ginagawa nilang feeds.

9. sa palay po ano yung mga fertilizer na ginagamit po Ninyo?

- unian 46-0-0 tsaka 14-14-14

**<u>DOMINADOR MALLARI</u>**

**Farmer**

1. Ano po ang pangalan nyo?
- Dominador Mallari
2. Ilang Years na po kayo nag tatanim ng tubo?
- 1231
3. Anong variety po ng sugarcane ang tinatanim nyo po?
- Alumni tsaka Badila
4. Sa isang hektarya ilang binhi po yung nagagamit nyo?
- Hindi ko alam e, ang ginagamit kase jan yung mismong tubo, puputulin yun e parang sa kamoting kahoy lang, yun lang din yung itatanim.
5. Sa isang tubo po ilan yung hati ginagawa po ninyo?
- Mga sampo ganon
6. Ginagamitan nyo pa po ba ng fertilizer?
- Oo naman
7. Anong fertilizer po ang ginagamit nyo sa tubo?
- Ano, urea tsaka 14-14-14
8. Gumagamit din po ba kayo ng spray para sa mga insekto?
- Oo
9. Ano pong spray o gamot yung ginagamit po ninyo?
- Ang ginagamit lang naming jan yung spray sa damo e
10. Sa mais naman po ilang years napo kayo nag tatanim?
- Ganon din katagal kagaya sa pagtatanim ko ng tubo
11. Anong Variety po ba yung mga tinatanim nyo?
- Marami ang mais e, may pang feeds, may pang kain.
12. Pang feeds tsaka kinakain na mais din po ba tinatanim Ninyo?
- Oo, pero mas madalas yung sa white corn tsaka sweet corn.
13. Sa pag tatanim po Ninyo ng mais ano po yung fertilizer ginagamit nyo?

- Ganon din sa sugarcane, sikat talaga yung urea tsaka 14-14.

**CARLOS B. TUAZON**

**Farmer**

Questions

- RICE

1. What are the lists of rice in the Philippines?
2. In all these lists, what are the common varieties of rice?
3. These common varieties are also used in 3 major island such as Luzon, Visayas, Mindanao?
4. What are the List of varieties of rice being used in Luzon, Visayas and Mindanao?
5. What are the Fertilizer for these varieties?
6. What are the pests/insects in those told varieties?
7. What is the land preparation for those told varieties?
8. What are the Diseases in those told varieties?
9. What are the pest controls or spray being used in those told varieties?
10. How do they manage the diseases in those told Diseases?

- Corn

1. What are the lists of corn in the Philippines?
2. In all these lists, what are the common varieties of corn?
3. These common varieties are also used in 3 major island such as Luzon, Visayas, Mindanao?
4. What are the List of varieties of corn being used in Luzon, Visayas and Mindanao?
5. What are the Fertilizer for these varieties?
6. What are the pests/insects in those told varieties?
7. What is the land preparation for those told varieties?
8. What are the Diseases in those told varieties?
9. What are the pest controls or spray being used in those told varieties?
10. How do they manage the diseases in those told Diseases?

- Sugarcane

1. What are the lists of sugarcane in the Philippines?
2. In all these lists, what are the common varieties of sugarcane?
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8. What are the Diseases in those told varieties?
9. What are the pest controls or spray being used in those told varieties?
10. How do they manage the diseases in those told Diseases?

- COW

1. What are the breeds of cows being raise in the Philippines?
2. What are the common breeds being raised in the Philippines?
3. Are these common breeds being raised in all 3 major island such as Luzon, Visayas, and Mindanao?
4. What are the breeds being raised in Luzon?
5. What are the breeds being raised in Visayas?
6. What are the breeds being raised in Mindanao?
7. In those told breeds what are the Diseases/Infections of it?
8. What are the cure methods and treatment for the Diseases/Infections of those told breeds?
9. What are the shelters of those told breeds?
10. What kind of food of those told breeds?
11. What are their Vaccines?
12. What are their Vitamins?

- PIG

1. What are the breeds of pigs being raise in the Philippines?
2. What are the common breeds being raised in the Philippines?
3. Are these common breeds being raised in all 3 major island such as Luzon, Visayas, and Mindanao?
4. What are the breeds being raised in Luzon?

5. What are the breeds being raised in Visayas?
6. What are the breeds being raised in Mindanao?
7. In those told breeds what are the Diseases/Infections of it?
8. What are the cure methods and treatment for the Diseases/Infections of those told breeds?
9. What are the shelters of those told breeds?
10. What kind of food of those told breeds?
11. What are their Vaccines?
12. What are their Vitamins?

- Chicken

1. What are the breeds of chicken being raise in the Philippines?
2. What are the common breeds being raised in the Philippines?
3. Are these common breeds being raised in all 3 major island such as Luzon, Visayas, and Mindanao?
4. What are the breeds being raised in Luzon?
5. What are the breeds being raised in Visayas?
6. What are the breeds being raised in Mindanao?
7. In those told breeds what are the Diseases/Infections of it?
8. What are the cure methods and treatment for the Diseases/Infections of those told breeds?
9. What are the shelters of those told breeds?
10. What kind of food of those told breeds?
11. What are their Vaccines?
12. What are their Vitamins?

Additional Questions

- How do they call a livestock raiser, is it also farmers?
- What are the traditional and modern techniques of farming? Can you provide a study of it?
- Do you have a definition of rural and urban farming?

**Proponents**: Good Morning po, mag iinterview po sana kami about sa farming, student po kami sa Dominican College of Tarlac at eto yung endorsement letter po namin. Need din po kasi naming ng conformation from professionals about sa information sa mga crops at livestock.

**Carlos B. Tuazon** – (sees the questions, name of crops and livestock its information) I would suggest that you use Philrice website, nandun lahat ng mga names ng mga rice, kumpleto na rin don pati pests.

**Carlos B. Tuazon -** Sa pests meron tayong bagong kalaban "Rice black bug at stem borer".

**Carlos B. Tuazon –** Sa land preparation naman dalawa yung way ng pag prepare ng lupa, una yung plowing, yung kalawa is yung harrowing.

**Carlos B. Tuazon –** Sa fertilizer naman need ng soil analysis bago gagamit ng ibat't ibang fertilizers. Sa information at mga pangalan may mga designated website naman about sa mga yun. Ang maganda kuha kayo ng information tapos balik kayo dito ichecheck ko.

**Proponents –** Sige po, bali iwan po naming yung isang copy ng endorsement letter sainyo then papirma nalang din po.

**Crops Information**

**Rice Varieties**

**PSB Rc18**

Pest/Insects

Stem Borer

Green Leafhopper

Brown Planthopper

Diseases

Blast

Bacterial Leaf Blight

Tungro

**PSB Rc 82**

Pest/Insects

Stem Borer

Green Leafhopper

Brown Planthopper

Diseases

Blast

Bacterial Leaf Blight

Tungro

**NSIC Rc 308**

Pest/Insects

Stem Borer

Green LeafHopper

**NSIC Rc 354**

Pest/Insects

Stem Borer

Green LeafHopper

Brown Planthopper

Diseases

Blast

Bacterial Leaf Blight

Sheat Blight

Tungro

**NSIC Rc 214**

Pest/Insects

Green Leafhopper

Brown Planthopper

Diseases

Blast

Bacterial Leaf Blight

Tungro

**NSIC Rc 298**

Pest/Insects

Green Leafhopper

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 352**

Pest/Insects

Stem Borer

Green Leafhopper

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 358**

Pest/Insects

Stem Borer

Green Leafhopper

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 400**

Pest/Insects

Green Leafhopper

Brown Planthopper

Diseases

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 402**

Pest/Insects

Green Leafhopper

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**PSB Rc10**

- Pest/Insects
  - Stem Borer
  - Green Leafhopper
  - Brown Planthopper
- Diseases
  - Blast
  - Sheath Blight
  - Bacterial Leaf Blight
  - Tungro

**NSIC Rc 218**

- Pest/Insects
  - Stem Borer
  - Green Leafhopper
- Diseases
  - Sheath Blight
  - Bacterial Leaf Blight
  - Tungro

**PSB Rc82**

- Pest/Insects
  - Stem Borer
  - Green Leafhopper
  - Brown Planthopper
- Diseases
  - Blast
  - Bacterial Leaf Blight
  - Tungro

**NSIC Rc 394**

- Pest/Insects
  - Brown Planthopper
  - Green Leafhopper
- Diseases
  - Sheath Blight
  - Bacterial Leaf Blight

Tungro

**NSIC Rc 302**

Pest/Insects

Brown Planthopper

Green Leafhopper

Diseases

Sheath Blight

Bacterial Leaf Blight

Tungro

Blast

**NSIC Rc 226**

Pest/Insects

Brown Planthopper

Green Leafhopper

Diseases

Blast

Bacterial Leaf Blight

Tungro

**NSIC Rc 360**

Pest/Insects

Stem Borer

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 356**

Pest/Insects

Stem Borer

Green Leafhopper

Brown Planthopper

Diseases

Blast

Sheath Blight

Bacterial Leaf Blight

Tungro

**NSIC Rc 224**

Pest/Insects

Green Leafhopper

Brown Planthopper

Diseases

Blast

Bacterial Leaf Blight

Tungro

**NSIC Rc 158**

**Pest/Insects**

Stem Borer

Green Leafhopper

Brown Planthopper

**Diseases**

Blast

Bacterial Leaf Blight

Tungro

**Land Preparation**

Dry Preparation

Dry preparation is typically practiced for upland rice, but can also be done for lowland fields.

Step 1:

- Construct no wider and taller than 50 cm x 30 cm bunds, around the field.
- Make sure that bunds are well compacted and properly sealed, with no cracks, holes, etc. This will minimize water losses through seepage (particularly in sloping lands).

- Adjust the spillway height to 3−5 cm for storing the same depth of water. Maintain this height to ensure sufficient water storage capacity especially during rainy or wet season

Step 2:

Plow the field.

Step 3:

Harrow and rototill the field.

Step 4:

A level field allows planters or drills to place seed more precisely and enables more uniform irrigation. A level field reduces weed pressure and helps control water.

Step 5:

Allow weeds to emerge for at least 2 weeks, then kill them with a non-selective herbicide, such as glyphosate. Plant your crop within 1−2 days, then spray a pre-emergence herbicide

**Wet Preparation**

Wet preparation is the most common way of preparing lowland fields. In this method, the soil is tilled in a saturated or flooded condition.

Step 1:

- Construct no wider and taller than 50 cm x 30 cm bunds, around the field.
- Make sure that bunds are well compacted and properly sealed, with no cracks, holes, etc. This will minimize water losses through seepage (particularly in sloping lands).
- Adjust the spillway height to 3−5 cm for storing the same depth of water. Maintain this height to ensure sufficient water storage capacity especially during rainy or wet season

Step 2:

Irrigate the field with 2−3 cm of water for about 3−7 days or until it is soft enough and suitable for an equipment to be used.

Step 3:

Plow the field.

Step 4:

Harrow the field 2−3 times within 5−7 days interval.

Step 5:

Allow weeds to emerge for at least 2 weeks, then kill them with a non-selective herbicide, such as glyphosate. Plant your crop within 1−2 days, then spray a pre-emergence herbicide

Step 6:

Leveling should be done two (2) days before planting.

**Pest/Insects**

**Stem borer**

Symptoms

Longitudinal white patches on leaf sheaths; central leaf whorl drying out and turning brown; tillers drying out without producing panicles; panicles may dry out or may produce no grain; adult insects are nocturnal moths which lay their eggs on the leaves or leaf sheaths of the rice plants; larvae are legless grubs which feed on leaf sheaths before entering the stem of the pant

Pest Control

Stem borers are difficult to control with insecticides as once they bore inside the stem they are protected from chemical sprays; in order for chemical control to be successful, repeated applications of appropriate insecticide must be made to the foliage; granular formulations give better control than sprays; clipping seedling prior to transplanting can successfully reduce moth numbers as eggs are laid at leaf tips; harvesting plants at ground level can remove the majority of larvae from the field; plowing or flooding the remaining stubble will kill off most of the remainder of the larvae in the field

**Brown planthopper**

Symptoms

Planthoppers cause rice leaves to turn yellow, then brown and dry (hopperburn), which can kill the plant. Infested plants may show eggs on leaf sheaths, honeydew, sooty molds, and virus symptoms like

Rice Ragged Stunt and Grassy Stunt.

Pest Control

Avoid overusing nitrogen and insecticides. Use resistant varieties, remove weeds, and monitor fields regularly. Control methods include flooding seedbeds, using light traps, and preserving natural enemies like spiders and parasitoids. Use insecticides only if planthoppers outnumber their predators and flooding isn't possible.

**Green leafhopper**

Symptoms

Plants may show no symptoms of leafhopper of planthopper damage; feeding punctures can leave the plants susceptible to bacterial or fungal infections; insects transmit many rice viruses; if infestations is severe, insects may cause plant to completely dry out; adults insects are pale green or brown winged insects with piercing-sucking mouthparts

Pest Control

Crop rotation for a year helps reduce hopper numbers. Preserve natural enemies by avoiding unnecessary insecticides. Use resistant varieties for effective control. Apply insecticides only when hopper levels reach the economic threshold.

**Diseases**

**Blast**

Symptoms

Lesions on all parts of shoot; white to green or gray diamond-shaped lesions with dark green borders; death of leaf blades; black necrotic patches on culm; rotting panicles

Management

Blast can be controlled by planting resistant rice varieties; avoid over-fertilizing crop with nitrogen as this increases the plant's susceptibility to the disease; utilize good water management to ensure plants do not suffer from drought stress; disease can be effectively controlled by the application of appropriate systemic fungicides

**Bacterial Leaf Blight**

Symptoms

Water-soaked stripes on leaf blades; yellow or white stripes on leaf blades; leaves appear grayish in color; plants wilting and rolling up; leaves turning yellow; stunted plants; plant death; youngest leaf on plant turning yellow.

Management

Bacterial blight can be effectively controlled by planting resistant rice varieties; avoid excessive nitrogen fertilization; plow stubble and straw into soil after harvest

**Tungro**

Symptoms

Plants are stunted with a yellow-orange discoloration; plants may have a reduced number of tillers and rust colored spots on leaves; leaves may be mottled, striped or exhibit interveinal necrosis

Management

Preventive measures are more effective for the control of tungro than direct disease control measures. Using insecticides to control leafhoppers is often not effective, because green leafhoppers continuously move to surrounding fields and spread tungro rapidly in very short feeding times.

**Sheath Blight**

Symptoms

Sheath blight symptoms appear from tillering to milk stage, starting as greenish-gray oval lesions on the lower leaf sheaths. These lesions spread upward and to nearby plants, turning gray-white with brown margins as they age.

Management

No resistant rice variety available for cultivation.

Corn Varieties

**IPB Var6**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Diseases

Downy mildew

**IPB Var8**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Diseases

downy mildew

**LB Lagkitan**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Aphids

Rootworms

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**IES Lagkitan**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**Native Corn**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Corn Earworm

White Grub

Seedling Maggot

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**IPB Var10**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Corn Earworm

White Grub

Seedling Maggot

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**IPB Var11**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Corn Earworm

White Grub

Seedling Maggot

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**IPB Var13**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Corn Earworm

White Grub

Seedling Maggot

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**Tiniguib**

Pest/Insects

Armyworms

Black Bug

Locusts

Rodents

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**Putian**

Pest/Insects

Fall Armyworms

Locusts

Rodents

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**IPB Var12**

Pest/Insects

Asian Corn Borer (ACB)

Fall armyworm

Corn Earworm

White Grub

Seedling Maggot

Diseases

Downy Mildew

Corn Smut

Leaf Blight

Stalk Rot

**Land Preparation**

Proper land preparation plays a crucial role in plant growth by promoting good root development and improving weed, pest, and disease management

**Plowing and Harrowing**

Plow at a depth of 15-20 cm when soil moisture is right. That is, when soil particles 15 cm below the surface separate and only thin portion sticks to the finger but no ball is formed. Harrow twice with 2-3 passing to break the clods.

If a disc plow is used, plow under corn stubbles at a depth of 18-20 cm. Use a disc plow to enable you to use corn stubbles as additional source of fertilizer. Clayey and weedy fields require two or more plowing and several harrowing.

**Furrowing**

Make furrows a day before or on the day of planting spaced at 75 cm and 8 cm deep.

**Pest/Insects**

**Asian Corn Borer (ACB)**

**Description**

ACB is the most destructive corn borer because it feeds on the core of the corn plant. When it occurs during flowering, it can cause breakage of the flower tassel and even the corn stalk.

**Symptoms**

Holes in the new leaf buds are one of the signs of ACB damage.

**Pest Control**

Removing the tassels from 75% of the plants, release of Trichogramma plasseyensis, an endemic species.

**Fall Armyworm**

**Description**

Fall Armyworm or "FAW" is a destructive caterpillar that feeds and causes holes in the leaves, flowers and fruits of corn, which can cause a 30-60% reduction in yield..

**Symptoms**

Feeding damage includes small circular to irregular holes and skeletonized leaves. Fruits may have shallow, dry wounds. Egg clusters (50–150 eggs) covered in whitish, cottony scales can be found on leaves. Young larvae are pale green to yellow, while older ones are darker green with side stripes and a pink or yellow underside.

**Pest Control**

Organic methods of controlling the armyworm include biological control by natural enemies which parasitize the larvae and the application of Bacillus thuringiensis..

**Aphids**

**Description**

Aphids are small sap-sucking insects and members of the superfamily Aphidoidea...

**Symptoms**

Heavy infestations can result in curled leaves and stunted plants; honeydew secretions promote growth of sooty mold; corn leaf aphids are blue-green in color, peach aphids are green-yellow in color; aphids may transmit viruses when feeding.

**Pest Control**

It is rare for aphids to reach levels that are damaging to the plant and no control is generally warranted as insecticide sprays will not prevent transmission of viruses.

**Corn Rootworms**

**Description**

The Corn Rootworm is one of the most damaging pests to corn, especially in the larval stage.

**Symptoms**

Larvae feed with chewing mouthparts on roots causing browning, lesions, tunneling, curved cornstalks (goosenecking), yield losses, and susceptibility to root and stalk diseases.

**Pest Control**

Crop rotation has been a major management strategy for corn rootworm.

**Corn Earworm**

**Description**

The corn earworm can cause serious economic damage to fresh market and processing sweet corn and hybrid dent seed corn..

**Symptoms**

Feeding damage to leaves, tassel and leaf whorls; preferred feeding site is the ear and insect produces extensive excrement at the tip of the ear; younger larvae feed on silks, severing them from the plant; young caterpillars are cream-white in color with a black head and black hairs; older larvae may be yellow-green to almost black in color with fine white lines along their body and black spots at the base of hairs; eggs are laid singly on both upper and lower leaf surfaces and are initially creamy white but develop a brown-red ring after 24 hours and darken prior to hatching.

**Pest Control**

Corn earworms are most problematic on sweet corn varieties and treatment should be applied at egg hatch; monitor plants for eggs and young larvae and also natural enemies that could be damaged by chemicals; Bacillus thuringiensis or Entrust SC may be applied to control insects on organically grown plants; appropriate chemical treatment may be required for control in commercial plantations.

**White Grub**

**Description**

White grubs are occasional pests of corn seedlings.

**Symptoms**

Damage typically appears as stunted, wilted, discolored, or dead seedlings and/or as gaps in rows where plants fail to emerge. White grubs prune roots and can feed on the mesocotyl causing plant death.

**Pest Control**

White grubs can be killed by use of certain kinds of insect parasitic nematodes that are commercially available

**Seedling Maggot**

**Description**

The Seedling Maggot is a common pest affecting germinating seeds and young seedlings, especially in cool, wet conditions.

**Symptoms**

Poor or uneven germination, hollowed-out seeds, wilting or stunted seedlings, and rotting roots or stems near the soil surface.

**Pest Control**

Planting in warm, dry soil for faster germination, using treated seeds or in-furrow insecticides, rotating crops, and avoiding fresh organic matter in the soil before planting.

**Locusts**

**Description**

Locusts are large, migratory grasshoppers that can cause significant damage to corn, especially during outbreaks.

**Symptoms**

Ragged holes in leaves, stripped foliage, damaged tassels and silks, and in severe cases, entire plants eaten down to the stalk.

**Pest Control**

Monitoring for early signs of infestation, using insecticides when populations are high, encouraging natural predators (like birds), and removing weeds or grasses near fields that serve as breeding grounds..

**Rodents**

**Description**

Rodents, including rats and mice, can cause significant damage to corn by feeding on seeds, stems, and ears. They may also cause indirect damage by creating tunnels in the soil.

**Symptoms**

Chewed or gnawed corn stalks, missing seeds, damaged ears, and visible burrows around the field.

**Pest Control**

Use traps or rodenticides, eliminate rodent nesting sites, and maintain clean field edges. Implementing rodent-proofing measures like fencing and using natural predators (e.g., owls or snakes) can also help manage populations.

## Diseases

### Downy Mildew

#### Description

Downy mildew is a fungal disease caused by Peronosclerospora sorghi, which infects the leaves and stems of corn. It is more prevalent in humid, warm conditions and can affect both young and mature plants.

#### Symptoms

Yellowing or chlorosis of leaves, often beginning at the leaf tips or margins. White to grayish, fuzzy growth on the undersides of leaves. In severe cases, leaves may die off.

#### Management

Plant resistant varieties, improve field drainage to avoid waterlogging, and apply fungicides if the disease is detected early.

### Corn Smut

#### Description

Corn smut is caused by the fungus Ustilago maydis, affecting the kernels, ears, and other plant parts. While it can be considered a delicacy in some regions, it significantly reduces yield and quality when it infects commercial corn.

#### Symptoms

Swelling and transformation of corn kernels into grayish, tumor-like growths. Infected ears may have swollen kernels that are white or grayish and can be soft and spongy.

#### Management

Remove and destroy infected plants. Use fungicide treatments if necessary and avoid planting corn in areas previously affected by smut.

### Leaf Blight

#### Description

Leaf blight is caused by several pathogens, including Helminthosporium and Cochliobolus species, which lead to the formation of lesions on leaves. These fungi thrive in warm, wet conditions and can affect both young seedlings and mature plants.

**Symptoms**

Brown or black lesions on leaves with yellow halos. Infected areas may coalesce, leading to large sections of the leaf turning necrotic. Severe cases can cause premature leaf death.

**Management**

Use disease-resistant varieties, practice crop rotation, and avoid excessive irrigation. Fungicide application may help if applied at the onset of symptoms.

Stalk Rot

**Description**

Stalk rot is caused by various fungal pathogens, such as Fusarium, Aspergillus, and Pythium. It typically affects corn during or after the flowering period and is worsened by stress factors like drought or poor soil conditions.

**Symptoms**

Discoloration of the corn stalk, often starting at the base and moving upwards. Affected plants may show wilting or lodging (falling over) due to weakened stems. The stalk may appear soft or rotted when cut open, with a distinct foul odor.

**Management**

Improve soil fertility, reduce plant stress (especially during pollination), and ensure proper irrigation practices. Remove and destroy infected plants. Crop rotation with non-corn crops helps reduce the buildup of pathogens in the soil.

**Fertilizers**

Ammonium

Urea 46-0-0

14-14-14

Sugarcane Information

Sugarcane Varieties

**Phil 99-1793**

Diseases

Downy Mildew

Smut
Yellow Spot
Leaf Scorch

**Phil 2000-2569**

Diseases

Downy Mildew
Smut
Yellow Spot
Leaf Scorch

**Phil 94-0913**

Diseases

Downy Mildew
Smut
Yellow Spot
Leaf Scorch

**Phil 92-1043**

Diseases

Downy Mildew
Smut
Yellow Spot
Leaf Scorch

**PHIL 56-226**

Diseases

Downy Mildew
Smut
Rust
Leaf Scorch

**VMC 70-150**

Diseases

Downy Mildew
Smut
Yellow Spot
Leaf Scorch

**VMC 76-505**

Diseases

Downy Mildew

Smut

Rust

Leaf Scorch

**Phil 63-17**

Diseases

Downy Mildew

Smut

Yellow Spot

Leaf Scorch

Rust

**PSR 97-41**

Diseases

Smut

Downy Mildew

Rust

**VMC 95-09**

Diseases

Downy Mildew

Smut

Yellow Spot

Leaf Scorch

Rust

Diseases

**Leaf scald**

Leaf scald is a bacterial disease of sugarcanes caused by Xanthomonas albilineans.

Symptoms

The disease manifests with white pencil lines on leaves and to which later turn red and brown eventually causes the leaves to scald and die the spread primarily through infected cuttings and harvesting tools, leaf scald is usually highly transmissible.

Management

Regularly disinfect the cutting and harvesting tools with bactericides to prevent the spread of the disease

**Downy Mildew**

Downy mildew is caused by the oomycete Peronosclerospora sacchari.

Symptoms

It is characterized by a white or yellowish leaf streaks, white powdery growth, and can lead to deep yellow or reddish-brown mottled streaks as it progresses. This disease is spread though airborne spores specifically during warm moist nights.

Management

Downy mildew can be managed with the use of resistant varieties and if necessary chemical control these strategies include using a disease free seed cane and isolating sugarcane from other healthy crops like corn then removing and destroying infected plants.

**Smut**

Sugarcane smut is a fungal disease of sugarcane caused by the fungus Sporisorium scitamineum.

Symptoms

Stunted growth of sugarcane stools; profuse production of tillers; shortened internodes; stems thin with narrow, erect leaves; black whip-like structure emerging from terminal bud.

Management

The disease can be successfully controlled by planting varieties of sugarcane which are resistant to the disease; disease can usually be eliminated from seed pieces by hot water treatment prior to planting; infected plants should be removed

**Yellow Spot**

Yellow spot is a fungal disease of sugarcane caused by the fungus Mycovellosiella koepkei.

Symptoms

Small, yellowish spots appear on the leaves, which later turn brown in the center with a yellow halo; in severe infections, the leaves may dry out and die prematurely, leading to reduced plant vigor.

Management

Use resistant varieties whenever possible; apply recommended fungicides if infection levels are high; maintain good field sanitation and avoid overcrowding to improve air circulation.

**Rust**

Rust is a fungal disease of sugarcane caused by Puccinia melanocephala (for common rust) or Puccinia kuehnii (for orange rust).

Symptoms

Small, elongated, reddish-brown pustules appear mainly on the undersides of leaves; affected leaves may turn yellow, dry up, and die prematurely, reducing photosynthesis and yield.

Management

Plant rust-resistant varieties; remove and destroy infected leaves; use fungicides if necessary; maintain proper field hygiene to minimize disease spread.

Land Preparation

**Ploughing**

The common method of tillage preparation is ploughing the land and bringing the soil to fine tilth.

Plough the field for 2 to 4 times at the depth of 50-60 cm with tractor drawn disc plough or victory plough

**Harrowing**

It is the secondary tillage operation in sugarcane cultivation which pulverizes, smoothens and compact the soil to conserve the moisture.

Harrowing is done at shallow depth of 12-15 cm to crush the clods by disc harrow or rotavator

**Levelling**

To ensure a uniform crop stand levelling is important also for easy movement of irrigation water.

Levelling can be carried out using a tractor operated leveller.

**Lay out of field**

Irrigation – cum – drainage channels along and across the slope of the field at 10-15m intervals.

Fertilizers

Urea

Super phosphate

Muriate of potash

Cow Information

Breeds

Philippine Native

Brahman

Holstein Friesian

Shelter

Proper housing is important in successful cattle fattening operation. Adequately protect animals against the adverse effects of weather when they are raised in relatively small areas. Animals in backyard cattle farms are usually tethered along roadsides and in backyards during the day and confined in a shed or corral at night. The permanent type of housing consisting of GI roofing, timber frames, concrete floor, feed trough and water troughs are used in most farms. The shelter is open-sided and is located near the farmer's house or under the shade trees. Building height ranges from 1.7 to 1.9 meters while the width varies from 2.1 to 2.7 meters. Each animal can be allocated with 1.5 to 4.5 sq. meters.

Food

Grass

Silage

Rumsol cattle feed

Diseases

**Mastitis**

Mastitis is an inflammation of the udder that reduces milk production and can affect food hygiene.

Symptoms

Mastitis can be caused by bacteria, viruses, or fungi. Symptoms include swelling, pain, and redness of the udder.

Cure

Intramammary antibiotics should be the first-line treatment for cows with mild uncomplicated mastitis in a single quarter

**Bovine spongiform encephalopathy**

Bovine spongiform encephalopathy is a neurodegenerative disease caused, according to the majority opinion, by abnormal proteins called prions that accumulate in the brain and spinal cord of animals.

**Symptoms**

A cow with BSE may have trouble walking or standing up. They may seem nervous or aggressive.

**Cure**:

There is no treatment available.

**Foot-and-mouth disease**

Foot-and-mouth disease (Aphthous fever) is a highly contagious viral disease that can affect cloven-hoofed animals, including cattle.

**Symptoms**

The disease can cause fever, blisters in the mouth, hooves, and weight loss. Prevention includes the use of vaccines and control of livestock movement. Efforts in early detection and rapid response are fundamental to containing any outbreak.

**Cure**

There is no cure but treatment is available, BIO-CEVIT or BIO-ADE+B.COMPLEX for effective treatment of secondary infection using BIO-TYLOSIN-PH or BIO-D.O.C. antibiotic

**Bovine pneumonia**

Bovine pneumonia is a common respiratory disease in cattle that can be caused by bacteria and viruses.

**Symptoms**

Symptoms include fever, coughing, and difficulty breathing. Prevention includes vaccination and management of sick animals to minimize their spread.

**Cure**

Vaccination Bovilis INtranasal RSP Live

**Anaplasmosis**

Anaplasmosis is a disease caused by anaplasma marginale transmitted by ticks that can cause anemia and coagulation problems in cattle.

**Symptoms**

Infected Cattle become anemic, weak, lethargic, go off feed, and run a fever.

**Cure**

Etracycline antibiotics (tetracycline, chlortetracycline, oxytetracyline)

**Bovine viral diarrhea**

Bovine viral diarrhea is a viral disease that affects calves and can cause diarrhea, dehydration, and weight loss.

**Symptoms**

It is usually seen as sudden onset depression, fever and anorexia, with excess salivation.

**Cure**

Treatment of BVD is limited primarily to supportive therapy.

**Vitamin**

Vitamin A

Vitamin E

Vitamin K

Vitamin D

B Vitamins

Pig Information

Breeds

Large White

Landrace

Duroc

Hampshire

Berkshire

Pietrain

Shelter

In Generally, boars should be four to six months old at the time of selection. ft whatever systems of operation, hog houses must be constructed properly to ensure maximum performance of the pigs. A good hog house may not improve the health conditions of the animals but a poor one will certainly increase disease problem easily.

For a small or backyard operations, cheap and locally available materials may be used such as bamboo and nipa.

Hog houses should be constructed on a slightly sloping and well- drained area so that it will not become too muddy and convenient to work in. t

Permanent hog houses should have concrete floors for easy cleaning and to minimize the occurrence of parasites and diseases. Concrete floors must not be too rough to cause foot and leg problems nor too smooth to be slippery when wet.

Feed Ingredients

Yellow Corn

Rice Bran

Copra Meal

Fish Meal

Soybean oil meal

Skimmed Milk

Ipil-Ipil Leaf Meal

Brown Sugar

Vitamins-Mineral

Vaccine

Autogenous vaccines

Autogenous vaccines are bacterial vaccines that are manufactured from the specific pathogenic bacteria isolated from the diseased pig.

Diseases/Infections

Swine Fever

Symptoms

Fever, loss of appetite

Increased thirst, chills and sometimes vomiting.

Constipation, later followed by diarrhea

Inflammation of the eye (conjunctivitis) thick discharges causing eyelids to stick together

Reddish, purple discoloration of skin at ears, abdomen, inner thighs or tail

Cure

Vaccinate all pigs against the disease using a reliable vaccine, weanling at one week before or after weaning-; sows and boars, every six months. Dispose all pigs known to have the disease. Disinfect contaminated pens and premises properly.

Swine Dysentery

Symptoms

Loss of appetite

Fever

Rough coat and weakness

Cure

Antibiotics in feed for two weeks when disease is prevalent

Quarantine new arrivals for a week and feed high level antibiotics

Brucellosis

Symptoms

An infectious disease that causes abortions, infertility, and low milk production in infected pigs.

Cure

There is no satisfactory or treatment.

Gastroenteritis Complex

Symptoms

Diarrhea starts 16 to 30 hours after the pig is exposed to virus.

Cure

There is not a specific treatment for transmissible gastroenteritis

MMA

Symptoms

Tenderness and warmth in mammary issue.

Cure

Antibiotic infuxion into udder;apply hot compress and mild antiseptic externally.

Roundworm Infection

Symptoms

Depends largely on the number of worms present in animals, kind of management and nutrition of pigs. Pigs manifest slow growth rate, thinness, thick growth of hair which is usually dull and lacking normal luster. Sometimes, pigs vomit worms or expel worms in the feces.

Cure

Oral administration of dewormer through feed or drinking water. Piperazine dewormers are usually effective.

Mange

Symptoms

Intense itchiness, forcing animal to rub vigorously affected portion of the body against wall of pen. At first, affected skin is reddened but, after sometime, skin becomes thickened, scaly, and wrinkled.

Cure

Spray animal with insecticidal preparations indicated for mange. Repeated spraying is necessary to attain satisfaction results. Like- wise, spray animal's quarters particularly floors and walls to kill mites hiding in cracks and crevices.

Chicken Information

Type of Native Chicken

Banaba

Bolinao

Darag

Joloano

Paraoakan

Type of Egg Layers

Lohmann Layers

Dekalb White Layers

Babcock White Layers

Shelter for Native Chicken

Native chickens should be provided with ample housing structures where they can roost during the night, find shelter during rainy weather, and build nests when they are of laying age.

Provide adequate range type housing for growers and breeders with 1-2 square meters per bird. Put birds of uniform age in a house to prevent fighting and disease outbreak. For enhanced breeder egg production, place the birds in separate cages with nests and range. They should also be fed with laying rations mixed with local feed materials such as rice bran, rice hull, copra, and corn.

Shelter for Egg Layers

A layer poultry house is a structure used for growing of layers. Proper planning of housing facilities for a flock of layer chickens requires knowledge of environmental requirements during various stages of growth. Major factors that should be considered are temperature, humidity and light. To control the aforementioned factors, the interior environment may be modified by use of ventilating and heating equipment. Structure may either be fully enclosed or may have curtains over part of a sidewall. A common method is to use mechanical ventilating system during cold weather and curtain ventilation during warm weather conditions. The aim is to provide conditions that ensure optimum performance of the birds.

Feed Ingredients

Ground Corn

Rice Hull

Rice Bran

Copra Meal

Rice Grits

Corn Bran

Vaccines

Nobilis AE + Pox

UNIVAX PLUS

CEVAC EDS K

Vitamins

Pampered Chicken Mama Vital Nutrients

Multivitamins and Calcium Lactate Tablets

Diseases/Infections

Respiratory Disease

Symptoms

Respiratory signs can include rales (fine crackles), snicking (sneezing), watery eyes, fluid or mucus from nostrils, swollen head, gaping (open-mouth breathing), gasping, or head shaking.

Cure

Confine sick birds for 1 week and put medication in the water. Deprive the birds of water for 2-3 hours before allowing them to drink.

Fowl Pox

Symptoms

Decrease in egg production, poor growth or weight gain and a loss of appetite.

Cure

Put tincture of iodine on the wounds every day. Provide the birds with medicated water. Administer vaccine at 2 months of age to prevent Fowl Pox.

Newcastle Disease

Symptoms

Chicken appears lethargic with ruffled feathers.

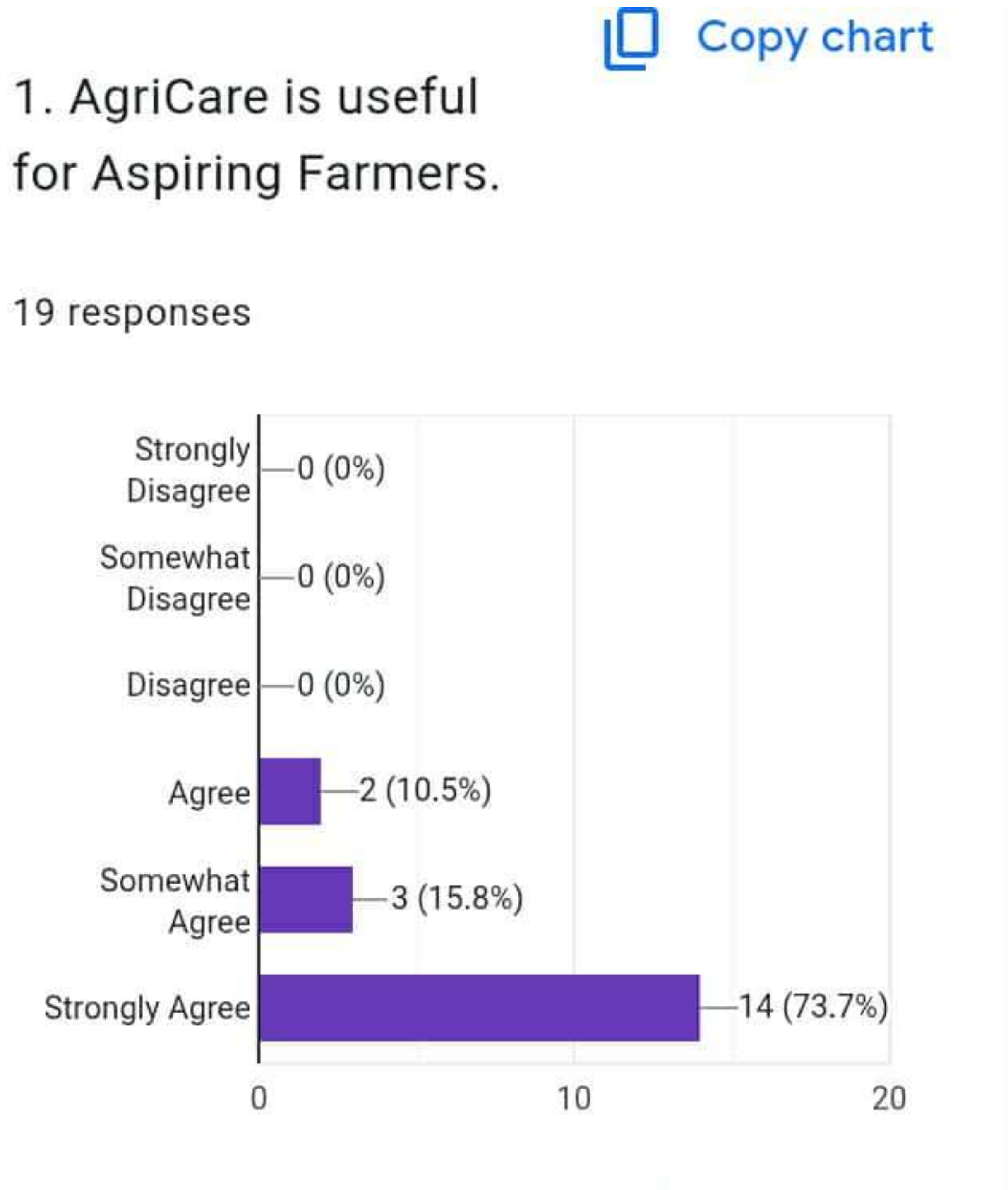
Cure

There is no treatment for Newcastle Disease, although treatment with antibiotics to control secondary infections may assist.

Carlos B. Tuazon

Signature

**Survey Forms/Questionnaires**



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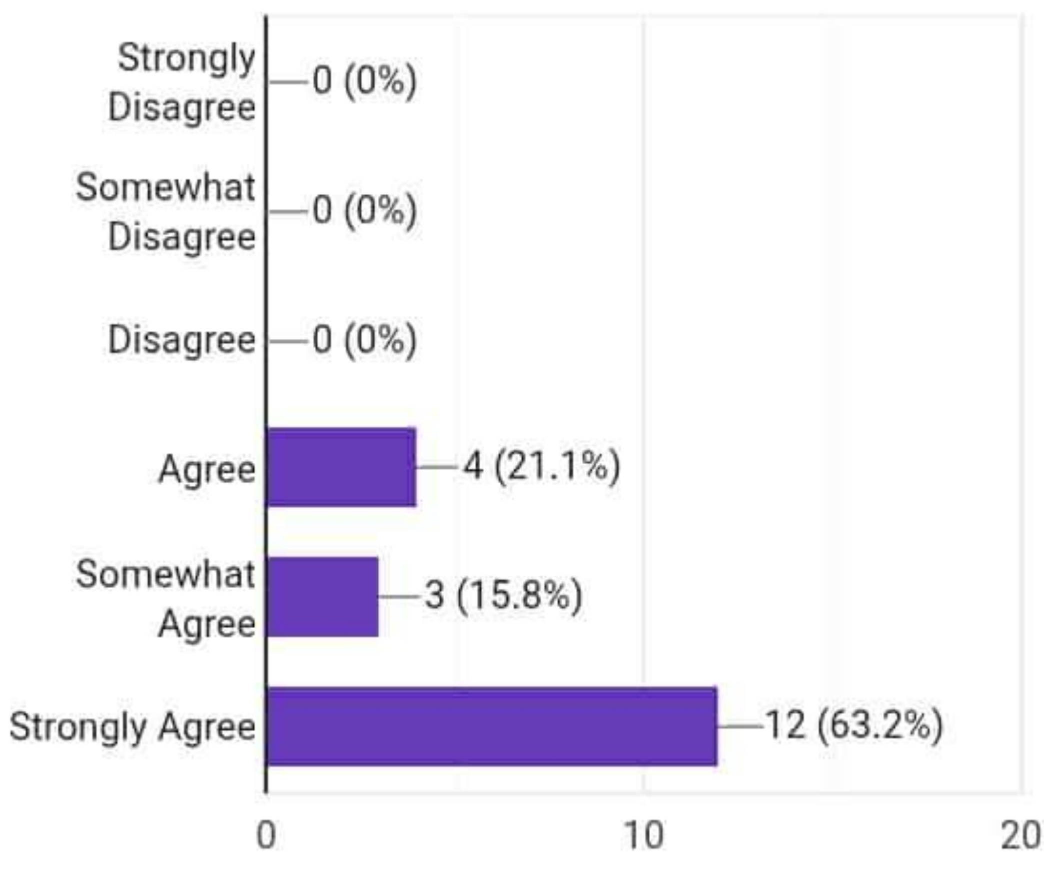
2. AgriCare provides usefulness for individuals with limited English language proficiency.

19 responses



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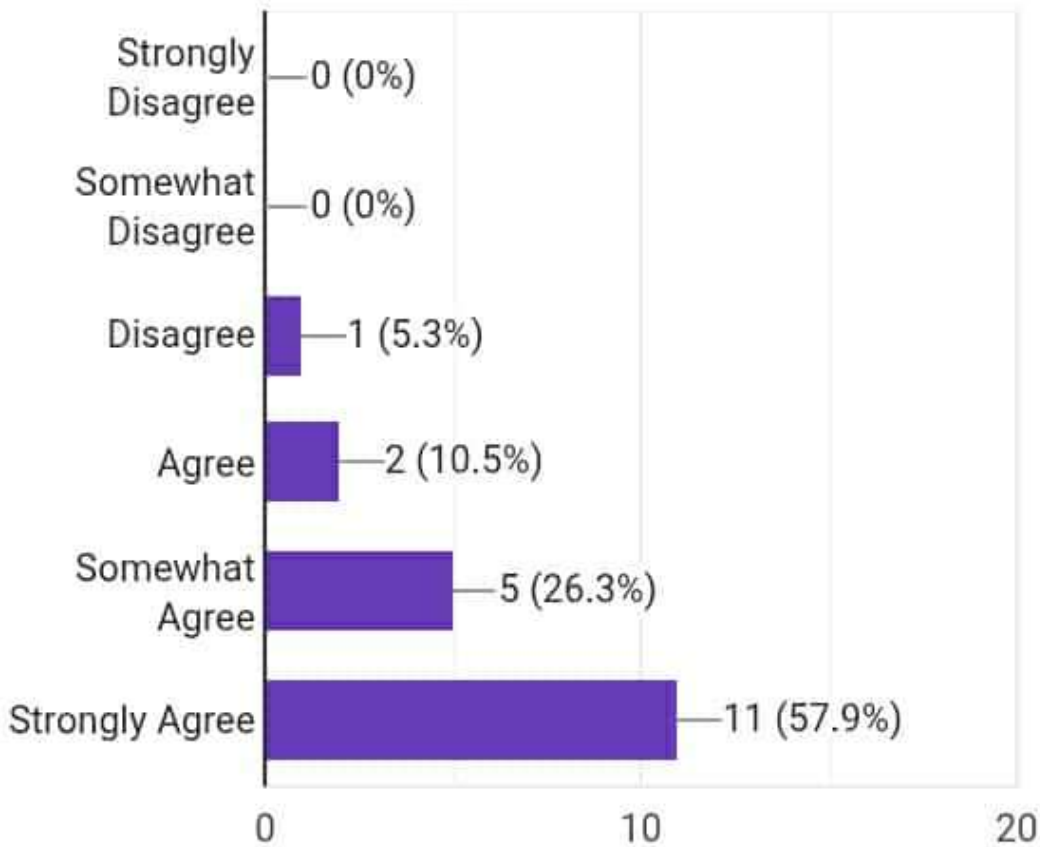
3. The number of buttons/links is reasonable and not overwhelming.

19 responses



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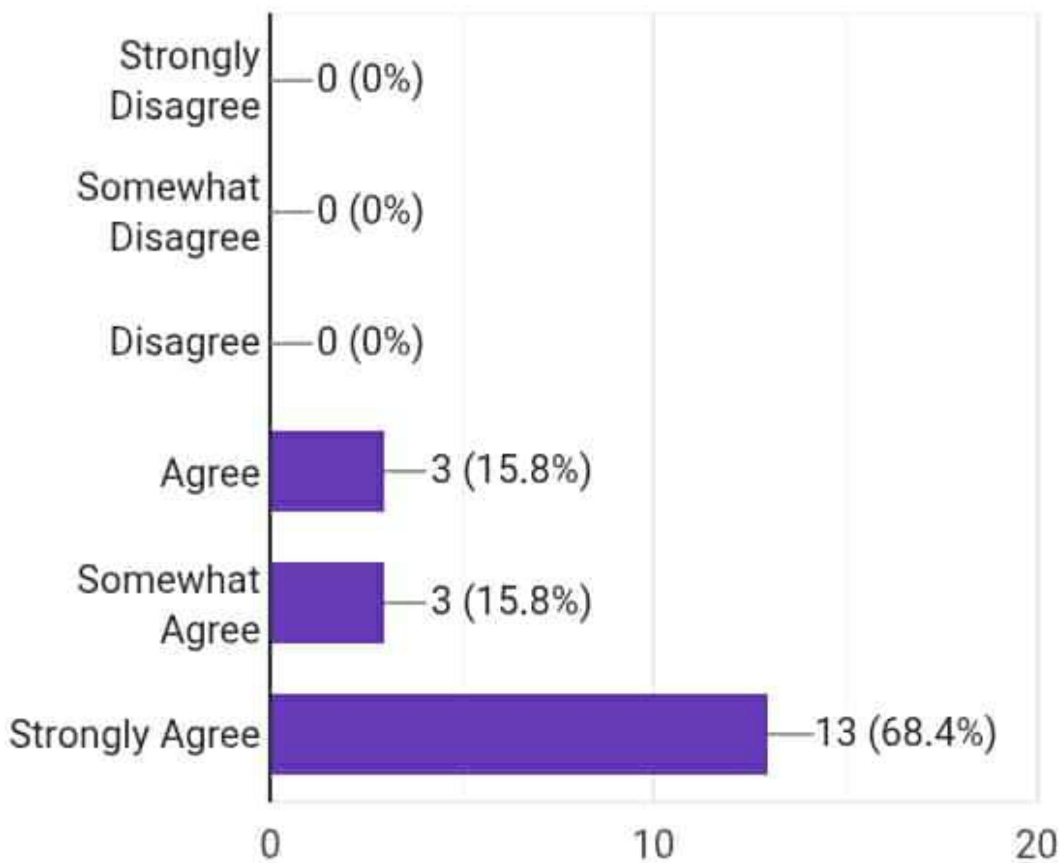
2. The navigation is clear and concise.

19 responses



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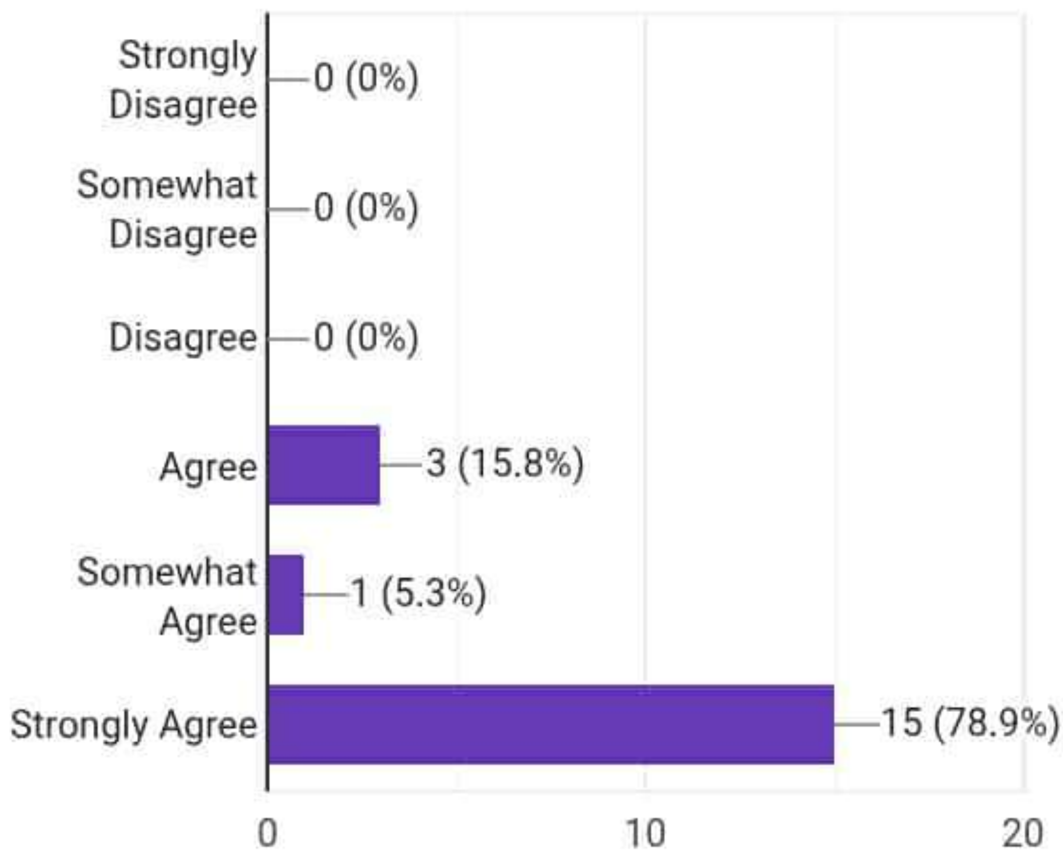
1. The navigation in AgriCare is easy to use.

19 responses



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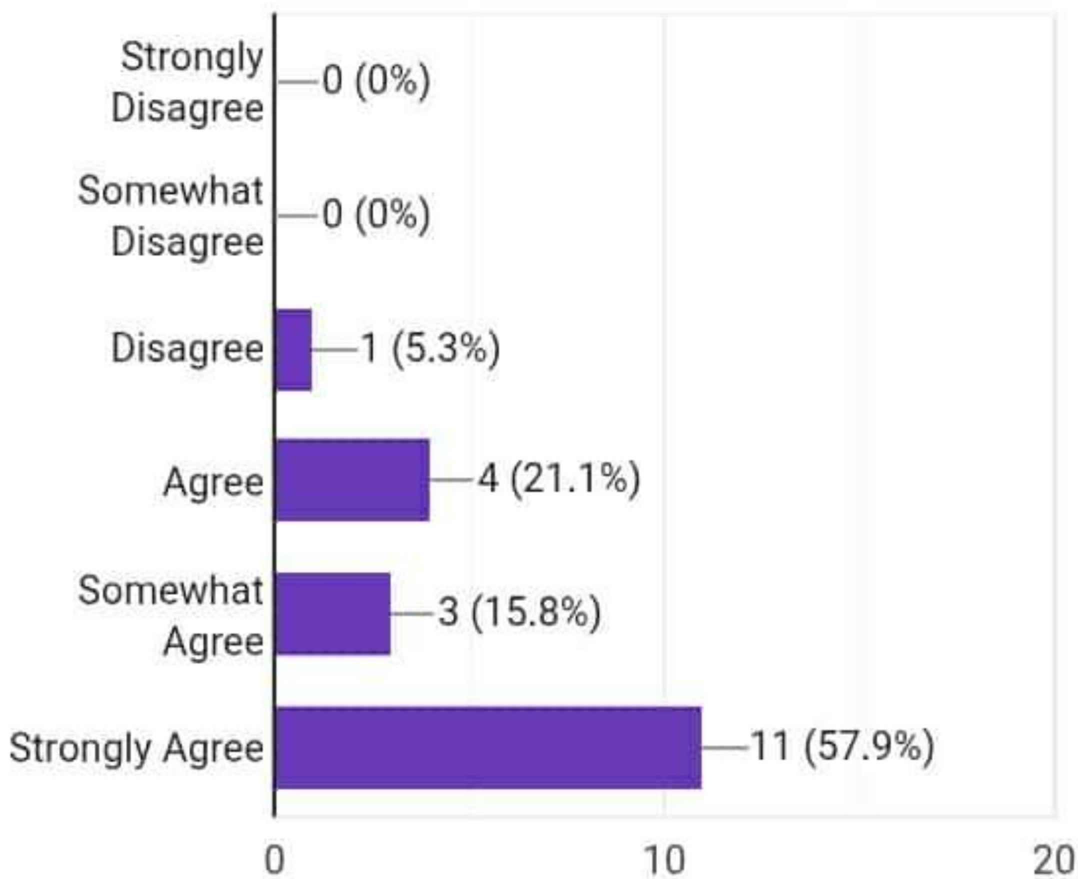
5. The **color scheme** is appropriate and visually pleasing.

19 responses



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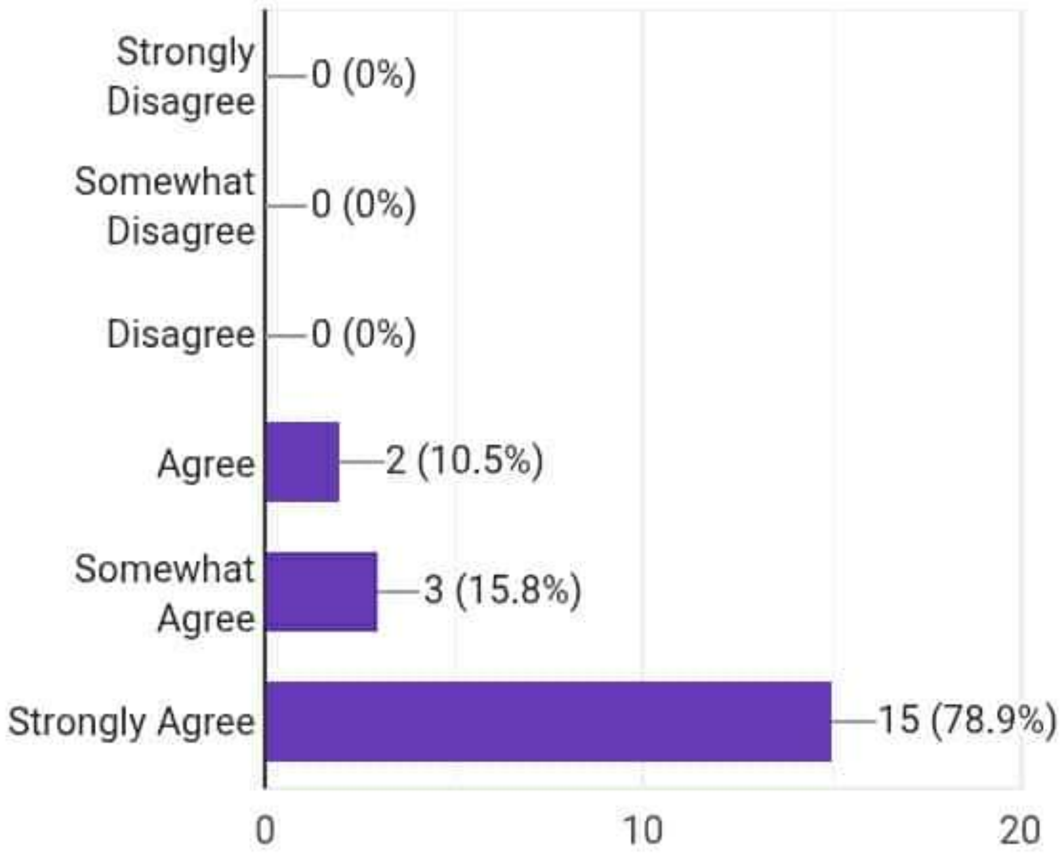
4. The **buttons** are appropriate and function as intended.

19 responses



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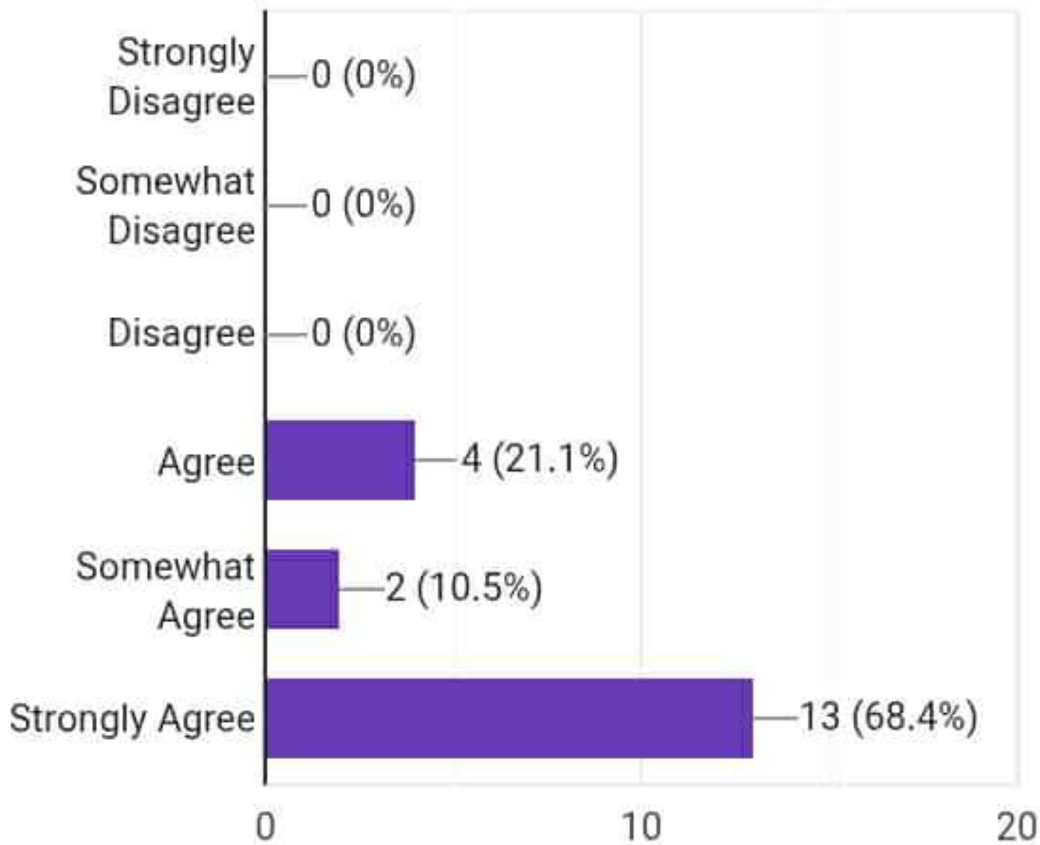
3. The **graphics** used in the app are appropriate and visually clear.

19 responses



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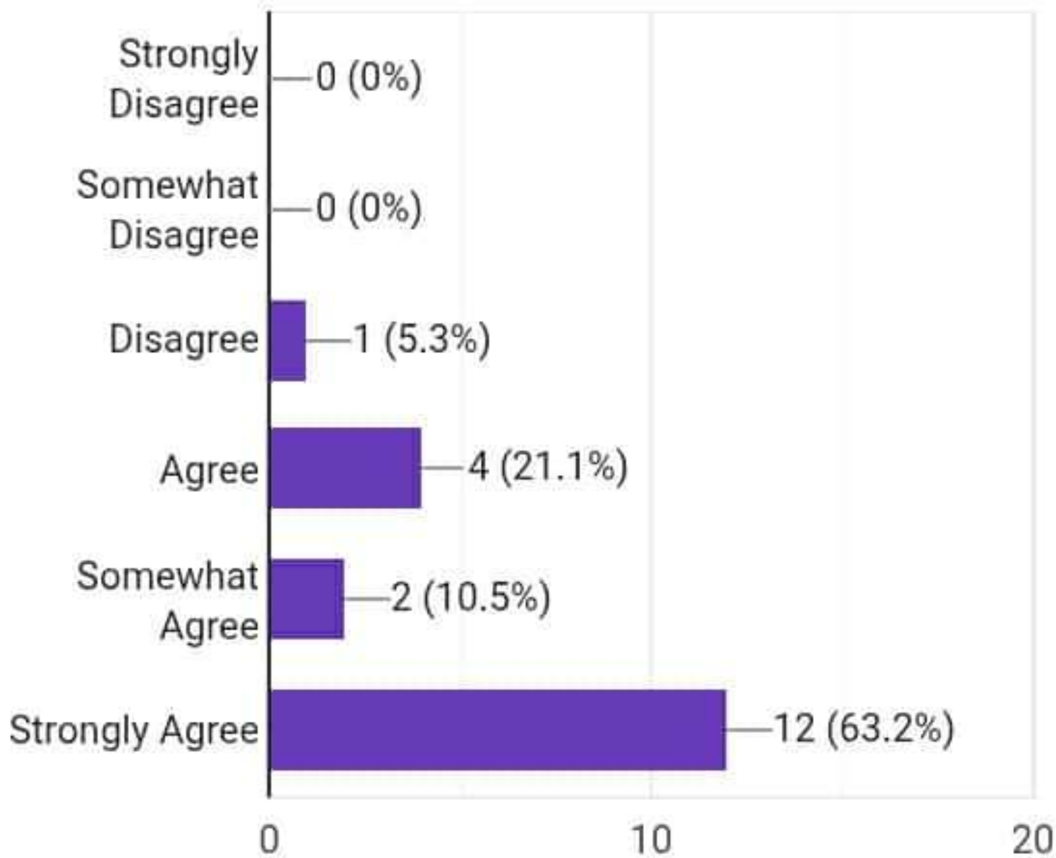
2. The **font size** is suitable and easy to read.

19 responses



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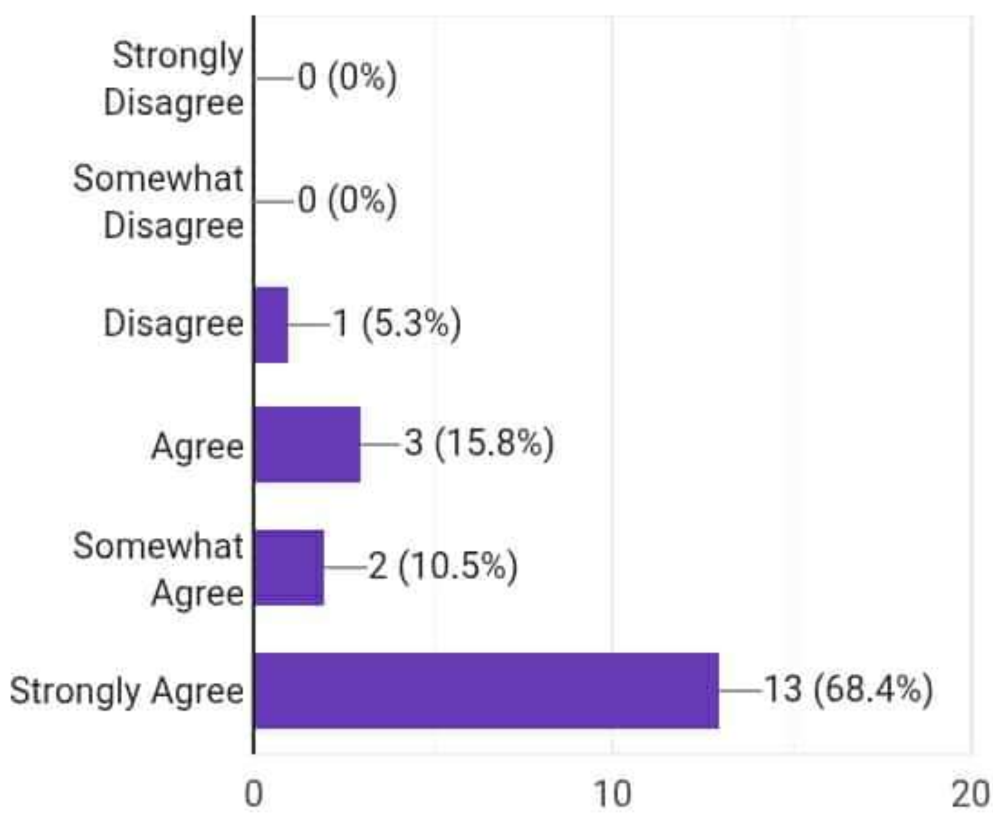
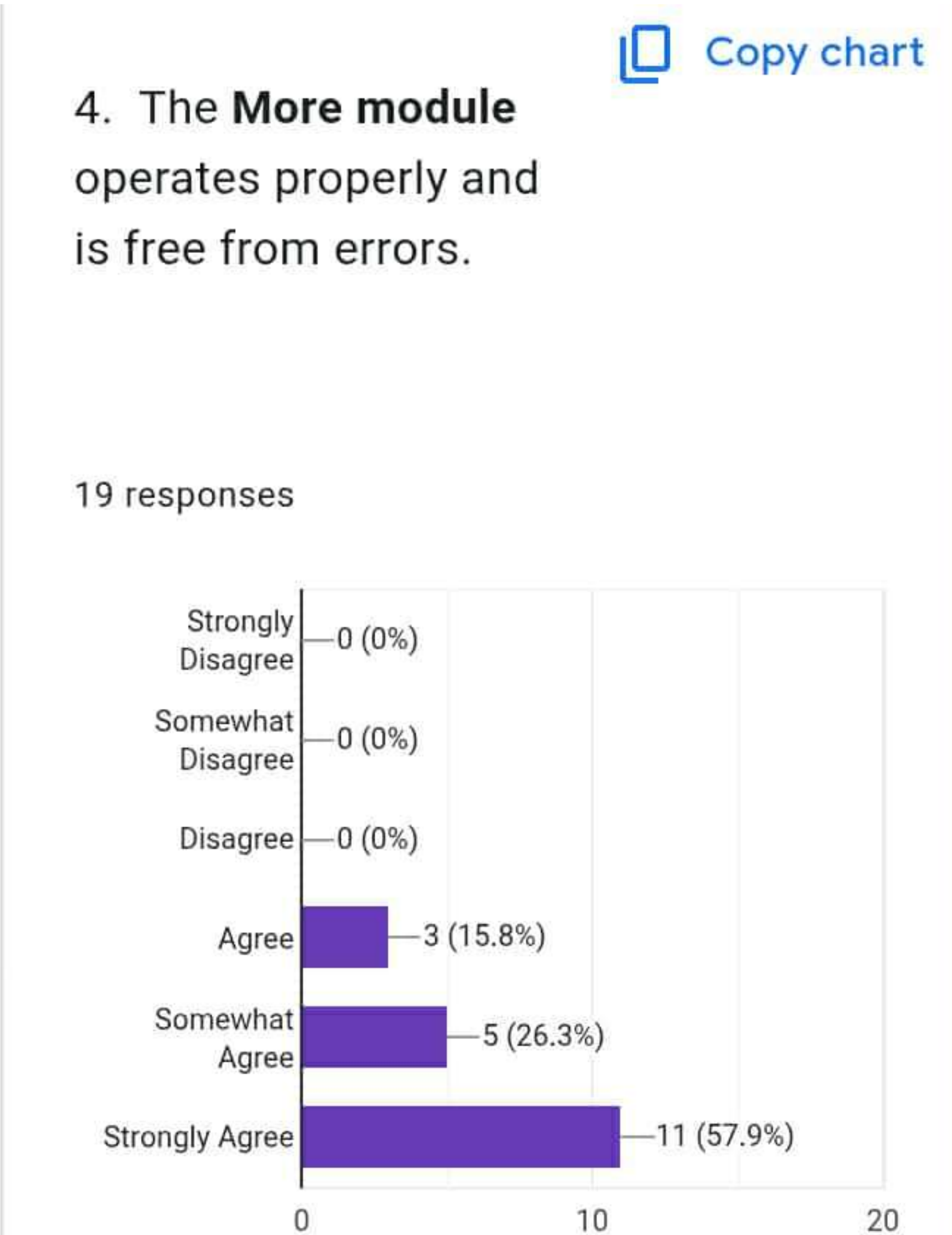
1. The **font type** used in AgriCare is appropriate.

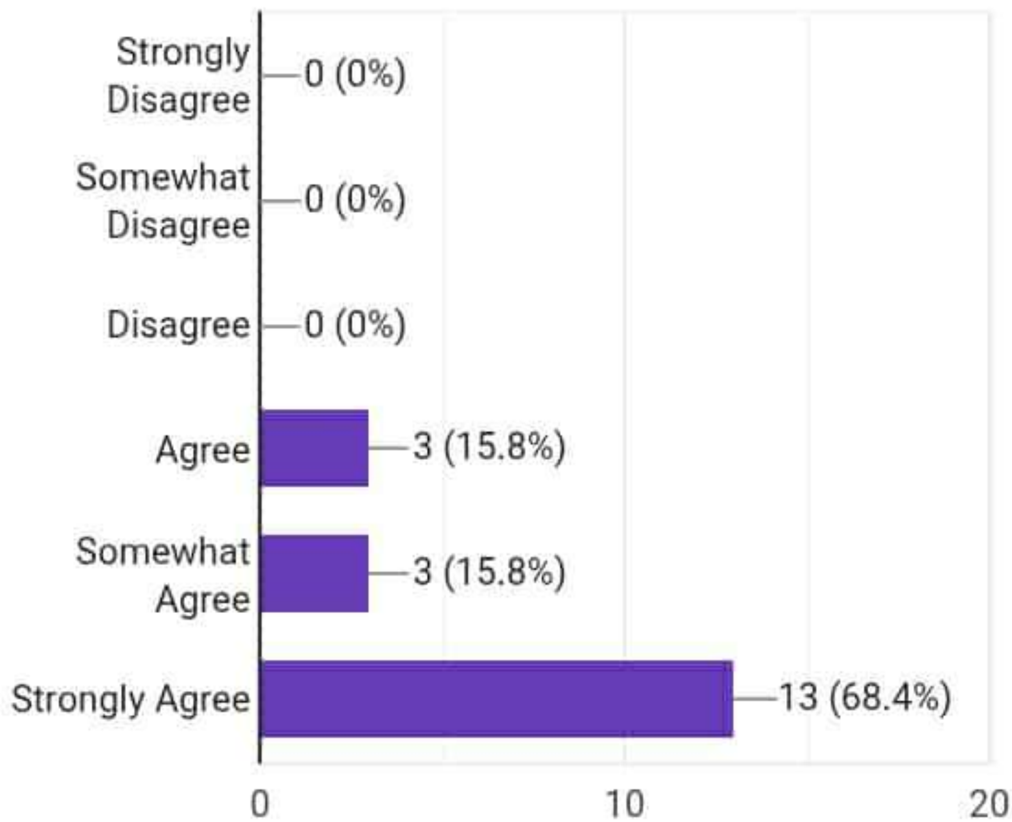
19 responses




Copy chart
4. The **More module** operates properly and is free from errors.
19 responses
Strongly Disagree
0 (0%)
Somewhat Disagree
0 (0%)
Disagree
0 (0%)
Agree
3 (15.8%)
Somewhat Agree
5 (26.3%)
Strongly Agree
11 (57.9%)
0
10
20

4. The **Farm Recording module** performs as expected and records data accurately without issues.

19 responses



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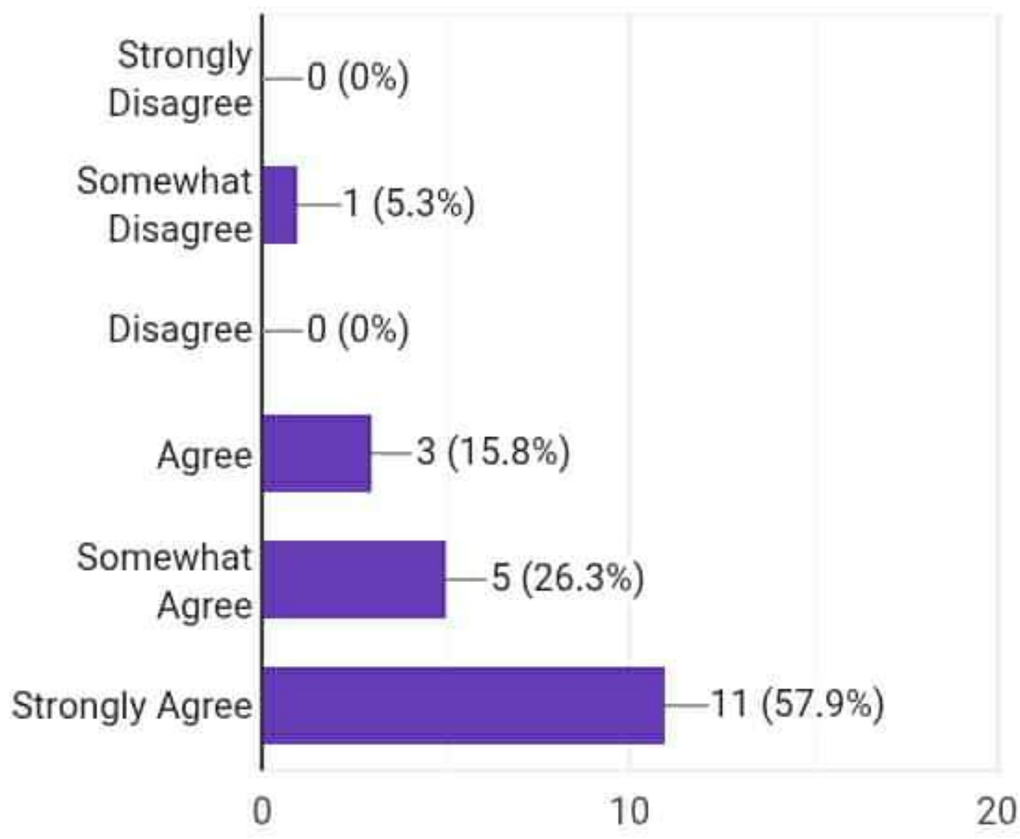
3. The **Weather module** works properly and provides accurate results without errors.

19 responses



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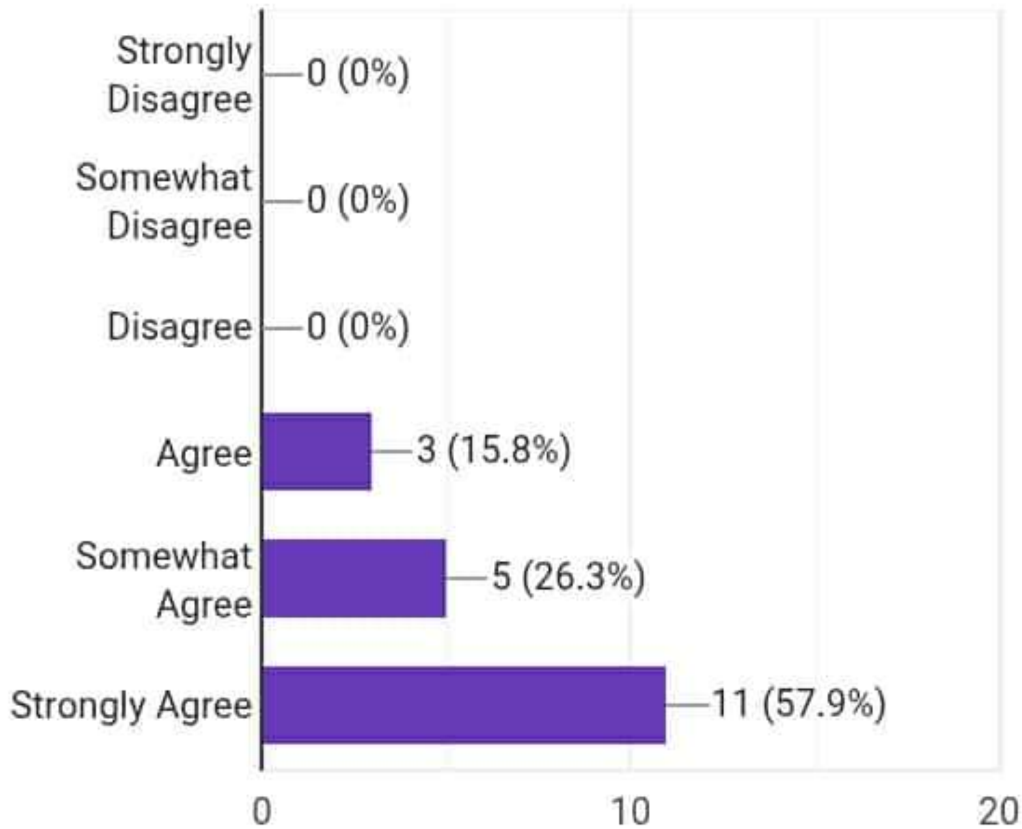
3.The **Livestock module** operates correctly and without errors.

19 responses



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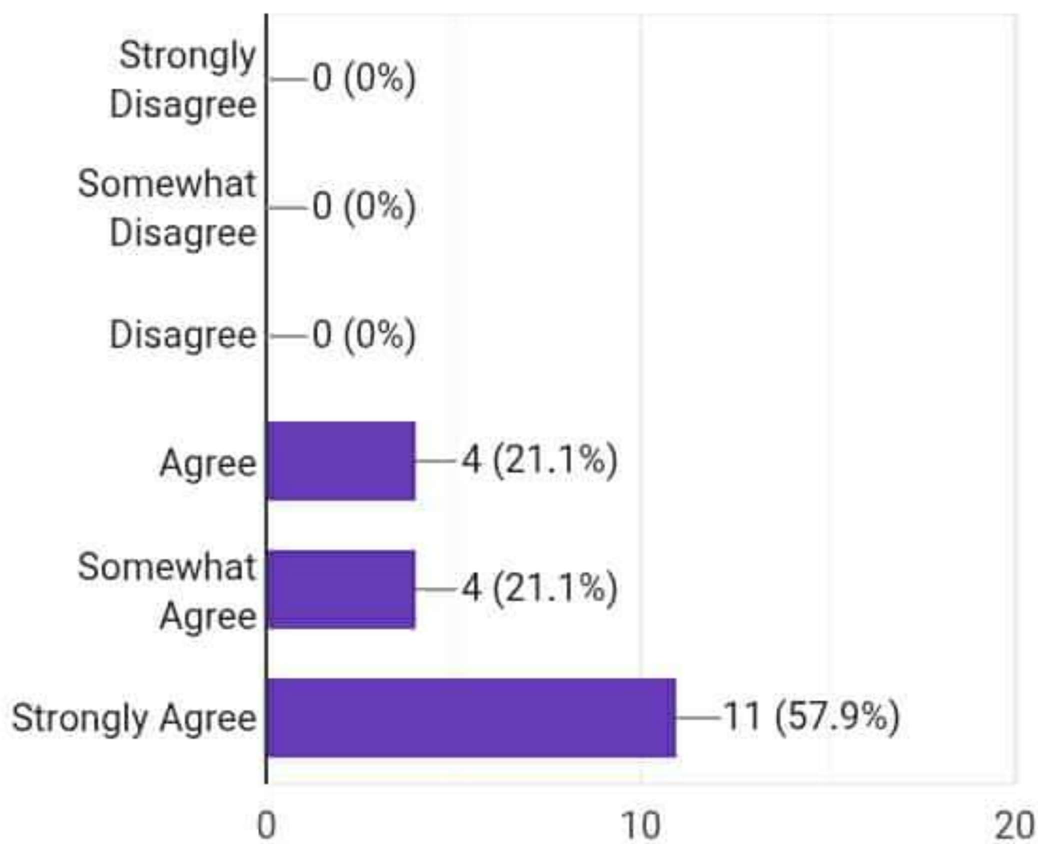
2.The **Livestock module** operates correctly and without errors.

19 responses



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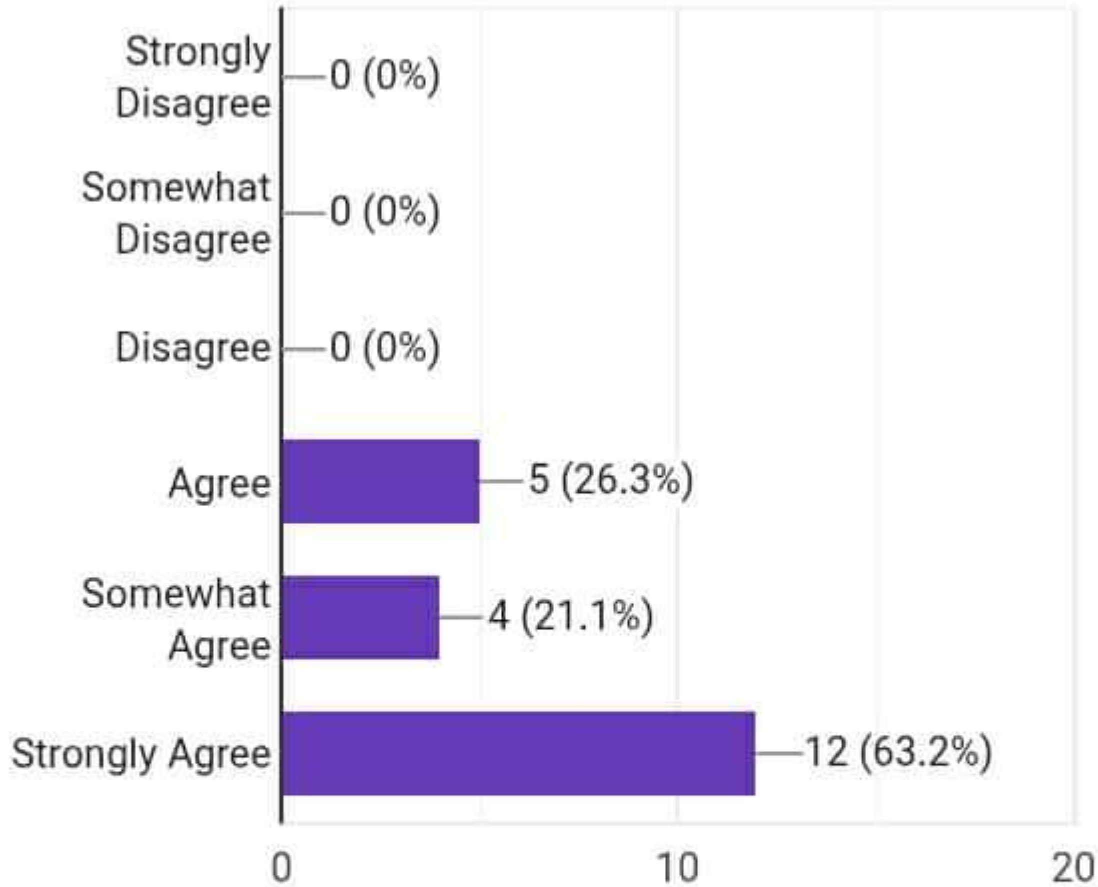
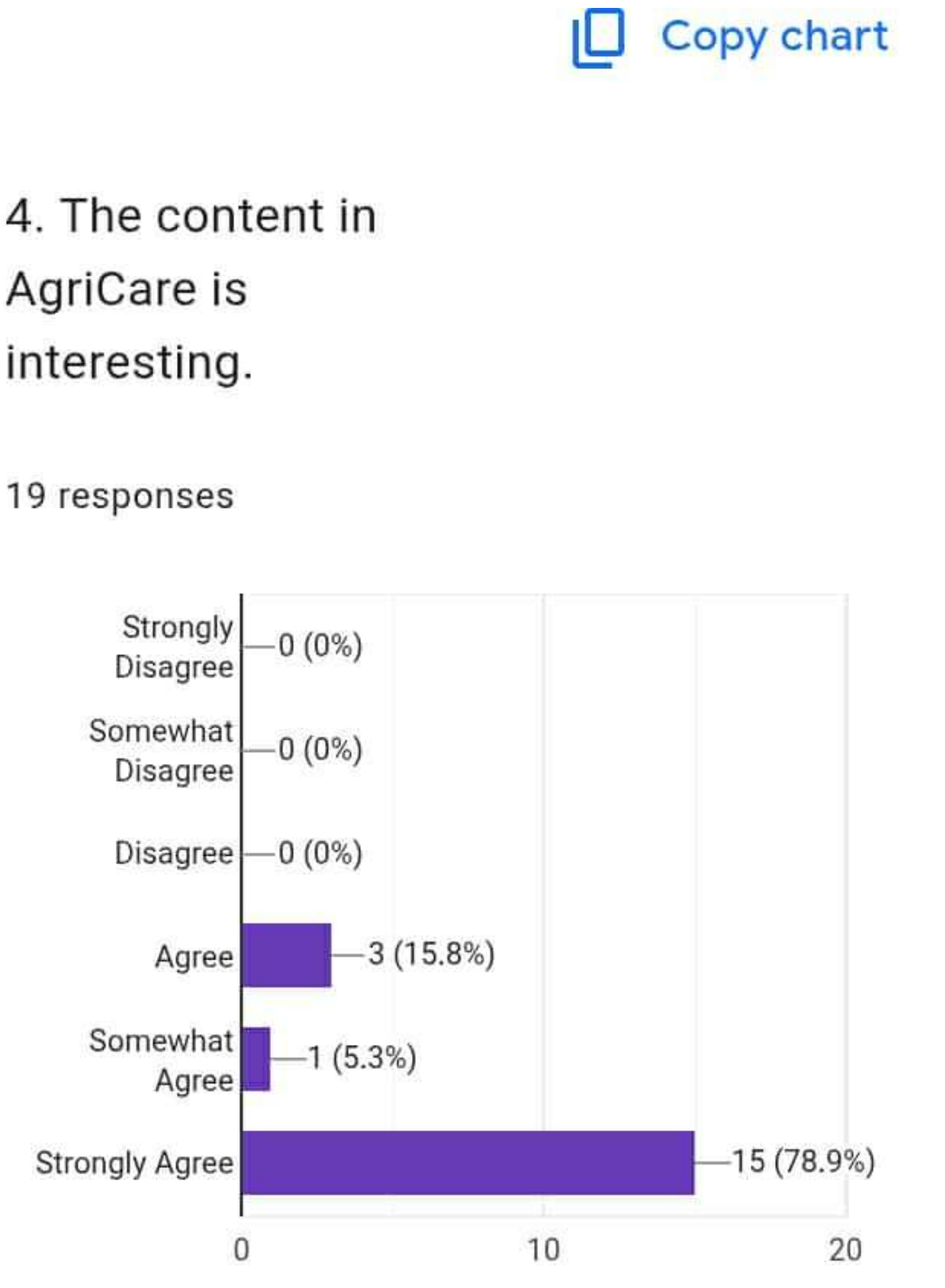
1. The **Crop module** functions smoothly without errors.

19 responses

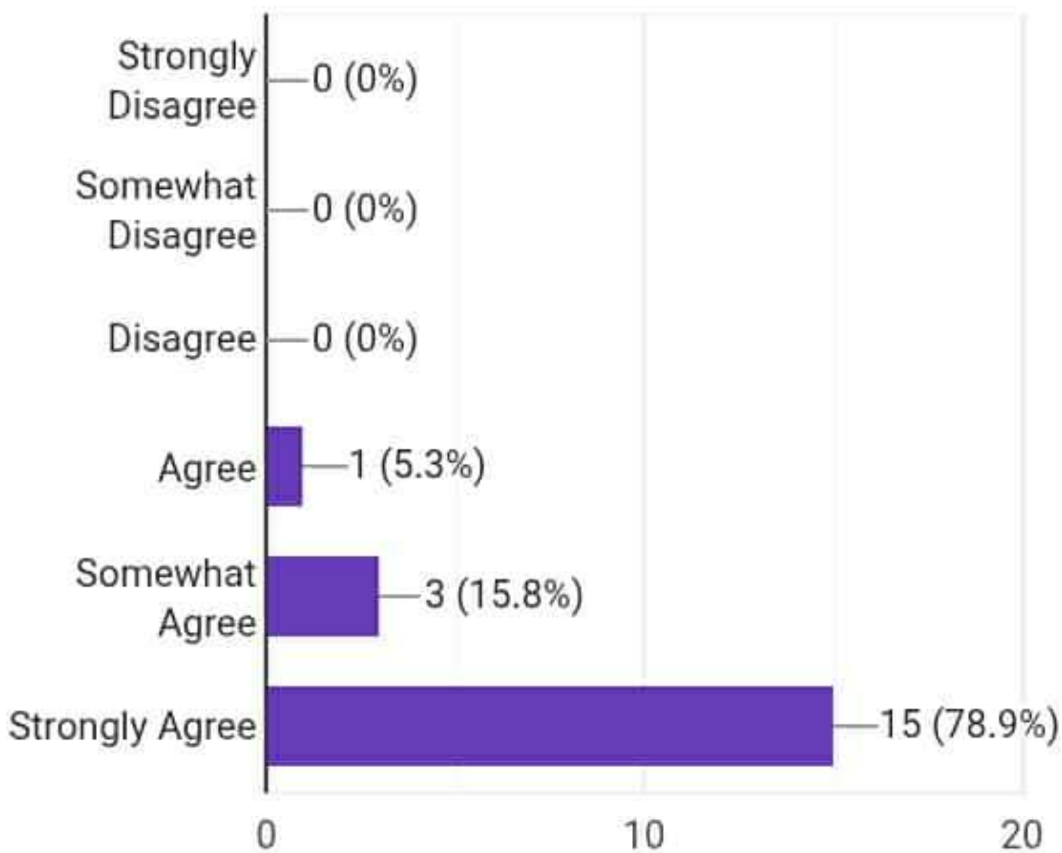


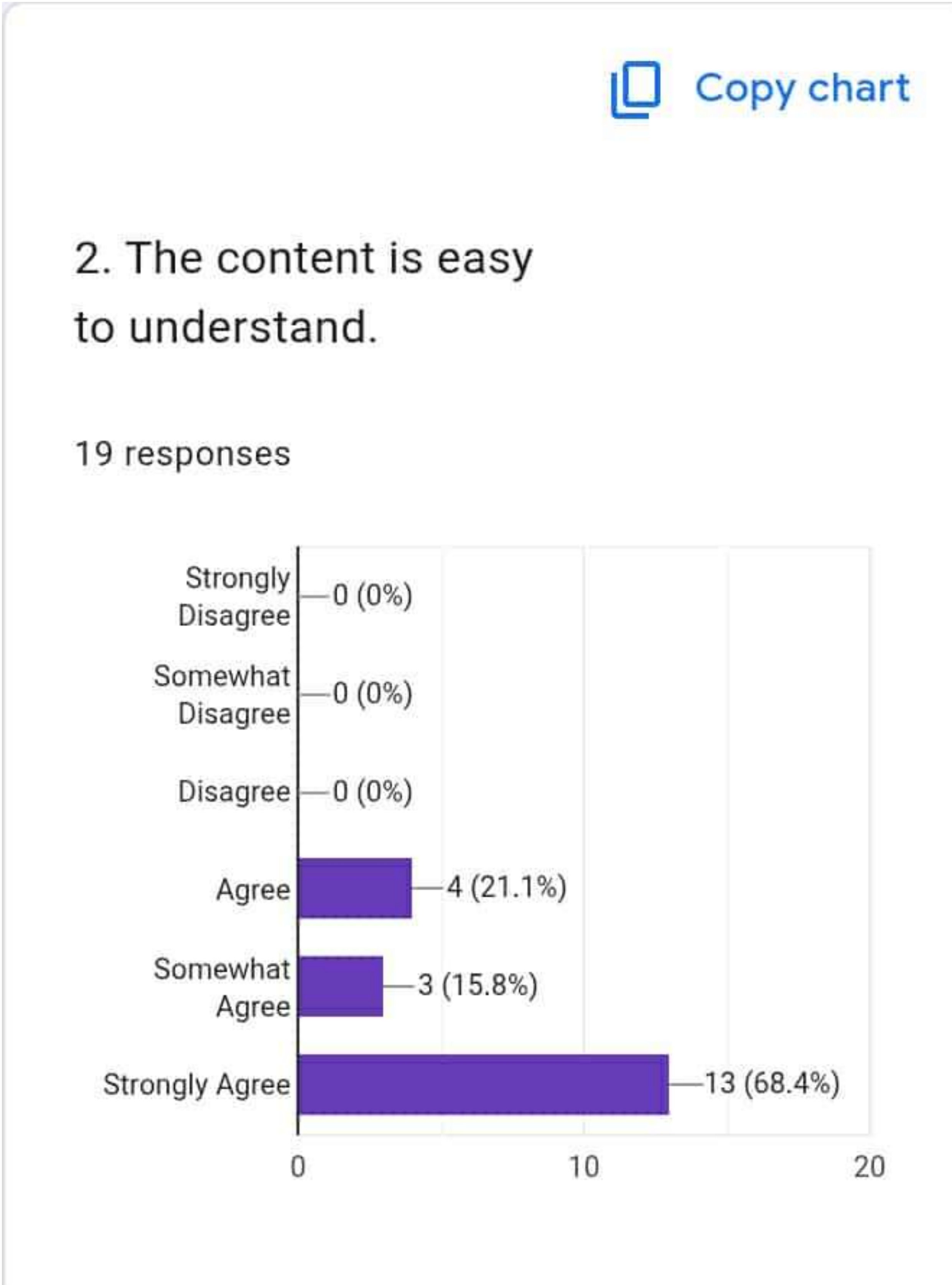

Copy chart
4. The content in AgriCare is interesting.
19 responses
Strongly Disagree
0 (0%)
Somewhat Disagree
0 (0%)
Disagree
0 (0%)
Agree
3 (15.8%)
Somewhat Agree
1 (5.3%)
Strongly Agree
15 (78.9%)
0
10
20

Copy chart

3. The content is related to Agriculture topic.

19 responses

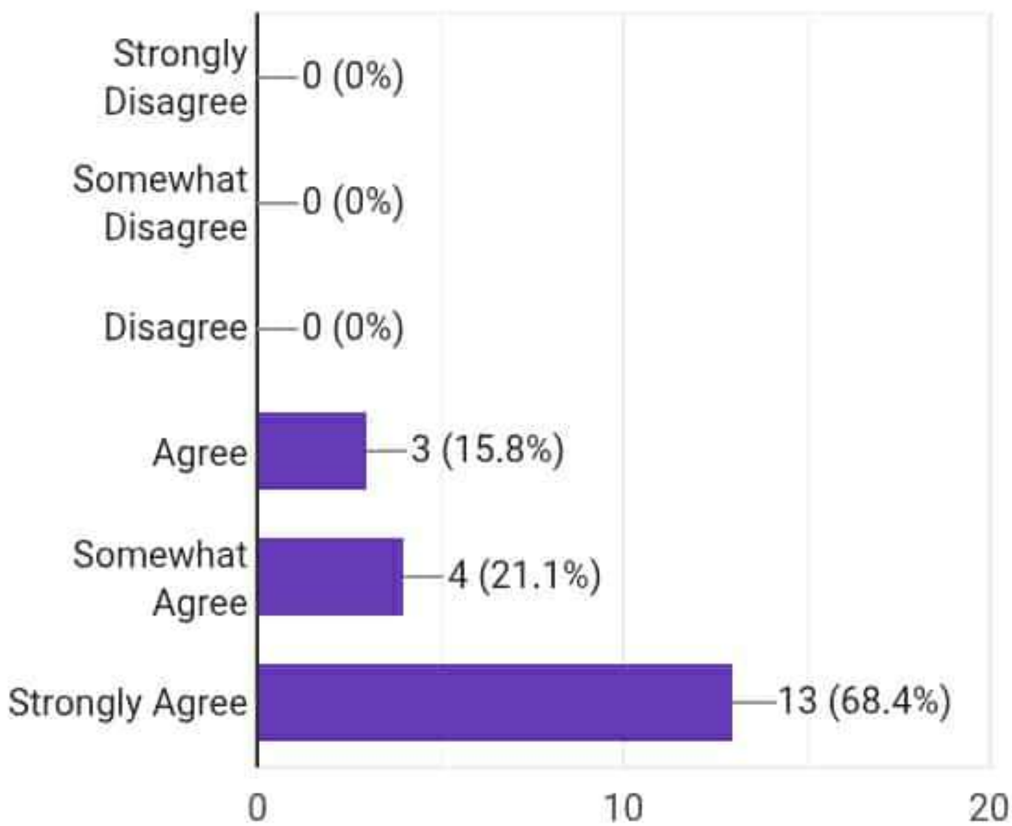



Copy chart
2. The content is easy to understand.
19 responses
Strongly Disagree
0 (0%)
Somewhat Disagree
0 (0%)
Disagree
0 (0%)
Agree
4 (21.1%)
Somewhat Agree
3 (15.8%)
Strongly Agree
13 (68.4%)
0
10
20

Copy chart

1. The content is clear.

19 responses



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3. AgriCare application is useful for limited or no internet connectivity.

19 responses